DELPHIX CONTINUOUS COMPLIANCE · DISCOVERY POLICY

# Brazil — LGPD (Lei 13.709/2018)

Discovery and masking of Brazilian personal data: CPF, CNPJ, PIS, voter card and CNS with valid check digits, quasi-identifiers, the sensitive personal data of art. 5, II, financial data and criminal records.

| 84 | 150 | 156 |
|---|---|---|
| DATA CLASSES | CLASSIFIERS | ALGORITHMS |

Discovery policy `BR – LGPD – v3`

Version 3 of October 12, 2026 · Assignment threshold 60%

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The first sections explain what the policy is for and how it works. The data class chapters then describe every data class: what it is, how its data is found and how it is masked, with examples. The appendices list every algorithm, file and classifier the policy contains.


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# About this discovery policy

This discovery policy finds the personal data in a Brazilian database and masks it, so that a non-production copy — for testing, development, training or analysis — no longer identifies anyone.

Masking names and CPFs alone is not enough: a birth date, a sex and a small municipality already point at one person. The policy works in the layers where the risk is:

- **Direct identifiers** — CPF, CNPJ, RG, CNH, voter card, PIS/NIS, work card, passport, foreigner documents, civil registry certificates, professional registrations, names, contact, address, accounts, cards and PIX key, devices, vehicles, property, free text.
- **Quasi-identifiers** — birth date, age, sex, municipality and IBGE code, CEP, neighbourhood, electoral zone, nationality, marital status, occupation, employer, education.
- **Sensitive personal data** — the categories of art. 5, II: racial or ethnic origin, religious conviction, political opinion, membership of a trade union or of an organization of religious, philosophical or political character, sexual life, genetic and biometric data — and, beyond the list, gender identity.
- **Health data** — sensitive under art. 5, II too, in a group of its own: National Health Card, medical records, ICD-10, medicines, clinical notes, tests, disability, occupational, reproductive and mental health.
- **Financial and credit data** — credit score and negative records, debt, income, assets, INSS benefits, social programmes and the Cadastro Único, income bracket, housing.
- **Criminal records and court cases** — records, prison status and case numbers.

In Data Control Tower the policy is one discovery policy with its classifiers, the data classes they vote for, the algorithms those data classes name — and the algorithms those lean on — and the lookup files they read. The appendices list each of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it on its own; an ambiguous one — `NOME`, `CARGO`, `RESULTADO`, `PROCESSO` — needs the values to agree. A column whose name means nothing (`CAMPO1`, `OBS`) is still found when the values have a recognizable format or vocabulary: a formatted CPF, a CNPJ, a CEP, a plate, a municipality.

Column names are recognized in Portuguese, in the prefixes of Brazilian data models (`NR_CPF`, `NM_CLIENTE`, `DT_NASCIMENTO`, `CO_MUNICIPIO_IBGE`, `CS_RACA`, `NU_CNS`, `SG_PARTIDO`) and in English.

> Before masking, inventory the binary columns (BLOB, images, PDF). Onboarding and proof-of-life processes often store a photo of the RG or CNH and a selfie: the document is still readable in the image, and the selfie is biometric data. Report those columns as residual risk.

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# The LGPD in brief

The General Personal Data Protection Law — Lei 13.709 of 14 August 2018 — has been in force since September 2020, and its administrative sanctions since 1 August 2021 (art. 65). It was amended by Laws 13.853/2019, 14.010/2020 and 14.460/2022 and by Lei 15.352 of 25 February 2026, which turned the authority into the **Agência Nacional de Proteção de Dados (ANPD)**, a special-regime agency with functional, technical, decision-making, administrative and financial autonomy (art. 55-A). Personal data protection has been a fundamental right since Constitutional Amendment 115/2022 (art. 5, LXXIX, of the Constitution).

What matters for a copy of production data:

- **Personal data** is information related to an identified or identifiable natural person (art. 5, I). The law protects natural persons; the data of a legal entity is outside it — but the CNPJ of an individual micro-entrepreneur leads to a person.
- **Sensitive personal data** is personal data on racial or ethnic origin, religious conviction, political opinion, membership of a trade union or of an organization of religious, philosophical or political character, data concerning health or sexual life, genetic or biometric data, when linked to a natural person (art. 5, II). The regime also covers any processing that **reveals** sensitive data and may cause harm (art. 11, § 1) — an INSS incapacity benefit reveals an illness.
- **Anonymized data** is not personal data, unless the anonymization can be reversed by the controller's own means or with reasonable effort, considering cost, time and the available technology (art. 12). See the next section.
- **Principles.** Purpose, adequacy and necessity (art. 6, I to III): using identified production data to test a system hardly passes the necessity principle when the test works with masked data. Security and prevention (art. 6, VII and VIII), and security measures from the design stage on (art. 46, § 2).
- **Children and adolescents.** Processing in their best interest, with specific consent of a parent for children (art. 14). The Digital Statute of Children and Adolescents (Lei 15.211/2025), in force since 17 March 2026, is regulated and enforced by the ANPD.
- **Security incidents** that may cause relevant risk or harm are reported to the ANPD and to the data subjects (art. 48) within **three business days** (Resolução CD/ANPD 15/2024). Having made the data unintelligible to those without access counts in the assessment of severity (art. 48, § 3): a masked copy that leaks is not the production database leaking.
- **Sanctions.** Warning, a fine of up to 2% of revenue in Brazil capped at R$ 50 million per infraction, daily fines, publicity, blocking and deletion of the data, suspension of the database or of the processing, and a ban on processing data (art. 52), calculated under Resolução CD/ANPD 4/2023.

Two subjects outside the text of the LGPD are in the policy too. **Financial data** is not sensitive under the law, but the operations of financial institutions are under secrecy (Lei Complementar 105/2001), credit databases have a law of their own (Lei 12.414/2011, the positive credit register) and credit protection is a legal basis of its own (art. 7, X). **Criminal records** are not on the list of art. 5, II either, but the Superior Labour Court held that requiring a job candidate's criminal record certificate without a legal or job-related reason gives rise to moral damages (IRR Tema 1): it is data that exposes people to discrimination, and gets the care of sensitive data.

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# Anonymization, pseudonymization and non-production copies

The LGPD tells two results apart. **Anonymization** is the use of reasonable technical means by which data loses the possibility of being associated, directly or indirectly, with an individual (art. 5, XI); anonymized data is outside the law (art. 12). **Pseudonymization** is processing by which data loses that possibility *except through additional information kept separately by the controller in a controlled and secure environment* (art. 13, § 4); pseudonymized data is still personal data.

The algorithms of this policy are deterministic for the Delphix masking key: the same CPF masks the same way in every table, which keeps joins working. So for whoever holds the key **and** production, the masked copy can be linked back — it is pseudonymized. For whoever holds only the copy, without the key or production, direct identifiers were replaced and quasi-identifiers generalized, and reversing the process is beyond reasonable effort.

- Keep the engine key out of reach of the users of the masked copy: it is the *additional information kept separately* of art. 13, § 4.

- Do not hand the copy to people who also access production without need.
- Review quasi-identifiers the policy does not know. The ANPD recalls that postal code, sex and birth date were enough to identify 87% of the population of the United States in a 2000 study.

In January 2024 the ANPD put a preliminary study and a guide on anonymization and pseudonymization to public consultation; in December 2025 the subject was still on the Regulatory Agenda 2025–2026 (item 9). The study describes techniques — suppression, generalization, noise addition, permutation, masking, substitution, tokenization — and warns that partial masking of the CPF ( `***.111.111-**` ), common in publications, can be reconstructed when each source hides different parts. This policy replaces the whole identifier with another valid one instead of hiding part of it.

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# How sensitive data is discovered

A discovery job looks at every column of its rule set and asks each classifier of the discovery policy how confident it is that the column belongs to its data class. The policy uses the four classifier frameworks of Delphix:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the data class out. It never assigns a data class on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per data class, and the column is assigned to the data class with the highest confidence when it reaches the assignment threshold of the discovery policy — **60%** here. Values that do not match a format or a list count against their data class by the classifier's reject strength.

These are the names Data Control Tower gives the objects: data classes, classifiers and a discovery policy, run by a discovery job. A Continuous Compliance Engine used on its own calls the same objects domains, classifiers and a profile set, run by a profiling job; everything in this document applies to both.

Databases pad values: a CHAR column fills them with spaces up to its length, and a numeric code kept as text often comes with zeros on the left. Discovery tests values as they come, so every format pattern first removes the spaces around the value, patterns of digits also accept zeros on the left, lists of single words are looked up word by word, and lists of names made of several words (places, categories) are held as a single value-format pattern. A column whose name says nothing — `CAMPO1` , `FLD_07` , the short names of packaged systems — is found by its values alone wherever their format is distinctive: documents with a check digit, e-mail addresses, telephone numbers, cards, places, names.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters: `rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA` . In them, `_?` is an optional underscore, `(a|b)` either form, `s?` an optional letter and `[a-z_]*` any continuation.

> Discovery proposes; it does not decide. Review the data classes assigned in the rule set before running a masking job, and look for columns the policy left unassigned.

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# How data is masked

Each data class names one algorithm, which a masking job applies to every column the rule set assigns to the data class. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each data class shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine and compliance node uses the same keys.

Where an algorithm that comes with Delphix already does the job — CM Numeric for amounts, Credit Card, the blood group lookup — the policy names it instead of creating one of its own. Those algorithms appear in the trees by their `dlpx-core:` name, the one an engine uses; Data Control Tower lists them without the prefix ( `CM Numeric` , `CreditCard` , `BloodType SL` ). They are on every engine and in every Data Control Tower, and are not listed in Appendix A.

Padding stays where it is: the spaces around a value are kept, and zeros on the left of a number are not masked — `0012345678` becomes `00` followed by what `12345678` becomes. A value kept in a zero-filled CHAR column still joins the same value kept as a number.

> The masked examples in this document were produced with one particular key. With other keys the masked values differ; their shape does not.

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# Essential and extended packs

The policy can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create just its data classes, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the discovery policy.

| PACK | DATA CLASSES | CLASSIFIERS | ALGORITHMS | FILES |
|---|---|---|---|---|
| **Essential** | 13 | 27 | 20 | 5 |
| **Extended** | 84 | 150 | 156 | 74 |

The essential pack holds these data classes, marked **Essential** in the tables and chapters:

- CPF — `BR_L1_CPF`
- RG — `BR_L1_RG`
- CNH — `BR_L1_CNH`
- Passport — `BR_L1_PASSAPORTE`
- Given name — `BR_L1_NOME`
- Surname — `BR_L1_SOBRENOME`
- Full name — `BR_L1_NOME_COMPLETO`
- E-mail — `BR_L1_EMAIL`
- Phone — `BR_L1_TELEFONE`
- Address — `BR_L1_ENDERECO`
- Number and complement — `BR_L1_ENDERECO_COMPLEMENTO`

- Birth date — `BR_L2_DATA_NASCIMENTO`
- CEP (postal code) — `BR_L2_CEP`

> With fewer data classes competing, a column that a data class of the extended pack would claim by its name can be proposed for an essential data class instead — a numeric code, for instance, can look like a telephone number. Review the discovery result.

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# Data classes at a glance

Every data class of the policy, how it is found and the algorithm that masks it.

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **DIRECT IDENTIFIERS** | | |
| **CPF** ESSENTIAL<br>BR_L1_CPF | COLUMN NAME · DATA TYPE · VALUE FORMAT | BR_CPF |
| **CNPJ**<br>BR_L1_CNPJ | COLUMN NAME · DATA TYPE · VALUE FORMAT | BR_CNPJ |
| **RG** ESSENTIAL<br>BR_L1_RG | COLUMN NAME · VALUE FORMAT | BR_CM_DIGITOS |
| **CNH** ESSENTIAL<br>BR_L1_CNH | COLUMN NAME | BR_CM_ALFANUM |
| **Voter card**<br>BR_L1_TITULO_ELEITOR | COLUMN NAME · VALUE FORMAT | BR_TITULO |
| **PIS, PASEP, NIT and NIS**<br>BR_L1_PIS_NIS | COLUMN NAME · VALUE FORMAT | BR_PIS |
| **Work card**<br>BR_L1_CTPS | COLUMN NAME | BR_CM_ALFANUM |
| **Passport** ESSENTIAL<br>BR_L1_PASSAPORTE | COLUMN NAME · VALUE FORMAT | BR_CM_ALFANUM |
| **Foreigner documents**<br>BR_L1_DOCUMENTO_ESTRANGEIRO | COLUMN NAME | BR_CM_ALFANUM |
| **Civil registry certificates**<br>BR_L1_CERTIDAO | COLUMN NAME · VALUE FORMAT | BR_CM_ALFANUM |
| **Professional registration**<br>BR_L1_REGISTRO_PROFISSIONAL | COLUMN NAME | BR_REGISTRO_PROFISSIONAL |
| **Given name** ESSENTIAL<br>BR_L1_NOME | COLUMN NAME · KNOWN VALUES | BR_NOMES |
| **Surname** ESSENTIAL<br>BR_L1_SOBRENOME | COLUMN NAME · KNOWN VALUES | BR_SOBRENOMES |
| **Full name** ESSENTIAL<br>BR_L1_NOME_COMPLETO | COLUMN NAME · KNOWN VALUES | BR_NOME_COMPLETO |
| **E-mail** ESSENTIAL<br>BR_L1_EMAIL | COLUMN NAME · DATA TYPE · VALUE FORMAT | BR_EMAIL |
| **Phone** ESSENTIAL<br>BR_L1_TELEFONE | COLUMN NAME · VALUE FORMAT | BR_TELEFONE |
| **Address** ESSENTIAL<br>BR_L1_ENDERECO | COLUMN NAME · DATA TYPE · VALUE FORMAT | BR_ENDERECO |
| **Number and complement** ESSENTIAL<br>BR_L1_ENDERECO_COMPLEMENTO | COLUMN NAME | BR_CM_ALFANUM |
| **Branch and account**<br>BR_L1_CONTA_BANCARIA | COLUMN NAME | BR_CM_DIGITOS |

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **Payment card** `BR_L1_CARTAO` | COLUMN NAME · VALUE FORMAT | `BR_CARTAO` |
| **PIX key** `BR_L1_CHAVE_PIX` | COLUMN NAME · VALUE FORMAT | `BR_CHAVE_PIX` |
| **IP address** `BR_L1_IP` | COLUMN NAME · VALUE FORMAT | `BR_IP` |
| **Device identifier** `BR_L1_DISPOSITIVO` | COLUMN NAME · VALUE FORMAT | `BR_CM_HEX` |
| **Licence plate** `BR_L1_PLACA` | COLUMN NAME · VALUE FORMAT | `BR_PLACA` |
| **RENAVAM** `BR_L1_RENAVAM` | COLUMN NAME | `BR_RENAVAM` |
| **Chassis** `BR_L1_CHASSI` | COLUMN NAME · VALUE FORMAT | `BR_CM_ALFANUM` |
| **Real estate** `BR_L1_IMOVEL` | COLUMN NAME | `BR_CM_ALFANUM` |
| **Contract and enrolment numbers** `BR_L1_CONTRATO` | COLUMN NAME | `BR_CM_ALFANUM` |
| **User and social media** `BR_L1_USUARIO` | COLUMN NAME | `BR_CM_ALFANUM` |
| **Credentials** `BR_L1_CREDENCIAL` | COLUMN NAME · VALUE FORMAT | `BR_REDIGIR` |
| **Geolocation** `BR_L1_GEOLOCALIZACAO` | COLUMN NAME · VALUE FORMAT | `BR_COORDENADA` |
| **Free text** `BR_L1_TEXTO_LIVRE` | COLUMN NAME · DATA TYPE · VALUE FORMAT | `BR_TEXTO_LIVRE` |
| **QUASI-IDENTIFIERS** | | |
| **Birth date** ESSENTIAL `BR_L2_DATA_NASCIMENTO` | COLUMN NAME · DATA TYPE | `BR_DATA_NASCIMENTO` |
| **Birth year** `BR_L2_ANO_NASCIMENTO` | COLUMN NAME · DATA TYPE | `BR_ANO` |
| **Age** `BR_L2_IDADE` | COLUMN NAME · DATA TYPE | `BR_IDADE` |
| **Personal event dates** `BR_L2_DATA_EVENTO` | COLUMN NAME · DATA TYPE | `BR_DATA_EVENTO` |
| **Sex** `BR_L2_SEXO` | COLUMN NAME · KNOWN VALUES | `BR_SEXO` |
| **Municipality** `BR_L2_MUNICIPIO` | COLUMN NAME · VALUE FORMAT | `BR_MUNICIPIO` |
| **Municipality code (IBGE)** `BR_L2_CODIGO_MUNICIPIO` | COLUMN NAME · VALUE FORMAT | `BR_CODIGO_MUNICIPIO` |
| **CEP (postal code)** ESSENTIAL `BR_L2_CEP` | COLUMN NAME · VALUE FORMAT | `BR_CEP` |
| **Neighbourhood** `BR_L2_BAIRRO` | COLUMN NAME | `BR_BAIRRO` |

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **Electoral zone and section**<br>BR_L2_ZONA_SECAO_ELEITORAL | COLUMN NAME | BR_CM_DIGITOS |
| **Nationality**<br>BR_L2_NACIONALIDADE | COLUMN NAME KNOWN VALUES | BR_NACIONALIDADE |
| **Marital status**<br>BR_L2_ESTADO_CIVIL | COLUMN NAME VALUE FORMAT | BR_ESTADO_CIVIL |
| **Occupation**<br>BR_L2_OCUPACAO | COLUMN NAME | BR_OCUPACAO_OU_CBO |
| **Employer**<br>BR_L2_EMPREGADOR | COLUMN NAME | BR_EMPREGADOR |
| **Education**<br>BR_L2_ESCOLARIDADE | COLUMN NAME VALUE FORMAT | BR_ESCOLARIDADE |
| **School or educational institution**<br>BR_L2_ESCOLA | COLUMN NAME | BR_ESCOLA |
| **Dependants and children**<br>BR_L2_DEPENDENTES | COLUMN NAME DATA TYPE | BR_CONTAGEM_LIMITADA |
| **SENSITIVE PERSONAL DATA** | | |
| **Colour or race**<br>BR_L3_RACA_COR | COLUMN NAME VALUE FORMAT | BR_RACA_COR |
| **Indigenous people and traditional community**<br>BR_L3_ETNIA | COLUMN NAME VALUE FORMAT | BR_ETNIA |
| **Religion**<br>BR_L3_RELIGIAO | COLUMN NAME VALUE FORMAT | BR_RELIGIAO |
| **Convictions and religious, philosophical or political organizations**<br>BR_L3_CONVICCAO_ORGANIZACAO | COLUMN NAME | BR_CATEGORIA_SUPRIMIDA |
| **Political opinion**<br>BR_L3_OPINIAO_POLITICA | COLUMN NAME VALUE FORMAT | BR_OPINIAO_POLITICA |
| **Party membership**<br>BR_L3_FILIACAO_PARTIDARIA | COLUMN NAME VALUE FORMAT | BR_PARTIDO |
| **Trade union membership**<br>BR_L3_FILIACAO_SINDICAL | COLUMN NAME VALUE FORMAT | BR_CATEGORIA_SUPRIMIDA |
| **Genetic data**<br>BR_L3_GENETICO | COLUMN NAME VALUE FORMAT | BR_REDIGIR |
| **Biometric data**<br>BR_L3_BIOMETRICO | COLUMN NAME DATA TYPE | BR_REDIGIR |
| **Sexual orientation**<br>BR_L3_ORIENTACAO_SEXUAL | COLUMN NAME VALUE FORMAT | BR_ORIENTACAO_SEXUAL |
| **Sexual life**<br>BR_L3_VIDA_SEXUAL | COLUMN NAME | BR_CATEGORIA_SUPRIMIDA |
| **Gender identity**<br>BR_L3_IDENTIDADE_GENERO | COLUMN NAME VALUE FORMAT | BR_IDENTIDADE_GENERO |
| **HEALTH DATA** | | |
| **National Health Card**<br>BR_SAUDE_CNS | COLUMN NAME VALUE FORMAT | BR_CNS |

| DATA CLASS | FOUND BY | ALGORITHM |
|---|---|---|
| **Health identifiers**<br>`BR_SAUDE_IDENTIFICADOR` | COLUMN NAME | `BR_CM_ALFANUM` |
| **Diagnosis and ICD**<br>`BR_SAUDE_DIAGNOSTICO` | COLUMN NAME · VALUE FORMAT | `BR_DIAGNOSTICO` |
| **Medicines**<br>`BR_SAUDE_MEDICAMENTO` | COLUMN NAME · VALUE FORMAT | `BR_MEDICAMENTO` |
| **Clinical text**<br>`BR_SAUDE_TEXTO_CLINICO` | COLUMN NAME · DATA TYPE | `BR_TEXTO_LIVRE` |
| **Test results**<br>`BR_SAUDE_RESULTADO_EXAME` | COLUMN NAME | `BR_CM_ALFANUM` |
| **Blood group**<br>`BR_SAUDE_TIPO_SANGUINEO` | COLUMN NAME · VALUE FORMAT | `dlpx-core:BloodTypeSL` |
| **Disability**<br>`BR_SAUDE_DEFICIENCIA` | COLUMN NAME · VALUE FORMAT | `BR_DEFICIENCIA` |
| **Occupational health**<br>`BR_SAUDE_OCUPACIONAL` | COLUMN NAME · VALUE FORMAT | `BR_SAUDE_OCUPACIONAL` |
| **Sexual and reproductive health**<br>`BR_SAUDE_REPRODUTIVA` | COLUMN NAME | `BR_CATEGORIA_SUPRIMIDA` |
| **Mental health**<br>`BR_SAUDE_MENTAL` | COLUMN NAME | `BR_CATEGORIA_SUPRIMIDA` |
| **FINANCIAL AND CREDIT DATA** | | |
| **Credit and negative records**<br>`BR_FIN_CREDITO` | COLUMN NAME | `BR_CREDITO` |
| **Debt**<br>`BR_FIN_DIVIDA` | COLUMN NAME · DATA TYPE | `dlpx-core:CM Numeric` |
| **Income and salary**<br>`BR_FIN_RENDA` | COLUMN NAME · DATA TYPE | `dlpx-core:CM Numeric` |
| **Assets**<br>`BR_FIN_PATRIMONIO` | COLUMN NAME · DATA TYPE | `dlpx-core:CM Numeric` |
| **Benefit and pension amount**<br>`BR_FIN_VALOR_BENEFICIO` | COLUMN NAME · DATA TYPE | `dlpx-core:CM Numeric` |
| **INSS benefit type**<br>`BR_FIN_BENEFICIO_INSS` | COLUMN NAME · VALUE FORMAT | `BR_BENEFICIO_INSS` |
| **Benefit number**<br>`BR_FIN_NUMERO_BENEFICIO` | COLUMN NAME · VALUE FORMAT | `BR_CM_DIGITOS` |
| **Social programmes**<br>`BR_FIN_PROGRAMA_SOCIAL` | COLUMN NAME · VALUE FORMAT | `BR_PROGRAMA_SOCIAL` |
| **Income bracket and social class**<br>`BR_FIN_FAIXA_RENDA` | COLUMN NAME · VALUE FORMAT | `BR_FAIXA_RENDA` |
| **Housing**<br>`BR_FIN_MORADIA` | COLUMN NAME · VALUE FORMAT | `BR_MORADIA` |
| **CRIMINAL RECORDS AND COURT CASES** | | |
| **Criminal records**<br>`BR_PENAL_ANTECEDENTES` | COLUMN NAME | `BR_ANTECEDENTES` |

| DATA CLASS | FOUND BY | ALGORITHM |
| --- | --- | --- |
| **Case number**<br>`BR_PENAL_PROCESSO` | COLUMN NAME VALUE FORMAT | `BR_PROCESSO` |

PART 8

# Direct identifiers

Data that identifies a person on its own. The CPF is the key of almost every Brazilian record — and, under Lei 14.534/2023, it is enough to identify a citizen in public services and is the number of the new National Identity Card. It is masked the same way everywhere and keeps valid check digits, because applications validate them. The same goes for CNPJ, PIS, voter card and RENAVAM.

## 8.1 CPF

BR_L1_CPF

ESSENTIAL | COLUMN NAME | DATA TYPE | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The taxpayer number of the Cadastro de Pessoas Físicas of the Receita Federal: nine digits and two check digits, `529.982.247-25`. The ninth digit is the tax region that issued the number. Under Lei 14.534/2023 it is enough to identify a citizen in public-service databases, and the National Identity Card uses the CPF as its number. It appears formatted, as bare digits or, in a numeric column, without its leading zeros.

`CPF_CNPJ` and `NR_DOCUMENTO` columns hold CPFs and CNPJs together, and fall here too.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_CPF — Path

| | |
|---|---|
| 90% | `cpf[a-z0-9_]*` `c_?p_?f` `n(r\|u\|um\|o\|ro\|umero)_?cpf` `cpf_?(cnpj\|cgc)` `cadastro_?(de_?)?pessoas?_?fisicas?` `documento_?fiscal` `(cliente\|cli\|clie\|pessoa\|pes\|func\|funcionario\|colaborador\|servidor\|empregado\|segurado\|beneficiario\|benef\|titular\|dependente\|paciente\|aluno\|contribuinte\|responsavel\|resp\|socio\|motorista\|conductor\|tomador\|devedor\|avalista\|fiador\|consumidor\|comprador\|eleitor\|candidato\|usuario)_?cpf` |
| 55% | `tax_?id` `taxpayer_?id` `documento_?(cliente\|pessoa)` `doc_?(cliente\|pessoa)` `nr_?documento` `num_?documento` `numero_?documento` `n(r\|u)_?doc` |

**DATA TYPE** BR_L1_CPF — Type

Accepts text (length 3 or more) or number (length 3 or more); any other type rules the data class out.

**VALUE FORMAT** BR_L1_CPF — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{3}\.\d{3}\.\d{3}-\d{2})` | 100% | |
| `0*(?:\d{11})` | 50% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

The first nine digits are mapped digit by digit — the tax region changes too — and the two check digits are recomputed: modulus 11, weights 10 to 2 for the first and 11 to 2 for the second, a remainder of 0 or 1 giving 0. Every masked CPF is valid, the format is kept and the mapping is one-to-one: a CPF used as a key does not collide.

A number of up to ten digits is taken for a CPF that lost its zeros and comes out with eleven. A value shaped like a CNPJ is masked as a CNPJ.

`BR_CPF` REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

`dlpx-core:Brazil CPF` CHECK DIGIT

Comes with Delphix and is used as it ships. Masks the nine digits of the CPF and recomputes both check digits; a CPF stored without its zeros on the left comes back with eleven digits.

`BR_CNPJ_VALIDO` **CHECK DIGIT**

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

> `BR_CNPJ_DV1` **CHECK DIGIT**
>
> Masks the digits with `BR_CNPJ_RAIZ` and recomputes a modulus 11 check digit with weights `5 4 3 2 9 8 7 6 5 4 3 2`.
>
> > `BR_CNPJ_RAIZ` **REGEX DECOMPOSE**
> >
> > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> >
> > > `BR_CM_DIGITOS_LETRAS_LARGURA_FIXA` **CHARACTER MAPPING**
> > >
> > > Maps each character to another of its group (2 groups), keeping length and format.

`BR_CNPJ_VALIDO_ZEROS` **CHECK DIGIT**

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

`BR_CM_DIGITOS` **CHARACTER MAPPING**

Maps each character to another of its group (1 groups), keeping length and format.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| 529.982.247-25 | → | **410.719.628-33** |
| 52998224725 | → | **41071962833** |
| 11.222.333/0001-81 | → | **21.137.125/0001-38** |

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## 8.2 CNPJ

BR_L1_CNPJ

**COLUMN NAME** **DATA TYPE** **VALUE FORMAT** **REGEX DECOMPOSE**

**WHAT IT IS**

The number of the National Register of Legal Entities: an eight-character root, a four-character establishment order ( `0001` is the head office) and two check digits, `11.222.333/0001-81`. Since July 2026 root and order may hold letters (Instrução Normativa RFB 2.229/2024): `12.ABC.345/01DE-35`.

Legal-entity data is outside the LGPD, but the CNPJ of an individual micro-entrepreneur (MEI) or sole trader leads to a person: the public CNPJ lookup shows their name.

**HOW IT IS FOUND**

**COLUMN NAME** BR_L1_CNPJ — Path

**90%** `cnpj[a-z0-9_]*` `c_?n_?p_?j` `cgc` `n(r|u|um|o|ro|umero)_?cnpj` `cadastro_?nacional_?(da_?)?pessoa_?juridica` `(empresa|emp|fornecedor|forn|cliente|cli|empregador|estabelecimento|estab|prestador|tomador|emitente|emissor|destinatario|dest|transportadora|operadora|mei|matriz|filial)_?cnpj`

**DATA TYPE** BR_L1_CNPJ — Type

Accepts text (length 3 or more) or number (length 3 or more); any other type rules the data class out.

VALUE FORMAT BR_L1_CNPJ — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:[0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2})` | 100% | |
| `0*(?:[0-9A-Z]{12}\d{2})` | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The root is mapped — digits to digits, letters to letters — and the establishment order stays, so the branches of a company remain branches of the same masked company. Both check digits are recomputed with weights 5 to 2 and 9 to 2, modulus 11, each character worth its ASCII code minus 48 — the rule of the alphanumeric CNPJ, which for digits is the usual one. Every masked CNPJ is valid, and the mapping is one-to-one.

`BR_CNPJ` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS` .

> `BR_CNPJ_VALIDO` CHECK DIGIT
>
> Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2` .
>
> > `BR_CNPJ_DV1` CHECK DIGIT
> >
> > Masks the digits with `BR_CNPJ_RAIZ` and recomputes a modulus 11 check digit with weights `5 4 3 2 9 8 7 6 5 4 3 2` .
> >
> > > `BR_CNPJ_RAIZ` REGEX DECOMPOSE
> > >
> > > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> > >
> > > > `BR_CM_DIGITOS_LETRAS_LARGURA_FIXA` CHARACTER MAPPING
> > > >
> > > > Maps each character to another of its group (2 groups), keeping length and format.
>
> `dlpx-core:Brazil CPF` CHECK DIGIT
>
> Comes with Delphix and is used as it ships. Masks the nine digits of the CPF and recomputes both check digits; a CPF stored without its zeros on the left comes back with eleven digits.
>
> `BR_CNPJ_VALIDO_ZEROS` CHECK DIGIT
>
> Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2` .
>
> `BR_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `11.222.333/0001-81` | → | **`21.137.125/0001-38`** |
| `12.ABC.345/01DE-35` | → | **`82.XPZ.728/01DE-28`** |

## 8.3 RG

BR_L1_RG

ESSENTIAL COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The identity card number issued by each state's identification institute, with a format and check digit of its own — in São Paulo `12.345.678-X` , with X as a digit. It appears with the issuing body ( `SSP/SP` ) or the state ( `MG-12.345.678` ).

HOW IT IS FOUND

COLUMN NAME BR_L1_RG — Path

90% `rg[a-z0-9_]*` `r_?g` `n(r|u|um|o|ro|umero)_?rg` `registro_?geral` `carteira_?(de_?)?identidade` `cedula_?(de_?)?identidade` `identidade` *(except when followed by `genero`, `visual`, `de_?genero`)* `doc_?identidade` `documento_?(de_?)?identidade` `ci_?(numero|num|nr)` `rg_?(cliente|pessoa|func|titular|dependente|mae|pai|responsavel|segurado)` `(cliente|pessoa|func|titular|responsavel)_?rg` `identity_?(card|document|no|number)` `id_?card`

VALUE FORMAT BR_L1_RG — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{1,2}\.\d{3}\.\d{3}-[\dxX])` | 70% | |
| `[A-Z]{2}-?\d{1,2}\.?\d{3}\.?\d{3}` | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Digits are mapped; letters, issuing body and separators stay. The check digit is not recomputed: each state has its own rule.

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `12.345.678-X` | → | **`46.916.122-X`** |
| `MG-12.345.678` | → | **`MG-46.916.122`** |

## 8.4 CNH

BR_L1_CNH

ESSENTIAL COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The registration number of the national driving licence: eleven digits, the last two check digits. RENACH and learner permit columns fall here too.

HOW IT IS FOUND

COLUMN NAME BR_L1_CNH — Path

90% `cnh[a-z0-9_]*` `c_?n_?h` `carteira_?(nacional_?)?(de_?)?habilitacao` `habilitacao(_?(numero|num|nr|registro))?` `registro_?cnh` `n(r|u|um|o|ro|umero)_?(registro_?)?cnh` `renach` `pgu` `permissao_?(para_?)?dirigir` `drivers?_?licen[cs]e(_?(no|num|number))?`

HOW IT IS MASKED

Digit by digit, keeping the length. The second check digit of the CNH depends on whether the first went past 9, a rule Check Digit cannot follow; it is not recomputed.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 02650306461 | → | **08913918799** |

## 8.5 Voter card

BR_L1_TITULO_ELEITOR

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

The voter registration number: eight sequential digits, two for the state (01 São Paulo, 02 Minas Gerais ... 28 abroad) and two check digits, `1023 8501 0671`.

HOW IT IS FOUND

COLUMN NAME BR_L1_TITULO_ELEITOR – Path

90% `titulo_?(de_?)?eleitor(_?(numero|num|nr))?` `titulo_?eleitoral` `tit_?eleitor` `n(r|u|um|o|ro|umero)_?titulo(_?eleitor)?` `inscricao_?eleitoral` `voter_?(id|card|registration)`

VALUE FORMAT BR_L1_TITULO_ELEITOR – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{4}\s?\d{4}\s?\d{4})` | 40% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The state stays, the sequence is mapped and both check digits are recomputed, including the São Paulo and Minas Gerais rule in which a remainder of 0 gives 1. Every masked voter card is valid, and a number that lost its zeros in a numeric column gets them back before the computation. See *Check digits computed by the policy*.

`BR_TITULO` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

> `BR_TITULO_VALIDO` STRING ALGORITHM CHAIN
>
> Applies `BR_TITULO_CORPO` → `BR_TITULO_DV`, in this order.
>
> > `BR_TITULO_CORPO` REGEX DECOMPOSE
> >
> > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> >
> > > `BR_CM_DIGITOS` CHARACTER MAPPING
> > >
> > > Maps each character to another of its group (1 groups), keeping length and format.
> >
> > `BR_TITULO_DV` STRING ALGORITHM CHAIN
> >
> > Applies `BR_TITULO_DV_P1` → `BR_TITULO_DV_P2` → `BR_TITULO_DV_P3` → `BR_TITULO_DV_P4` → `BR_TITULO_DV_P5`, in this order.
> >
> > > `BR_TITULO_DV_P1` REGEX DECOMPOSE
> > >
> > > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> > >
> > > > `BR_TITULO_DV_T1` DATA CLEANSING
> > > >
> > > > Replaces each value found in `br-titulo-dv-1.txt` (100 pairs) by its counterpart; values not in the file stay as they are.
> > >
> > > `BR_TITULO_DV_P2` REGEX DECOMPOSE
> > >
> > > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> > >
> > > > `BR_TITULO_DV_T2` DATA CLEANSING
> > > >
> > > > Replaces each value found in `br-titulo-dv-2.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.
> > >
> > > `BR_TITULO_DV_P3` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_TITULO_DV_T3` DATA CLEANSING

Replaces each value found in `br-titulo-dv-3.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

`BR_TITULO_DV_P4` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_TITULO_DV_T4` DATA CLEANSING

Replaces each value found in `br-titulo-dv-4.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

`BR_TITULO_DV_P5` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_TITULO_DV_T5` DATA CLEANSING

Replaces each value found in `br-titulo-dv-5.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

`BR_TITULO_VALIDO_1_ZERO` STRING ALGORITHM CHAIN

Applies `BR_TITULO_1_ZERO` → `BR_TITULO_VALIDO`, in this order.

`BR_TITULO_1_ZERO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_TITULO_1_ZERO_TABELA` DATA CLEANSING

Replaces each value found in `br-zeros-1.txt` (10 pairs) by its counterpart; values not in the file stay as they are.

`BR_TITULO_VALIDO_2_ZEROS` STRING ALGORITHM CHAIN

Applies `BR_TITULO_2_ZEROS` → `BR_TITULO_VALIDO`, in this order.

`BR_TITULO_2_ZEROS` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_TITULO_2_ZEROS_TABELA` DATA CLEANSING

Replaces each value found in `br-zeros-2.txt` (10 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 102385010671 | → | **330614820671** |
| 004356870906 | → | **001187220981** |

## 8.6 PIS, PASEP, NIT and NIS

BR_L1_PIS_NIS

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The worker's number in PIS (Caixa) or PASEP (Banco do Brasil), which is also the INSS NIT and the NIS of the Cadastro Único and Bolsa Família: ten digits and a check digit, `120.56412.54-7`.

HOW IT IS FOUND

COLUMN NAME BR_L1_PIS_NIS – Path

90% `pis[a-z0-9_]*` `pasep` `pis_?pasep` `nit` `nis` `n(r|u|um|o|ro|umero)_?(pis|nit|nis|pasep)` `nis_?(titular|responsavel|beneficiario)` `numero_?identificacao_?(do_?)?trabalhador` `inscricao_?(inss|previdencia)` `nit_?(inss|contribuinte)`

VALUE FORMAT BR_L1_PIS_NIS — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{3}\.\d{5}\.\d{2}-\d)` | 90% | |
| `0*(?:\d{11})` | 50% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The first digit stays, the other nine are mapped and the check digit is recomputed with weights 3 2 9 8 7 6 5 4 3 2, modulus 11. Every masked PIS is valid and the mapping is one-to-one. The same algorithm masks the PIS inside the National Health Card.

`BR_PIS` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

> `BR_PIS_VALIDO` CHECK DIGIT
>
> Masks the digits with `BR_PIS_CORPO` and recomputes a modulus 11 check digit with weights `3 2 9 8 7 6 5 4 3 2`.
>
> > `BR_PIS_CORPO` REGEX DECOMPOSE
> >
> > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> >
> > > `BR_CM_DIGITOS` CHARACTER MAPPING
> > >
> > > Maps each character to another of its group (1 groups), keeping length and format.
>
> `BR_PIS_VALIDO_ZEROS` CHECK DIGIT
>
> Masks the digits with `BR_PIS_CORPO` and recomputes a modulus 11 check digit with weights `3 2 9 8 7 6 5 4 3 2`.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `120.56412.54-7` | → | **`113.81104.27-9`** |

## 8.7 Work card

BR_L1_CTPS

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Number and series of the paper Carteira de Trabalho e Previdência Social. The digital card uses the CPF.

HOW IT IS FOUND

COLUMN NAME BR_L1_CTPS — Path

| | |
|---|---|
| 90% | `ctps[a-z0-9_]*` `carteira_?(de_?)?trabalho` `serie_?ctps` `n(r\|u\|um\|o\|ro\|umero)_?ctps` `work_?permit` |

HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `1234567/001` | → | **`7162897/064`** |

## 8.8 Passport

BR_L1_PASSAPORTE

ESSENTIAL · COLUMN NAME · VALUE FORMAT · CHARACTER MAPPING

### WHAT IT IS

Brazilian passport numbers — two letters and six digits, `FZ123456` — and foreign ones.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_PASSAPORTE – Path

90% `passaporte[a-z0-9_]*` `passport[a-z0-9_]*` `n(r|u|um|o|ro|umero)_?passaporte` `pasaporte`

**VALUE FORMAT** BR_L1_PASSAPORTE – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-Z]{2}\d{6}` | 40% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| FZ123456 | → | **RS124760** |

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## 8.9 Foreigner documents

BR_L1_DOCUMENTO_ESTRANGEIRO

COLUMN NAME · CHARACTER MAPPING

### WHAT IT IS

The National Migration Registry number (RNM, formerly RNE) of the migration card, the refugee protocol and the visa number.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_DOCUMENTO_ESTRANGEIRO – Path

85% `rnm` `crnm` `rne` `registro_?nacional_?(migratorio|estrangeiro)` `carteira_?(de_?)?registro_?(nacional_?)?migratorio` `protocolo_?(de_?)?refugio` `refugiado_?(protocolo|numero)` `visto(_?(numero|num|nr))?` `n(r|u|um|o|ro|umero)_?visto` `visa_?(no|number)` `dpri`

### HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| V123456-7 | → | **P220047-2** |

## 8.10 Civil registry certificates

BR_L1_CERTIDAO

COLUMN NAME | VALUE FORMAT | CHARACTER MAPPING

### WHAT IT IS

The registration number (matrícula) of birth, marriage and death certificates: 32 digits with the registry office, collection, year, book type, book, page, entry and two check digits. Also the Declaration of Live Birth and the Death Declaration.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_CERTIDAO — Path

85% `certidao_?(de_?)?(nascimento|casamento|obito)(_?(numero|num|nr|matricula))?` `matricula_?(da_?)?certidao` `certidao_?matricula` `termo_?(de_?)?nascimento` `registro_?(de_?)?nascimento` `livro_?folha_?termo` `dnv` `declaracao_?(de_?)?nascido_?vivo` `declaracao_?(de_?)?obito` `n(r|u)_?(dnv|do)`

**VALUE FORMAT** BR_L1_CERTIDAO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{6}\s?\d{2}\s?\d{2}\s?\d{4}\s?\d\s?\d{5}\s?\d{3}\s?\d{7}\s?\d{2})` | 80% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Digit by digit, keeping the length. The check digit is not recomputed.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `10453901552013100012021000012321` | → | **`82431996553020383891084411419170`** |

## 8.11 Professional registration

BR_L1_REGISTRO_PROFISSIONAL

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

The registration number with a professional council — CRM, OAB, CREA, COREN, CRO, CRP, CRC and others —, `CRM-SP 123456`. The councils publish these registers with the professional's name.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_REGISTRO_PROFISSIONAL — Path

85% `crm` `cro` `crp` `crf` `crn` `crefito` `coren` `crea` `cau` `crc` `oab` `cress` `crmv` `cfm` `conselho_?(de_?)?classe(_?(numero|num|nr))?` `registro_?(profissional|conselho|de_?classe)` `n(r|u|um|o|ro|umero)_?(crm|oab|crea|coren|cro|crp|conselho|registro_?profissional)` `inscricao_?(oab|crm|crea|conselho)`

### HOW IT IS MASKED

Digits are mapped; council, state and separators stay.

`BR_REGISTRO_PROFISSIONAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| CRM–SP 123456 | → | **CRM–SP 904292** |

## 8.12 Given name

BR_L1_NOME

ESSENTIAL · COLUMN NAME · KNOWN VALUES · REGEX DECOMPOSE

WHAT IT IS

The given name, one or more: `Maria`, `João Pedro`, `Maria de Fátima`. The detection list holds the most frequent names of the 2022 census.

HOW IT IS FOUND

COLUMN NAME BR_L1_NOME – Path

| | |
|---|---|
| 85% | `primeiro_?nome` `prenome` `nome_?proprio` `nomes?_?(de_?)?batismo` `nome_?1` `first_?names?` `given_?names?` `fname` `middle_?names?` |
| 50% | `nome` *(except when followed by list E1)* `nm` *(except when followed by list E2)* `name` *(except when followed by list E3)* |

KNOWN VALUES BR_L1_NOME – List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `br-detectar-nomes.txt` | 1,046 | 70% | Maria, Ana, Francisca, Júlia, Julia, Antônia |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Word by word, each name replaced with one from the 2022 census list. `de`, `da`, `dos` and `e` stay in place. More than three words are masked character by character.

`BR_NOMES` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

`BR_PALAVRA_NOME` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_NOME`.

`BR_NOME` NAME

Replaces the name with one of the 895 names in `br-nomes.txt`.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Maria de Fátima | → | **Joelson de Camila** |
| JOÃO PEDRO | → | **ALEXANDRE VANESSA** |

## 8.13 Surname

BR_L1_SOBRENOME

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

### WHAT IT IS

The surnames, usually two — the mother's and the father's —, with particles: `da Silva Santos`. Also the maiden name. The detection list holds the most frequent surnames of the 2022 census.

### HOW IT IS FOUND

> COLUMN NAME BR_L1_SOBRENOME – Path
>
> 85% `sobrenomes?` `ultimo_?nome` `nome_?(de_?)?familia` `last_?names?` `surnames?` `family_?names?` `lname` `apelido_?familia` `sobrenome_?(pai|mae|paterno|materno|solteira|casada)` `nome_?(de_?)?solteira` `sobrenome_?(de_?)?casada`

> KNOWN VALUES BR_L1_SOBRENOME – List
>
| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `br-detectar-sobrenomes.txt` | 636 | 70% | Silva, Santos, Oliveira, Souza, Pereira, Ferreira |
>
> Values are split into words, and each word is looked up.
>
> A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Word by word, each surname replaced with one from the list. Particles and agnomens — Júnior, Filho, Neto, Sobrinho — stay in place.

`BR_SOBRENOMES` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

> `BR_PALAVRA_SOBRENOME` REGEX DECOMPOSE
>
> Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SOBRENOME`.
>
> > `BR_SOBRENOME` NAME
> >
> > Replaces the name with one of the 535 names in `br-sobrenomes.txt`.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `da Silva Santos` | → | **`da Brás Fortunato`** |

## 8.14 Full name

BR_L1_NOME_COMPLETO

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

### WHAT IT IS

Given names and surnames in one column: `José Carlos da Silva Júnior`. Also the mother's and father's names — asked in almost every Brazilian register and used as a security question —, the guardian's, the spouse's, and the **social name**, which reveals a trans person's gender identity.

A column called just `NOME` may hold given names or full names: the values decide between the two data classes.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_NOME_COMPLETO — Path

**90%** `nome_?(completo|compl|civil|social|registro|guerra|abreviado)` `nomecompleto` `(nm|no|nome|ds)_?(cliente|cli|paciente|pac|funcionario|func|empregado|colaborador|servidor|segurado|beneficiario|benef|titular|dependente|dep|mae|pai|responsavel|resp|conjuge|companheiro|contato|avalista|fiador|devedor|tomador|requerente|solicitante|testemunha|representante|procurador|proprietario|locatario|locador|inquilino|vitima|reu|autor|acusado|investigado|doador|herdeiro|pessoa|pes|motorista|condutor|cidadao|eleitor|contribuinte|aluno|estudante|crianca|medico|enfermeiro|professor|socio|tutor|curador|pagador|favorecido|destinatario|remetente|candidato|associado|membro|cooperado|consumidor|comprador|corretor|interessado|parte|advogado|usuario|atendente|vendedor|gerente|supervisor|avo|filho|filha|irmao|parente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)` `(cliente|cli|paciente|pac|funcionario|func|empregado|colaborador|servidor|segurado|beneficiario|benef|titular|dependente|dep|mae|pai|responsavel|resp|conjuge|companheiro|contato|avalista|fiador|devedor|tomador|requerente|solicitante|testemunha|representante|procurador|proprietario|locatario|locador|inquilino|vitima|reu|autor|acusado|investigado|doador|herdeiro|pessoa|pes|motorista|condutor|cidadao|eleitor|contribuinte|aluno|estudante|crianca|medico|enfermeiro|professor|socio|tutor|curador|pagador|favorecido|destinatario|remetente|candidato|associado|membro|cooperado|consumidor|comprador|corretor|interessado|parte|advogado|usuario|atendente|vendedor|gerente|supervisor|avo|filho|filha|irmao|parente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)_?(nome|nm)` `nome_?(d[aoe]_?)?(cliente|cli|paciente|pac|funcionario|func|empregado|colaborador|servidor|segurado|beneficiario|benef|titular|dependente|dep|mae|pai|responsavel|resp|conjuge|companheiro|contato|avalista|fiador|devedor|tomador|requerente|solicitante|testemunha|representante|procurador|proprietario|locatario|locador|inquilino|vitima|reu|autor|acusado|investigado|doador|herdeiro|pessoa|pes|motorista|condutor|cidadao|eleitor|contribuinte|aluno|estudante|crianca|medico|enfermeiro|professor|socio|tutor|curador|pagador|favorecido|destinatario|remetente|candidato|associado|membro|cooperado|consumidor|comprador|corretor|interessado|parte|advogado|usuario|atendente|vendedor|gerente|supervisor|avo|filho|filha|irmao|parente|customer|patient|employee|member|holder|mother|father|spouse|contact|person|student|driver|owner)` `filiacao(_?[12])?` `full_?name` `person_?name`

**60%** `mae` `pai` `genitora?` `conjuge` `responsavel_?legal` `representante_?legal` `procurador` `avalista` `fiador` `testemunha` `beneficiario` `herdeiro` `curador` `tutor` `titular`

**50%** `nome` *(except when followed by list E1)* `nm` *(except when followed by list E2)* `name` *(except when followed by list E3)*

**KNOWN VALUES** BR_L1_NOME_COMPLETO — List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| br-detectar-nomes.txt | 1,046 | 65% | Maria, Ana, Francisca, Júlia, Julia, Antônia |
| br-detectar-sobrenomes.txt | 636 | 65% | Silva, Santos, Oliveira, Souza, Pereira, Ferreira |

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the data class.

#### HOW IT IS MASKED

In Brazilian order: two words are a given name and a surname; three, one given name and two surnames (`Maria da Silva` too); from four on, two given names and the rest surnames. Particles and agnomens stay. `SURNAMES, GIVEN NAMES` with a comma, as in lists, is recognized too. The result is a plausible Brazilian name, never the original.

`BR_NOME_COMPLETO` **REGEX DECOMPOSE**

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

> `BR_SOBRENOMES` **REGEX DECOMPOSE**
>
> Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.
>
> > `BR_PALAVRA_SOBRENOME` **REGEX DECOMPOSE**
> >
> > Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SOBRENOME`.
> >
> > > `BR_SOBRENOME` **NAME**
> > >
> > > Replaces the name with one of the 535 names in `br-sobrenomes.txt`.
> >
> > `BR_CM_ALFANUM` **CHARACTER MAPPING**
> >
> > Maps each character to another of its group (7 groups), keeping length and format.
>
> `BR_NOMES` **REGEX DECOMPOSE**
>
> Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.
>
> > `BR_PALAVRA_NOME` **REGEX DECOMPOSE**
> >
> > Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_NOME`.
> >
> > > `BR_NOME` **NAME**
> > >
> > > Replaces the name with one of the 895 names in `br-nomes.txt`.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `José Carlos da Silva Júnior` | → | **`Renato Neusa da Brás Júnior`** |
| `Ana Paula Souza Lima` | → | **`Cristiane Gerson Cipriano Tobias`** |
| `SILVA, MARIA` | → | **`BRÁS, JOELSON`** |

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## 8.15 E-mail

`BR_L1_EMAIL`

**ESSENTIAL** **COLUMN NAME** **DATA TYPE** **VALUE FORMAT** **REGEX DECOMPOSE**

#### WHAT IT IS

Personal and corporate e-mail addresses.

#### HOW IT IS FOUND

> **COLUMN NAME** `BR_L1_EMAIL – Path`
>
> 85% `e_?mail[a-z0-9_]*` `mail[a-z0-9_]*` `correio_?eletronico` `endereco_?eletronico` `(ds|de|tx)_?email`

> **DATA TYPE** `BR_L1_EMAIL – Type`
>
> Accepts text (length 5 or more); any other type rules the data class out.

> **VALUE FORMAT** `BR_L1_EMAIL – Regex`
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[\p{L}\p{N}._%+'\-]+@[\p{L}\p{N}.\-]+\.[A-Za-z]{2,24}` | 100% | |
>
> A value that matches none counts 20% against the data class.

#### HOW IT IS MASKED

User and domain are mapped character by character, and the top-level domain — `.com`, `.com.br` — becomes `.test`, reserved for testing: a masked address never delivers a message.

`BR_EMAIL` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

> `BR_CM_ALFANUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (7 groups), keeping length and format.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `maria.silva@gmail.com` | → | **`cazua.zotro@pgoeb.test`** |

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## 8.16 Phone

`BR_L1_TELEFONE`

**ESSENTIAL** **COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

#### WHAT IT IS

Numbers with the country code (55), the area code (DDD) and eight digits for a landline (starting 2 to 5) or nine for a mobile (starting 9): `(11) 98765-4321`, `+55 21 3456-7890`, `11987654321`.

#### HOW IT IS FOUND

> **COLUMN NAME** `BR_L1_TELEFONE – Path`
>
> **85%** `tel` `telefone[a-z0-9_]*` `fone[a-z0-9_]*` `celular[a-z0-9_]*` `cel` `movel` `whats_?app` `zap` `contato_?(tel|telefone|celular)` `(nr|nu|num|numero|ds|tx)_?(tel|telefone|fone|celular|cel)` `ddd_?(tel|telefone|celular|fone)` `fax` `ramal` `phone[a-z0-9_]*` `mobile[a-z0-9_]*`

> **VALUE FORMAT** `BR_L1_TELEFONE – Regex`
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(\+?55\s?)?\(?0?[1-9]\d\)?\s?9\d{4}[\s\-]?\d{4})` | 100% | |
| `0*(?:(\+?55\s?)?\(?0?[1-9]\d\)?\s?[2-5]\d{3}[\s\-]?\d{4})` | 80% | |
| `0*(?:9\d{4}-\d{4})` | 80% | |
>
> A value that matches none counts 30% against the data class.

#### HOW IT IS MASKED

Country code, DDD and the first digit stay — the DDD covers a whole region, and the first digit tells a mobile from a landline —; the other digits are mapped, keeping length and separators.

`BR_TELEFONE` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

> `BR_CM_DIGITOS` **CHARACTER MAPPING**
>
> Maps each character to another of its group (1 groups), keeping length and format.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `(11) 98765-4321` | → | **`(11) 93134-0400`** |
| `+55 21 3456-7890` | → | **`+55 21 3047-9607`** |

---

## 8.17 Address

BR_L1_ENDERECO

ESSENTIAL | COLUMN NAME | DATA TYPE | VALUE FORMAT | SECURE LOOKUP

### WHAT IT IS

The address line: street type and name, number, complement, sometimes the neighbourhood. `Av. Paulista, 1578, 12º andar – Bela Vista`.

### HOW IT IS FOUND

**COLUMN NAME** BR_L1_ENDERECO – Path

**80%** `endereco[a-z0-9_]*` *(except when preceded by `mac`, `ip`, `e_?mail`, `web`, `url`, `eletronico`; except when followed by list E4)* `end_?(residencial|comercial|cobranca|entrega|correspondencia|completo)` *(except when preceded by `mac`, `ip`, `e_?mail`, `web`, `url`)* `logradouro[a-z0-9_]*` `lograd(ds|no|nm|tx)_?` `(logradouro|endereco|rua)` `rua` `residencia` `domicilio[a-z0-9_]*` `address[a-z0-9_]*` *(except when preceded by `mac`, `ip`, `e_?mail`, `web`, `url`; except when followed by list E5)* `street(_?(address|name))?`

**DATA TYPE** BR_L1_ENDERECO – Type

Accepts text (length 6 or more); any other type rules the data class out.

**VALUE FORMAT** BR_L1_ENDERECO – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `(?i)(rua\|r\.\|avenida\|av\.?\|travessa\|tv\.?\|alameda\|al\.\|estrada\|est\.\|rodovia\|rod\.\|praca\|praça\|pc\.\|largo\|viela\|servidao\|quadra\|qd\.?\|setor\|conjunto\|beco\|ladeira\|vila)\s+.+` | 80% | |
| `(?i).*,\s*\d+.*` | 30% | |

A value that matches none counts 20% against the data class.

### HOW IT IS MASKED

The whole line is replaced with a fictitious Brazilian address — a common street name, number, apartment or house, neighbourhood. Municipality, state and CEP in their own columns are handled by their data classes.

`BR_ENDERECO` SECURE LOOKUP

Replaces the value with one of the 4,000 values in `br-enderecos.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Av. Paulista, 1578, 12º andar – Bela Vista` | → | **`Rua do Comércio, 980`** |

## 8.18 Number and complement

BR_L1_ENDERECO_COMPLEMENTO

ESSENTIAL | COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

House number, complement, apartment, block and lot in their own columns.

HOW IT IS FOUND

COLUMN NAME BR_L1_ENDERECO_COMPLEMENTO — Path

70% `complemento(_?endereco)?` `compl` `n(r|u|um|o|ro|umero)_?(endereco|casa|residencia|imovel|logradouro)` `numero_?(end|lograd)` `num_?casa` `apto` `apartamento` `bloco` `lote` `quadra` `casa_?(numero|num|nr)` `house_?number` `apartment` `unit_?number`

HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Apto 42 Bloco B | → | **Offi 87 Nsaso W** |

## 8.19 Branch and account

BR_L1_CONTA_BANCARIA

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

Bank branch and account, with each bank's check digits (sometimes X). The bank code (COMPE, ISPB) is not personal data and has no data class.

HOW IT IS FOUND

COLUMN NAME BR_L1_CONTA_BANCARIA — Path

85% `conta_?(corrente|poupanca|bancaria|banco|salario|pagamento|deposito|credito|debito)` `cc_?(numero|num|nr)` `n(r|u|um|o|ro|umero)_?conta` `num_?cta` `nr_?cta` `cta_?(corrente|bancaria)` `agencia(_?(conta|bancaria|numero|num|nr|dv))?` `ag_?(conta|bancaria)` `dv_?(conta|agencia)` `digito_?(conta|agencia)` `iban` `account_?(no|num|number)` `bank_?account` `operacao_?conta`

HOW IT IS MASKED

Digits are mapped; X and separators stay. Check digits are not recomputed: each bank has its own rule.

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 12345-X | → | **31564-X** |

## 8.20 Payment card

BR_L1_CARTAO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Credit and debit card numbers, recognized by the Luhn algorithm.

HOW IT IS FOUND

COLUMN NAME BR_L1_CARTAO – Path

85% `cartao(_?(credito|debito|numero|num|nr|pan))?` `n(r|u|um|o|ro|umero)_?cartao` `pan` `card_?(no|num|number)` `credit_?card` `cc_?number`

VALUE FORMAT BR_L1_CARTAO – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:3[47]\d{13}\|[1-9]\d{12,13}\|[1-9]\d{15,18})` | 90% | LUHN check digit; after removing `[\s\-]` |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Credit Card, which comes with Delphix: another number that passes the Luhn check, keeping the first six digits — the issuer, not the holder. Only values shaped like a card reach it; the others pass unchanged instead of failing the row.

`BR_CARTAO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`dlpx-core:Credit Card` PAYMENT CARD

Comes with Delphix and is used as it ships. Replaces the card number with another that passes the Luhn check, keeping the first six digits — the issuer, not the holder.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 5162 3062 1937 8829 | → | **5162 3076 6490 1610** |

## 8.21 PIX key

BR_L1_CHAVE_PIX

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

A PIX key is a CPF, a CNPJ, an e-mail, a phone number in the `+5511987654321` format or a random key (a UUID).

HOW IT IS FOUND

COLUMN NAME BR_L1_CHAVE_PIX – Path

90% `chave_?pix` `pix(_?(chave|key|destino|recebedor|pagador|favorecido))?` `chave_?(aleatoria|evp)` `evp` `pix_?key`

VALUE FORMAT BR_L1_CHAVE_PIX – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:[0-9a-f]{8}-[0-9a-f]{4}-4[0-9a-f]{3}-[89ab][0-9a-f]{3}-[0-9a-f]{12})` | 30% | |

A value that matches none counts 20% against the data class.

HOW IT IS MASKED

Each kind of key is masked as its own data class: CPF and CNPJ with valid check digits, e-mail with `.test`, phone keeping the DDD, random key keeping its hexadecimal format. A CPF key and the same person's CPF still agree.

`BR_CHAVE_PIX` REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

`BR_EMAIL` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_TELEFONE` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

`BR_CPF` REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

`dlpx-core:Brazil CPF` CHECK DIGIT

Comes with Delphix and is used as it ships. Masks the nine digits of the CPF and recomputes both check digits; a CPF stored without its zeros on the left comes back with eleven digits.

`BR_CNPJ_VALIDO` CHECK DIGIT

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

`BR_CNPJ_DV1` CHECK DIGIT

Masks the digits with `BR_CNPJ_RAIZ` and recomputes a modulus 11 check digit with weights `5 4 3 2 9 8 7 6 5 4 3 2`.

`BR_CNPJ_RAIZ` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

`BR_CM_DIGITOS_LETRAS_LARGURA_FIXA` CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

`BR_CNPJ_VALIDO_ZEROS` CHECK DIGIT

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

`BR_CNPJ` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

`BR_CM_HEX` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `529.982.247-25` | → | **`410.719.628-33`** |
| `123e4567-e89b-12d3-a456-426614174000` | → | **`951b0298-d33a-44c2-b413-745894460718`** |

## 8.22 IP address

BR_L1_IP

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses. The Marco Civil da Internet requires providers to keep connection and access logs, with the IP, which lead to the subscriber.

HOW IT IS FOUND

COLUMN NAME BR_L1_IP — Path

85% `ip` `ip_?(address|addr|origem|destino|cliente|usuario|acesso|login|remoto|publico)` `endereco_?ip` `ipv[46]` `remote_?addr(ess)?` `client_?ip` `host_?ip`

VALUE FORMAT BR_L1_IP — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:((25[0-5]\|2[0-4]\d\|1?\d?\d)\.){3}(25[0-5]\|2[0-4]\d\|1?\d?\d))` | 100% | |
| `0*(?:[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7})` | 90% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Each IPv4 octet is mapped to 1–254; an IPv6 is mapped keeping its hexadecimal characters.

`BR_IP` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_HEX`.

> `BR_OCTETO` NUMERIC MAPPING
>
> Maps the number to another between 1 and 254.
>
> `BR_CM_HEX` CHARACTER MAPPING
>
> Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `200.147.67.142` | → | **`146.137.44.231`** |

## 8.23 Device identifier

BR_L1_DISPOSITIVO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

IMEI, IMSI, ICCID, MAC address, advertising identifiers, cookies and sessions.

HOW IT IS FOUND

COLUMN NAME BR_L1_DISPOSITIVO — Path

80% `imei` `imsi` `iccid` `mac` `mac_?address` `endereco_?mac` `device_?(id|uuid|serial)` `id_?(dispositivo|aparelho)` `dispositivo_?id` `advertising_?id` `idfa` `gaid` `cookie(_?id)?` `session_?id` `id_?sessao` `serial_?(aparelho|celular)`

VALUE FORMAT BR_L1_DISPOSITIVO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2})` | 100% | |
| `0*(?:\d{15})` | 80% | LUHN check digit |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, with hexadecimal digits kept hexadecimal: a MAC stays a MAC.

`BR_CM_HEX` CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `3c:22:fb:9a:10:e4` | → | **`5e:83:bb:3c:31:f9`** |

## 8.24 Licence plate

BR_L1_PLACA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Plates in the old pattern, three letters and four digits ( `ABC-1234` ), and in the Mercosur pattern, with a letter in the fifth position ( `BRA2E19` ).

### HOW IT IS FOUND

> **COLUMN NAME** BR_L1_PLACA — Path
>
> **85%** `placa(_?(veiculo|carro|moto|numero))?` `n(r|u|um|o|ro|umero)_?placa` `placa_?mercosul` `license_?plate` `plate(_?(no|number))?`

> **VALUE FORMAT** BR_L1_PLACA — Regex
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-Z]{3}\d[A-Z]\d{2}` | 90% | |
| `[A-Z]{3}-?\d{4}` | 80% | |
>
> A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Letters become letters and digits become digits, position by position, with the hyphen: the plate pattern stays.

`BR_PLACA` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM` .

> `BR_CM_DIGITOS_LETRAS` CHARACTER MAPPING
>
> Maps each character to another of its group (2 groups), keeping length and format.
>
> `BR_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| BRA2E19 | → | **WHM3Z25** |
| ABC-1234 | → | **VLD-6328** |

---

## 8.25 RENAVAM

BR_L1_RENAVAM

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

The code of the National Motor Vehicle Registry: eleven digits, the last a check digit. Old nine-digit numbers are padded with zeros.

### HOW IT IS FOUND

> **COLUMN NAME** BR_L1_RENAVAM — Path
>
> **90%** `renavam` `cod_?renavam` `codigo_?renavam` `n(r|u|um|o|ro|umero)_?renavam` `registro_?nacional_?(de_?)?veiculos`

HOW IT IS MASKED

The first ten digits are mapped and the check digit recomputed with weights 3 2 9 8 7 6 5 4 3 2, modulus 11. Every masked RENAVAM is valid, with eleven digits.

`BR_RENAVAM` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS` .

> `BR_RENAVAM_VALIDO` CHECK DIGIT
>
> Masks the digits with `BR_CM_DIGITOS_LARGURA_FIXA` and recomputes a modulus 11 check digit with weights `3 2 9 8 7 6 5 4 3 2` .
>
> > `BR_CM_DIGITOS_LARGURA_FIXA` CHARACTER MAPPING
> >
> > Maps each character to another of its group (1 groups), keeping length and format.
>
> `BR_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 00639884962 | → | **05594949808** |

## 8.26 Chassis

BR_L1_CHASSI

COLUMN NAME | VALUE FORMAT | CHARACTER MAPPING

WHAT IT IS

The 17-character vehicle identification number (VIN) engraved on the chassis, and the engine number.

HOW IT IS FOUND

> COLUMN NAME BR_L1_CHASSI – Path
>
> 85% `chassi(_?(numero|num|nr))?` `n(r|u|um|o|ro|umero)_?chassi` `vin` `numero_?motor` `n(r|u)_?motor`

> VALUE FORMAT BR_L1_CHASSI – Regex
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-HJ-NPR-Z0-9]{17}` | 70% | |
>
> A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 9BWZZZ377VT004251 | → | **3KVRMX584XM353424** |

## 8.27 Real estate

BR_L1_IMOVEL

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

The property registration number (matrícula), the municipal property tax (IPTU) number and the rural property codes — CAR, CCIR, NIRF, SNCR. Each leads to the owner through public registers.

### HOW IT IS FOUND

COLUMN NAME — BR_L1_IMOVEL — Path

85% `matricula_?(do_?)?imovel` `matricula_?(cartorio|cri|registro)` `inscricao_?(imobiliaria|municipal_?imovel|iptu|cadastral)` `iptu(_?(numero|num|nr|inscricao))?` `indicacao_?fiscal` `codigo_?(car|ccir|nirf|sncr|incra)` `n(r|u)_?(car|ccir|nirf)` `car_?(imovel|rural)` `ccir` `nirf` `cib` `sncr`

### HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 123.456.7890-1 | → | **061.766.2692-2** |

## 8.28 Contract and enrolment numbers

BR_L1_CONTRATO

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

Contract, policy, proposal, protocol, consumer unit and installation numbers, employee, civil servant (SIAPE) or student enrolment, member cards and customer codes.

### HOW IT IS FOUND

COLUMN NAME — BR_L1_CONTRATO — Path

70% `n(r|u|um|o|ro|umero)_?(contrato|apolice|proposta|matricula|carteirinha|protocolo|instalacao|uc|unidade_?consumidora|cliente|pedido_?cliente)` `contrato_?(numero|num|nr)` `matricula(_?(funcionario|func|empregado|servidor|aluno|associado))?` *(except when followed by* `imovel`, `certidao`, `cartorio`, `cri`*)* `matr` `cod_?(cliente|funcionario|servidor|aluno|associado|segurado|beneficiario)` `codigo_?(cliente|funcionario|servidor|aluno|associado)` `id_?(cliente|funcionario|servidor|aluno|associado)` `unidade_?consumidora` `instalacao` `apolice` `carteirinha` `ra_?aluno` `registro_?academico` `siape`

### HOW IT IS MASKED

Character by character, keeping length and format. The mapping is deterministic, so joins keep working.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| CT-123456 | → | **NG-557411** |

## 8.29 User and social media

BR_L1_USUARIO

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Login, nickname and social media profiles.

HOW IT IS FOUND

> COLUMN NAME BR_L1_USUARIO — Path
>
> 70% `usuario` `user_?name` `username` `login(_?(nome|usuario))?` `apelido` `nick(_?name)?` `alias` `perfil_?(instagram|facebook|twitter|tiktok|linkedin)` `instagram` `facebook` `twitter` `tiktok` `linkedin` `rede_?social` `arroba` `handle`

HOW IT IS MASKED

Character by character, keeping length and format.

BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| @maria_silva | → | **@cazua_zotro** |

## 8.30 Credentials

BR_L1_CREDENCIAL

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Passwords and their hashes, PINs, security codes, CVV, tokens, API keys and secret answers.

HOW IT IS FOUND

> COLUMN NAME BR_L1_CREDENCIAL — Path
>
> 90% `senha` `password` `passwd` `pwd` `pass` `hash_?senha` `senha_?hash` `pin` `codigo_?(seguranca|verificacao|acesso)` `cvv` `cvc` `token(_?(acesso|refresh|api))?` `access_?token` `refresh_?token` `api_?key` `chave_?api` `segredo` `secret` `otp` `frase_?secreta` `pergunta_?secreta` `resposta_?secreta`

VALUE FORMAT BR_L1_CREDENCIAL — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `\$2[abxy]?\$\d{2}\$[./A-Za-z0-9]{53}` | 95% | |
| `\$(1\|5\|6)\$(rounds=\d+\$)?[./A-Za-z0-9]{1,16}\$[./A-Za-z0-9]{22,86}` | 95% | |
| `\$argon2(id\|i\|d)\$v=\d+\$m=\d+,t=\d+,p=\d+\$\S+` | 95% | |
| `pbkdf2_sha(1\|256\|512)\$\d+\$\S+\$\S+` | 95% | |
| `\{(SSHA\|SHA\|SSHA256\|SSHA512\|SMD5\|MD5\|CRYPT)\}\S+` | 95% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Every character becomes `X`. A credential has no use in a test copy, and a hash masked character by character could still be attacked.

`BR_REDIGIR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Senha#2024 | → | XXXXXXXXXX |

## 8.31 Geolocation

BR_L1_GEOLOCALIZACAO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Latitude and longitude, together or in separate columns, from apps, deliveries and visits.

HOW IT IS FOUND

COLUMN NAME BR_L1_GEOLOCALIZACAO — Path

| | |
|---|---|
| 85% | `lat` `latitude` `lon` `lng` `longitude` `coordenadas?` `coord_?[xy]` `geo_?(lat\|lon\|lng\|loc\|localizacao\|posicao)` `geolocalizacao` `gps(_?(localizacao\|posicao\|coordenadas))?` `localizacao_?gps` |
| 50% | `long` |

VALUE FORMAT BR_L1_GEOLOCALIZACAO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `-?([0-2]?\d\|3[0-3])\.\d{3,}\s*,\s*-(3[4-9]\|[4-6]\d\|7[0-4])\.\d{3,}` | 90% | |
| `-(3[4-9]\|[4-6]\d\|7[0-4])\.\d{4,}` | 60% | |
| `-?([0-2]?\d\|3[0-3])\.\d{4,}` | 30% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The integer part and the first decimal stay — about 11 km — and the other decimals are mapped: the place is generalized, not moved far away.

`BR_COORDENADA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_CM_DIGITOS` CHARACTER MAPPING

> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `-23.561414,-46.655881` | → | **`-23.522235,-46.651532`** |

## 8.32 Free text

BR_L1_TEXTO_LIVRE

COLUMN NAME · DATA TYPE · VALUE FORMAT · FREE TEXT REDACTION

WHAT IT IS

Remarks, service histories, complaints, justifications and messages where anything gets written — CPF, phone and name included.

HOW IT IS FOUND

> **COLUMN NAME** BR_L1_TEXTO_LIVRE — Path
>
> **60%** `observac(ao|oes)` `obs` `comentarios?` `notas?` `anotac(ao|oes)` `descricao_?` `(ocorrencia|atendimento|chamado|reclamacao|solicitacao|caso|fato)` `historico_?` `(atendimento|chamado|contato)` `relato` `parecer` `justificativa` `mensagem` *(except when preceded by list E6; except when followed by list E7)* `texto(_?livre)?` `detalhes` `resumo_?(atendimento|caso)` `reclamacao` `ocorrencia_?(descricao|texto)` `remarks?` `comments?` `notes?` `free_?text` `memo`

> **DATA TYPE** BR_L1_TEXTO_LIVRE — Type
>
> Accepts text (length 30 or more); any other type rules the data class out.

> **VALUE FORMAT** BR_L1_TEXTO_LIVRE — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{3}\.\d{3}\.\d{3}-\d{2})` | 70% | anywhere in the value |
| `[\w.+\-]+@[\w\-]+\.[A-Za-z]{2,}` | 70% | anywhere in the value |
| `\(?\d{2}\)?\s?9\d{4}-\d{4}` | 60% | anywhere in the value |

> A value that matches none does not count against the data class.

HOW IT IS MASKED

Free Text Redaction: CPF and CNPJ (formatted or not), e-mail, phones, cards, CEP, plates and IPs become `[DADO]`; given names and surnames of the 2022 census list become `[NOME]`. The rest of the text stays.

`BR_TEXTO_LIVRE` **FREE TEXT REDACTION**

Replaces the text matching 12 patterns with `[DADO]`, and the 3,302 terms of `br-texto-nomes.txt` with `[NOME]`.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Cliente Souza, CPF 529.982.247-25, e-mail m.souza@gmail.com, tel (11) 98765-4321.` | → | **`Cliente [NOME], CPF [DADO], e-mail [DADO], tel (11) [DADO].`** |

PART 9

# Quasi-identifiers

Data that identifies no one on its own, but does in combination. Brazil has 5,570 municipalities, and 3,860 of them had fewer than 20,000 inhabitants in the 2022 census; the smallest, Serra da Saudade (MG), had 833. A birth date, a sex and one of those municipalities point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

## 9.1 Birth date

BR_L2_DATA_NASCIMENTO

ESSENTIAL · COLUMN NAME · DATA TYPE · DATE SHIFT

### WHAT IT IS

The birth date, as a date, timestamp or text.

### HOW IT IS FOUND

**COLUMN NAME** BR_L2_DATA_NASCIMENTO – Path

90% `dt_?nasc(imento)?` `data_?(de_?)?nasc(imento)?` `dat_?nasc` `d_?nasc` `nascimento` `dn` `dta_?nasc` `dtnasc` `datanasc(imento)?` `aniversario` `dob` `date_?of_?birth` `birth_?date` `birthday`

**DATA TYPE** BR_L2_DATA_NASCIMENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

### HOW IT IS MASKED

Shifted up to 60 days either way. Ages and school years change for few people; the date stays plausible and the same birth shifts the same way in every table.

`BR_DATA_NASCIMENTO` DATE SHIFT

Moves the date by -60 to 60 days.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1985-07-20 | → | **1985-08-05T00:00** |

## 9.2 Birth year

BR_L2_ANO_NASCIMENTO

COLUMN NAME · DATA TYPE · REGEX DECOMPOSE

### WHAT IT IS

The birth year in a column of its own.

### HOW IT IS FOUND

**COLUMN NAME** BR_L2_ANO_NASCIMENTO – Path

90% `ano_?(de_?)?nasc(imento)?` `an_?nasc` `nu_?ano_?nasc` `year_?of_?birth` `birth_?year` `yob`

**DATA TYPE** BR_L2_ANO_NASCIMENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

### HOW IT IS MASKED

The decade stays and the last digit is mapped: `1985` becomes a year from 1980 to 1989.

`BR_ANO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `BR_CM_DIGITOS` CHARACTER MAPPING
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1985 | → | **1980** |

## 9.3 Age

BR_L2_IDADE

COLUMN NAME | DATA TYPE | REGEX DECOMPOSE

WHAT IT IS

Age in years, also age group.

HOW IT IS FOUND

> COLUMN NAME BR_L2_IDADE – Path
>
> 85% `idade(_?(anos|atual|paciente|cliente|obito|diagnostico|admissao|na_?data))?` `nu_?idade` `qt_?idade` `faixa_?etaria` `anos_?de_?idade` `age(_?(years|group))?`

> DATA TYPE BR_L2_IDADE – Type
> Accepts number (length up to 6) or text (length up to 6); any other type rules the data class out.

HOW IT IS MASKED

The tens digit stays and the unit is mapped (37 becomes 30 to 39). A hundred or more becomes 90: rare ages identify.

`BR_IDADE` REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

> `BR_CM_DIGITOS` CHARACTER MAPPING
> Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 37 | → | **35** |
| 104 | → | **90** |

## 9.4 Personal event dates

BR_L2_DATA_EVENTO

COLUMN NAME | DATA TYPE | DATE SHIFT

WHAT IT IS

Dates that say something about the person: death, marriage, hiring and dismissal, retirement, admission and discharge, diagnosis, childbirth, arrest, document issuance.

HOW IT IS FOUND

COLUMN NAME BR_L2_DATA_EVENTO — Path

80% `dt_?(obito|falecimento|casamento|divorcio|admissao|demissao|desligamento|contratacao|aposentadoria|internacao|alta|diagnostico|atendimento|consulta|parto|prisao|condenacao|expedicao|emissao_?rg|emissao_?(cnh|documento)|naturalizacao|vacinacao|cirurgia|acidente|afastamento|posse|exoneracao) data_?(do_?|de_?)?(obito|falecimento|casamento|divorcio|admissao|demissao|desligamento|contratacao|aposentadoria|internacao|alta|diagnostico|atendimento|consulta|parto|prisao|condenacao|expedicao|naturalizacao|vacinacao|cirurgia|acidente|afastamento|posse|exoneracao) date_?of_?(death|marriage|hire|admission|discharge)`

DATA TYPE BR_L2_DATA_EVENTO — Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the data class out.

HOW IT IS MASKED

Shifted up to 30 days either way.

`BR_DATA_EVENTO` DATE SHIFT

Moves the date by -30 to 30 days.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 2021-03-15 | → | **2021-03-05T00:00** |

## 9.5 Sex

BR_L2_SEXO

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Sex as the record holds it: `F` / `M`, `Feminino` / `Masculino`, `Mulher` / `Homem` or the codes 1 and 2 of IBGE, SUS and eSocial.

HOW IT IS FOUND

COLUMN NAME BR_L2_SEXO — Path

85% `sexo(_?(biologico|nascimento|registro|paciente|cliente))?` `(tp|cs|co|cd|ds|sg|in|fl|id)_?sexo` `genero` *(except when followed by* `identidade`, `autodeclarado`, `social`*)* `sex` `gender` *(except when followed by* `identity`*)*

KNOWN VALUES BR_L2_SEXO — List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `br-detectar-sexo.txt` | 8 | 80% | feminino, masculino, fem, masc, mulher, homem |
| `br-detectar-sexo-letra.txt` | 2 | 60% | f, m |

Values are split into words, and each word is looked up.

A value that matches none counts 50% against the data class.

#### HOW IT IS MASKED

Replaced within its own vocabulary: `F` stays `F` or `M`, `1` stays `1` or `2`.

`BR_SEXO` **REGEX DECOMPOSE**

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

> `BR_SEXO_FEMININO_MASCULINO` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-sexo-feminino-masculino.txt`, chosen by a keyed hash.
>
> `BR_SEXO_MULHER_HOMEM` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-sexo-mulher-homem.txt`, chosen by a keyed hash.
>
> `BR_SEXO_FEM_MASC` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-sexo-fem-masc.txt`, chosen by a keyed hash.
>
> `BR_SEXO_FEMALE_MALE` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-sexo-female-male.txt`, chosen by a keyed hash.
>
> `BR_SEXO_F_M` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-sexo-f-m.txt`, chosen by a keyed hash.
>
> `BR_SEXO_1_2` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-sexo-1-2.txt`, chosen by a keyed hash.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| F | → | **M** |
| Masculino | → | **Feminino** |

---

## 9.6 Municipality

`BR_L2_MUNICIPIO`

**COLUMN NAME** **VALUE FORMAT** **DATA CLEANSING**

#### WHAT IT IS

The municipality of residence, birth (naturalidade) or occurrence, by name — on its own or with the state, `Bonito – MS`, `Bonito/MS`.

#### HOW IT IS FOUND

> **COLUMN NAME** `BR_L2_MUNICIPIO – Path`
>
> **85%** `municipio(_?(residencia|nascimento|endereco|domicilio|paciente|cliente|ocorrencia|notificacao|atendimento)?` *(except when followed by* ***`ibge`****,* ***`cod`****,* ***`codigo`****,* ***`id`****)* `(no|nm|ds|nome)_?(municipio|cidade)` `cidade(_?(residencia|nascimento|natal|endereco|cliente))?` `naturalidade` `local_?(de_?)?nascimento` `mun_?(res|resid|nasc|nascimento)` `city` `town` `localidade`

> **VALUE FORMAT** `BR_L2_MUNICIPIO – Values`
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 5,165 known values, ignoring case — Acrelândia, Assis Brasil, Brasiléia, Bujari, Capixaba, Cruzeiro do Sul, … | 80% | |
>
> A value that matches none counts 40% against the data class.

#### HOW IT IS MASKED

A municipality with fewer than 20,000 inhabitants becomes the nearest municipality of the same state with at least 20,000; larger ones stay. See *Where people live*.

`BR_MUNICIPIO` **DATA CLEANSING**

Replaces each value found in `br-municipios-generalizados.txt` (55,203 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Serra da Saudade | → | **São Gotardo** |
| Borá – SP | → | **Paraguaçu Paulista – SP** |
| Campinas | → | **Campinas** |

## 9.7 Municipality code (IBGE)

BR_L2_CODIGO_MUNICIPIO

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The IBGE municipality code, with seven digits, or six without the check digit, as in DATASUS.

HOW IT IS FOUND

COLUMN NAME BR_L2_CODIGO_MUNICIPIO – Path

85% `(cd|co|cod|codigo|id|nu)_?(municipio|mun|cidade)(_?(ibge|res|resid|residencia|nasc|nascimento|ocorrencia|notificacao))?` `(cd|co|cod|codigo)_?ibge(_?(municipio|mun|cidade))?` `ibge(_?(municipio|mun|cidade|codigo|cod))?` `municipio_?ibge` `geocodigo` `id_?mn_?resi` `id_?municip` `mun_?res_?ibge`

VALUE FORMAT BR_L2_CODIGO_MUNICIPIO – Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:(1[1-7]\|2[1-9]\|3[1-35]\|4[1-3]\|5[0-3])\d{4,5})` | 30% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The same generalization as the name: a small municipality's code becomes the code of the municipality that replaces its name. The two still agree.

`BR_CODIGO_MUNICIPIO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

> `BR_CODIGO_MUNICIPIO_TABELA` DATA CLEANSING
>
> Replaces each value found in `br-codigos-municipio-generalizados.txt` (7,720 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 3166600 | → | **3162104** |
| 350720 | → | **353550** |

## 9.8 CEP (postal code)

BR_L2_CEP

ESSENTIAL · COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The postal code: region, sub-region, sector, sub-sector, sub-sector division and a three-digit suffix that identifies the street or the locality. In a large city, a CEP is a stretch of street.

HOW IT IS FOUND

COLUMN NAME BR_L2_CEP — Path

90% `cep(_?(residencia|residencial|comercial|cobranca|entrega|cliente|endereco|paciente|origem|destino))?` `(nr|nu|num|numero|cd|co|cod)_?cep` `codigo_?(de_?)?enderecamento_?postal` `codigo_?postal` `zip_?code` `zip` `postal_?code`

VALUE FORMAT BR_L2_CEP — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{5}-\d{3})` | 80% | |
| `0*(?:\d{2}\.\d{3}-\d{3})` | 90% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The first three digits — the postal sector — stay, and the others become zero: `01310-100` becomes `01300-000`. In a numeric column, without the leading zero, too.

`BR_CEP` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 01310-100 | → | **01300-000** |

## 9.9 Neighbourhood

BR_L2_BAIRRO

COLUMN NAME SECURE LOOKUP

WHAT IT IS

The neighbourhood of residence.

HOW IT IS FOUND

COLUMN NAME BR_L2_BAIRRO — Path

85% `bairro(_?(residencia|endereco|cliente|entrega|cobranca))?` `(no|nm|ds|nome)_?bairro` `distrito_?residencia` `localidade_?bairro` `vila_?residencia` `neighbou?rhood`

HOW IT IS MASKED

Replaced with a neighbourhood name common in Brazil — Centro, Jardim América, Vila Nova —, unrelated to the original.

`BR_BAIRRO` SECURE LOOKUP

Replaces the value with one of the 40 values in `br-bairros.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Vila Madalena | → | **Jardim Esperança** |

## 9.10 Electoral zone and section

BR_L2_ZONA_SECAO_ELEITORAL

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

The zone and section where the person votes. A section gathers a few hundred voters of one polling place.

### HOW IT IS FOUND

COLUMN NAME — BR_L2_ZONA_SECAO_ELEITORAL — Path

80% `zona_?(eleitoral)?` `secao_?(eleitoral)?` `zona_?secao` `nr_?(zona|secao)` `nu_?(zona|secao)` `local_?(de_?)?votacao`

### HOW IT IS MASKED

Digit by digit, keeping the length.

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 0245 | → | **0409** |

---

## 9.11 Nationality

BR_L2_NACIONALIDADE

COLUMN NAME | KNOWN VALUES | SECURE LOOKUP

### WHAT IT IS

Nationality or country of birth. Among immigrants, Venezuelans, Haitians, Bolivians and Portuguese are the largest groups.

### HOW IT IS FOUND

COLUMN NAME — BR_L2_NACIONALIDADE — Path

80% `nacionalidade` `pais_?(de_?)?(nascimento|origem|nacionalidade)` `cidadania` `nationality` `citizenship` `country_?of_?birth`

KNOWN VALUES — BR_L2_NACIONALIDADE — List

| LIST | VALUES | STRENGTH | FIRST VALUES |
|---|---|---|---|
| `br-detectar-nacionalidades.txt` | 58 | 70% | Brasileira, Venezuelana, Haitiana, Boliviana, Portuguesa, Paraguaia |

Values are split into words, and each word is looked up.

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

Replaced with a nationality from the list: Brazilian and the largest foreign communities.

`BR_NACIONALIDADE` SECURE LOOKUP

Replaces the value with one of the 16 values in `br-nacionalidades.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Senegalesa | → | **Venezuelana** |

---

## 9.12 Marital status

BR_L2_ESTADO_CIVIL

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Marital status as IBGE and eSocial record it — single, married, stable union, divorced, separated, widowed —, in words or as a code.

### HOW IT IS FOUND

> **COLUMN NAME** BR_L2_ESTADO_CIVIL – Path
>
> 85% `estado_?civil` `est_?civil` `(cd|co|tp|ds|in)_?estado_?civil` `situacao_?conjugal` `estciv` `marital_?status` `civil_?status`

> **VALUE FORMAT** BR_L2_ESTADO_CIVIL – Values
>
| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 21 known values, ignoring case — Solteiro a, Casado a, União estável, Divorciado a, Separado a, Viúvo a, ... | 70% | |
>
> A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

Replaced with another status from the list; a code from 1 to 5 stays a code from 1 to 5.

`BR_ESTADO_CIVIL` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_ESTADO_CIVIL_LISTA` .

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.
>
> `BR_ESTADO_CIVIL_CODIGO` SECURE LOOKUP
>
> Replaces the value with one of the 5 values in `br-estado-civil-codigos.txt` , chosen by a keyed hash.
>
> `BR_ESTADO_CIVIL_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 6 values in `br-estado-civil.txt` , chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Separado judicialmente` | → | **`União estável`** |
| 2 | → | **4** |

## 9.13 Occupation

BR_L2_OCUPACAO

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

Occupation, profession or position, in words or by the code of the Brazilian Classification of Occupations (CBO), `2235-10`. A rare occupation in a small municipality identifies.

### HOW IT IS FOUND

COLUMN NAME — BR_L2_OCUPACAO — Path

**80%** `ocupacao` `profissao` `cbo` `(cd|co|cod|codigo)_?(cbo|ocupacao|profissao)` `cargo_?(funcionario|atual|ocupado)` `funcao_?(exercida|funcionario)` `atividade_?profissional` `occupation` `profession` `job_?title`

**50%** `cargo` `funcao`

### HOW IT IS MASKED

A text is replaced with a common occupation; a six-digit CBO code with a common CBO code; `2235-10` is mapped digit by digit.

`BR_OCUPACAO_OU_CBO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_OCUPACAO`.

> `BR_CBO` SECURE LOOKUP
>
> Replaces the value with one of the 34 values in `br-cbo.txt`, chosen by a keyed hash.
>
> `BR_CM_DIGITOS` CHARACTER MAPPING
>
> Maps each character to another of its group (1 groups), keeping length and format.
>
> `BR_OCUPACAO` SECURE LOOKUP
>
> Replaces the value with one of the 40 values in `br-ocupacoes.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Coordenadora de operações regionais | → | **Do lar** |
| 2235-10 | → | **0208-89** |

## 9.14 Employer

BR_L2_EMPREGADOR

COLUMN NAME | SECURE LOOKUP

### WHAT IT IS

The name of the company or public body the person works for.

### HOW IT IS FOUND

COLUMN NAME — BR_L2_EMPREGADOR — Path

**80%** `empregador(_?(nome|razao_?social))?` `(nome|nm|no)_?empregador` `empresa_?(onde_?trabalha|trabalho|empregadora|atual)` `local_?(de_?)?trabalho` `empregadora` `razao_?social_?empregador` `employer(_?name)?` `workplace`

### HOW IT IS MASKED

Replaced with a fictitious company.

`BR_EMPREGADOR` SECURE LOOKUP

Replaces the value with one of the 24 values in `br-empregadores.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Petróleo Brasileiro S.A. | → | **Associação Mãos Solidárias** |

## 9.15 Education

BR_L2_ESCOLARIDADE

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Education level as IBGE and eSocial record it, in words or as a code.

HOW IT IS FOUND

COLUMN NAME BR_L2_ESCOLARIDADE — Path

80% `escolaridade` `grau_?(de_?)?instrucao` `grau_?instr` `nivel_?(de_?)?(escolaridade|instrucao|ensino)` `(cd|co|tp|ds)_?escolaridade` `instrucao` `formacao_?academica` `education(_?level)?`

VALUE FORMAT BR_L2_ESCOLARIDADE — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 22 known values, ignoring case — Sem instrução, Fundamental incompleto, Fundamental completo, Médio incompleto, Médio completo, Superior incompleto, … | 60% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced with another level from the list; a code stays a code.

`BR_ESCOLARIDADE` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_ESCOLARIDADE_LISTA`.

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_ESCOLARIDADE_CODIGO` SECURE LOOKUP
>
> Replaces the value with one of the 12 values in `br-escolaridade-codigos.txt`, chosen by a keyed hash.
>
> `BR_ESCOLARIDADE_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 10 values in `br-escolaridade.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Pós-doutorado | → | **Sem instrução** |

## 9.16 School or educational institution

BR_L2_ESCOLA

COLUMN NAME · SECURE LOOKUP

WHAT IT IS

The school, college or university, by name or by its INEP code.

HOW IT IS FOUND

> COLUMN NAME BR_L2_ESCOLA – Path
>
> 75% `escola(_?(nome|atual|origem))?` `(nome|nm|no)_?(escola|instituicao_?ensino|faculdade|universidade)` `instituicao_?(de_?)?ensino` `faculdade` `universidade` `colegio` `(cd|co)_?(inep|entidade|escola)` `school(_?name)?` `university`

HOW IT IS MASKED

Replaced with a fictitious institution.

`BR_ESCOLA` SECURE LOOKUP

Replaces the value with one of the 12 values in `br-escolas.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Escola Estadual Doutor Arnaldo | → | **Escola Estadual Professora Maria José** |

## 9.17 Dependants and children

BR_L2_DEPENDENTES

COLUMN NAME · DATA TYPE · REGEX DECOMPOSE

WHAT IT IS

Number of dependants, children or household members.

HOW IT IS FOUND

> COLUMN NAME BR_L2_DEPENDENTES – Path
>
> 80% `dependentes` `qt_?(dependentes|filhos|dep)` `qtd_?(dependentes|filhos)` `quantidade_?(de_?)?(dependentes|filhos)` `n(r|u|um|o|ro|umero)_?(de_?)?(dependentes|filhos)` `filhos` `moradores_?(no_?)?domicilio` `qt_?moradores` `number_?of_?children` `dependents`

> DATA TYPE BR_L2_DEPENDENTES – Type
>
> Accepts number (length up to 4) or text (length up to 4); any other type rules the data class out.

HOW IT IS MASKED

Values from 0 to 5 stay; above 5 they become 5. Very large families are rare, and a rare count identifies.

`BR_CONTAGEM_LIMITADA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| 9 | → | **5** |

PART 10

# Sensitive personal data

The categories of art. 5, II that are not health data — which has a group of its own: racial or ethnic origin, religious conviction, political opinion, membership of a trade union or of an organization of religious, philosophical or political character, sexual life, genetic and biometric data — and, beyond the list, sexual orientation and gender identity, which expose people to the same discrimination. Categorical columns keep their shape: an S/N flag stays a flag, a numeric code is replaced by another code of the same catalog and any other value is replaced by one from the list of its category. Where the values share no vocabulary, they are suppressed to `Não informado`.

## 10.1 Colour or race

BR_L3_RACA_COR

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Self-declared colour or race in the IBGE categories — in the 2022 census, brown (parda) 45.3%, white 43.5%, black 10.2%, indigenous 0.8% and yellow 0.4% —, in words or as the codes 1 to 5 and 9 used in SUS and eSocial. Racial origin is sensitive data (art. 5, II).

### HOW IT IS FOUND

COLUMN NAME — BR_L3_RACA_COR — Path

95% `raca(_?cor)?` `cor(_?raca)?` `raca_?etnia` `(cd|co|cs|tp|ds|id|in)_?(raca|cor|raca_?cor)` `autodeclaracao_?(racial|raca|cor)` `cor_?da_?pele` `etnia_?racial` `race` `ethnicity`

VALUE FORMAT — BR_L3_RACA_COR — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 19 known values, ignoring case — Branca, Preta, Parda, Amarela, Indígena, Não declarada, … | 90% | |
| One of 7 known values, ignoring case — 1, 2, 3, 4, 5, 6, … | 30% | |

A value that matches none counts 40% against the data class.

### HOW IT IS MASKED

A code is replaced with another code of the catalog; any other value with an IBGE category.

`BR_RACA_COR` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_RACA_COR_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_RACA_COR_CODIGO` SECURE LOOKUP

Replaces the value with one of the 6 values in `br-raca-cor-codigos.txt`, chosen by a keyed hash.

`BR_RACA_COR_LISTA` SECURE LOOKUP

Replaces the value with one of the 6 values in `br-raca-cor.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Parda | → | **Não declarada** |
| 3 | → | **5** |

## 10.2 Indigenous people and traditional community

BR_L3_ETNIA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The indigenous people — the 2022 census counted 1,694,836 indigenous people of 391 ethnicities; the most numerous are the Tikúna, Kokama, Makuxí and Guarani Kaiowá —, the quilombola or traditional community and the village, often as an S/N flag.

HOW IT IS FOUND

COLUMN NAME BR_L3_ETNIA — Path

**90%** `etnia` *(except when followed by* `racial` *)* `povo(_?indigena)?` `etnia_?indigena` `(cd|co|ds|no)_?etnia` `comunidade_?(tradicional|quilombola|indigena)` `quilombola` `aldeia` `terra_?indigena` `povo_?tradicional` `indigena` `lingua_?indigena` `ethnic_?group` `indigenous`

VALUE FORMAT BR_L3_ETNIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 47 known values, ignoring case — Tikúna, Kokama, Makuxí, Guarani Kaiowá, Guajajara, Terena, … | 90% | |
| One of 7 known values, ignoring case — s, n, sim, não, y, 0, … | 30% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag in its vocabulary and any other value is replaced with one from the list.

`BR_ETNIA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ETNIA_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_ETNIA_LISTA` SECURE LOOKUP

Replaces the value with one of the 30 values in `br-etnias.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Krenak | → | **Guarani Nhandeva** |
| S | → | **S** |

## 10.3 Religion

BR_L3_RELIGIAO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Religion or church. In the 2022 census, among people aged 10 or over, 56.7% declared themselves Catholic, 26.9% Evangelical, 9.3% of no religion, 1.8% Spiritist and 1.0% of Umbanda or Candomblé.

HOW IT IS FOUND

COLUMN NAME BR_L3_RELIGIAO — Path

95% `religiao` `crenca_?religiosa` `convicc?ao_?religiosa` `confissao_?religiosa` `(cd|co|ds|tp)_?religiao` `igreja(_?(frequenta|membro|nome))?` `denominacao_?religiosa` `culto` `religion` `church`

VALUE FORMAT BR_L3_RELIGIAO — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 45 known values, ignoring case — Católica Apostólica Romana, Evangélica, Evangélica pentecostal, Evangélica de missão, Espírita, Umbanda, … | 90% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced with a religion from the list.

`BR_RELIGIAO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_RELIGIAO_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_RELIGIAO_LISTA` SECURE LOOKUP

Replaces the value with one of the 13 values in `br-religioes.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Adventista do Sétimo Dia | → | **Candomblé** |

## 10.4 Convictions and religious, philosophical or political organizations

BR_L3_CONVICCAO_ORGANIZACAO

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Philosophical and moral convictions, conscientious objection and membership of an organization of religious, philosophical or political character — a religious order, freemasonry, a social movement —, which the LGPD lists apart from religion and party.

HOW IT IS FOUND

COLUMN NAME BR_L3_CONVICCAO_ORGANIZACAO — Path

90% `convicc?ao_?(filosofica|moral|politica)` `crenca_?(filosofica|pessoal)` `escusa_?(de_?)?consciencia` `objecao_?(de_?)?consciencia` `maconaria` `loja_?maconica` `ordem_?(religiosa|filosofica)` `organizacao_?(religiosa|filosofica|politica)` `filiacao_?(religiosa|filosofica)` `associacao_?(religiosa|filosofica)` `movimento_?(social|politico|religioso)` `grupo_?(religioso|filosofico)` `beliefs?` `philosophical_?beliefs?`

HOW IT IS MASKED

The values share no vocabulary: they are suppressed to `Não informado`. Flags and codes keep their shape.

`BR_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_SUPRIMIR` SECURE LOOKUP

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Recusa transfusão de sangue` | → | **`Não informado`** |

## 10.5 Political opinion

BR_L3_OPINIAO_POLITICA

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Political position, ideology and voting intention, from surveys and registers.

HOW IT IS FOUND

COLUMN NAME BR_L3_OPINIAO_POLITICA — Path

90% `opiniao_?politica` `posicionamento_?politico` `orientacao_?politica` `preferencia_?(politica|partidaria)` `ideologia(_?politica)?` `intencao_?(de_?)?voto` `voto_?(pretendido|intencao)` `espectro_?politico` `political_?(opinion|view|orientation)` `voting_?intention`

VALUE FORMAT BR_L3_OPINIAO_POLITICA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 21 known values, ignoring case — Esquerda, Centro-esquerda, Centro, Centro-direita, Direita, Sem preferência, … | 70% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced with a position from the list — left to right, no preference.

`BR_OPINIAO_POLITICA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_OPINIAO_POLITICA_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_OPINIAO_POLITICA_LISTA` SECURE LOOKUP

Replaces the value with one of the 7 values in `br-opinioes-politicas.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Conservador | → | **Centro-esquerda** |

## 10.6 Party membership

BR_L3_FILIACAO_PARTIDARIA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The party the person is a member of, by acronym, name or ballot number. The TSE had 30 registered parties in September 2026.

HOW IT IS FOUND

COLUMN NAME BR_L3_FILIACAO_PARTIDARIA — Path

95% `filiacao_?partidaria` `partido(_?politico)?(_?(filiacao|sigla|nome|numero))?` `sigla_?partido` `(cd|co|sg|nm|nr)_?partido` `filiado_?(a_?)?partido` `militancia` `militante` `political_?party` `party_?membership`

VALUE FORMAT BR_L3_FILIACAO_PARTIDARIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 69 known values, ignoring case — MDB, Movimento Democrático Brasileiro, PDT, Partido Democrático Trabalhista, PT, Partido dos Trabalhadores, … | 80% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

An acronym becomes another acronym, a ballot number another number and a name another name from the list of registered parties; a flag stays a flag.

`BR_PARTIDO` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_PARTIDO_NOME`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_PARTIDO_NUMERO` SECURE LOOKUP

Replaces the value with one of the 30 values in `br-partidos-numeros.txt`, chosen by a keyed hash.

`BR_PARTIDO_SIGLA` SECURE LOOKUP

Replaces the value with one of the 30 values in `br-partidos-siglas.txt`, chosen by a keyed hash.

`BR_PARTIDO_NOME` SECURE LOOKUP

Replaces the value with one of the 31 values in `br-partidos-nomes.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| PT | → | **MISSÃO** |
| Partido Trabalhista Brasileiro | → | **Democracia Cristã** |

## 10.7 Trade union membership

BR_L3_FILIACAO_SINDICAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The union, being a union member, the union contribution or fee deducted from payroll and the union confederation.

HOW IT IS FOUND

COLUMN NAME BR_L3_FILIACAO_SINDICAL — Path

95% `sindicato(_?(nome|filiado|filiacao|categoria|sigla|cnpj))?` `sindicalizado` `filiacao_?sindical` `filiado_?sindicato` `contribuicao_?(sindical|assistencial|confederativa)` `mensalidade_?sindical` `desconto_?sindical` `imposto_?sindical` `central_?sindical` `entidade_?sindical` `delegado_?sindical` `dirigente_?sindical` `trade_?union` `union_?member(ship)?`

VALUE FORMAT BR_L3_FILIACAO_SINDICAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 23 known values, ignoring case — CUT, Força Sindical, UGT, CTB, NCST, CSB, … | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Não informado`.

`BR_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`BR_SUPRIMIR` **SECURE LOOKUP**

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| `Sindicato dos Bancários de SP` | → | **`Não informado`** |
| `S` | → | **`S`** |

## 10.8 Genetic data

BR_L3_GENETICO

**COLUMN NAME** **VALUE FORMAT** **REGEX DECOMPOSE**

**WHAT IT IS**

Genetic and DNA test results, genetic profiles, mutations, karyotype, newborn screening.

**HOW IT IS FOUND**

**COLUMN NAME** BR_L3_GENETICO — Path

**90%** `dna` `adn` `genetic[oa]s?` `dados?_?geneticos?` `genoma` `genotip[a-z_]*` `exame_?genetico` `teste_?(genetico|de_?dna|paternidade)` `perfil_?genetico` `mutacao(_?genetica)?` `brca[12]?` `cariotipo` `triagem_?neonatal` `teste_?do_?pezinho` `painel_?genetico` `snp` `haplotipo` `genetic[a-z_]*` `genome`

**VALUE FORMAT** BR_L3_GENETICO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[ACGTUN]{24,}` | 90% | |

A value that matches none counts 30% against the data class.

**HOW IT IS MASKED**

Every character becomes `X`.

`BR_REDIGIR` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| `BRCA1 positivo` | → | **`XXXXXXXXXXXXXX`** |

## 10.9 Biometric data

BR_L3_BIOMETRICO

**COLUMN NAME** **DATA TYPE** **REGEX DECOMPOSE**

**WHAT IT IS**

Fingerprint, face, iris and voice templates from time clocks, proof of life, account opening and turnstiles.

HOW IT IS FOUND

COLUMN NAME BR_L3_BIOMETRICO — Path

90% `biometri[a-z_]*` `digita(l|is)(_?(template|polegar|indicador|hash|imagem))?` `impressao_?digital` `template_?(facial|digital|biometrico)` `face_?(template|hash|id|encoding)` `reconhecimento_?facial` `facial_?(template|hash|embedding)` `iris(_?(template|scan))?` `retina` `voz_?(template|biometrica)` `assinatura_?(biometrica|digitalizada)` `palma_?mao` `fingerprints?` `biometric[a-z_]*`

DATA TYPE BR_L3_BIOMETRICO — Type

Accepts text; any other type rules the data class out.

HOW IT IS MASKED

Every character becomes `X`. A masked template authenticates no one.

`BR_REDIGIR` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| A1B2C3D4E5F6 | → | XXXXXXXXXXXX |

## 10.10 Sexual orientation

BR_L3_ORIENTACAO_SEXUAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Self-declared sexual orientation, in health and diversity registers.

HOW IT IS FOUND

COLUMN NAME BR_L3_ORIENTACAO_SEXUAL — Path

95% `orientacao_?sexual` `orient_?sexual` `(cd|co|ds|tp)_?orientacao_?sexual` `preferencia_?sexual` `sexual_?orientation`

VALUE FORMAT BR_L3_ORIENTACAO_SEXUAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 15 known values, ignoring case — Heterossexual, Gay, Lésbica, Bissexual, Pansexual, Assexual, … | 80% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced with a category from the list.

`BR_ORIENTACAO_SEXUAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ORIENTACAO_SEXUAL_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_ORIENTACAO_SEXUAL_LISTA` SECURE LOOKUP

Replaces the value with one of the 8 values in `br-orientacoes-sexuais.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Homossexual | → | **Pansexual** |

## 10.11 Sexual life

BR_L3_VIDA_SEXUAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Sexual activity and behaviour, number of partners, sexual debut — data concerning sexual life, sensitive under art. 5, II.

HOW IT IS FOUND

COLUMN NAME BR_L3_VIDA_SEXUAL — Path

90% `vida_?sexual` `atividade_?sexual` `comportamento_?sexual` `pratica_?sexual` `parceiros?_?` `sexua(l|is)` `n(r|u|um)_?parceiros` `sexualmente_?ativ[oa]` `inicio_?(da_?)?vida_?sexual` `relacao_?sexual` `sex(ual)?_?life` `sexual_?(activity|behavio(u)?r|partners)`

HOW IT IS MASKED

A flag stays a flag; any other value becomes `Não informado`.

`BR_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_SUPRIMIR` SECURE LOOKUP

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Ativa | → | **Não informado** |

## 10.12 Gender identity

BR_L3_IDENTIDADE_GENERO

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Self-declared gender identity. Not on the list of art. 5, II, but it exposes people to the same discrimination. The social name falls under the full-name data class.

HOW IT IS FOUND

COLUMN NAME BR_L3_IDENTIDADE_GENERO — Path

95% `identidade_?(de_?)?genero` `(cd|co|ds|tp)_?identidade_?genero` `genero_?(autodeclarado|social)` `transgenero` `pessoa_?trans` `pronomes?` `gender_?identity`

VALUE FORMAT BR_L3_IDENTIDADE_GENERO — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 19 known values, ignoring case — Mulher cisgênero, Homem cisgênero, Mulher trans, Homem trans, Travesti, Não binário, … | 70% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced with a category from the list.

`BR_IDENTIDADE_GENERO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_IDENTIDADE_GENERO_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

> `BR_IDENTIDADE_GENERO_LISTA` **SECURE LOOKUP**
>
> Replaces the value with one of the 8 values in `br-identidades-genero.txt`, chosen by a keyed hash.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| Transgênero | → | **Prefere não informar** |

PART 11

# Health data

Data concerning health is sensitive personal data (art. 5, II). The law forbids sharing health data between controllers to obtain an economic advantage, except in health services and pharmaceutical care, and forbids health plan operators to use it for risk selection (art. 11, §§ 4 and 5). Public health studies must anonymize or pseudonymize the data whenever possible (art. 13). The group covers what health systems, health plans, laboratories and HR (occupational health) keep.

## 11.1 National Health Card

BR_SAUDE_CNS

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

### WHAT IT IS

The number of the Cartão Nacional de Saúde (SUS card): fifteen digits whose weighted sum, with weights 15 to 1, is a multiple of 11. A definitive number starts with 1 or 2 and is the person's PIS followed by `000` or `001` and a digit; a provisional one starts with 7, 8 or 9.

### HOW IT IS FOUND

**COLUMN NAME** BR_SAUDE_CNS — Path

90% `cns[a-z0-9_]*` `cartao_?(nacional_?)?(de_?)?saude` `cartao_?sus` `n(r|u|um|o|ro|umero)_?(cns|cartao_?sus)` `(co|cd|nu)_?cns_?(paciente|usuario|cidadao|profissional)?` `cns_?(paciente|usuario|cidadao|profissional)` `carteira_?sus`

**VALUE FORMAT** BR_SAUDE_CNS — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:[12789]\d{2}\s?\d{4}\s?\d{4}\s?\d{4})` | 50% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

In a definitive number, the first eleven digits are masked by the PIS algorithm — the masked CNS still carries the masked PIS — and the tail is recomputed. Every masked definitive CNS is valid. A provisional one is masked digit by digit, keeping the first.

`BR_CNS` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

> `BR_CNS_VALIDO` STRING ALGORITHM CHAIN
>
> Applies `BR_CNS_CORPO` → `BR_CNS_DV`, in this order.
>
> > `BR_CNS_CORPO` REGEX DECOMPOSE
> >
> > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> >
> > > `BR_PIS_VALIDO` CHECK DIGIT
> > >
> > > Masks the digits with `BR_PIS_CORPO` and recomputes a modulus 11 check digit with weights `3 2 9 8 7 6 5 4 3 2`.
> > >
> > > > `BR_PIS_CORPO` REGEX DECOMPOSE
> > > >
> > > > Matches one pattern and gives each part of the value its own action; when none matches: keep.
> > > >
> > > > > `BR_CM_DIGITOS` CHARACTER MAPPING
> > > > >
> > > > > Maps each character to another of its group (1 groups), keeping length and format.
> >
> > `BR_CNS_DV` STRING ALGORITHM CHAIN
> >
> > Applies `BR_CNS_DV_P1` → `BR_CNS_DV_P2` → `BR_CNS_DV_P3` → `BR_CNS_DV_P4` → `BR_CNS_DV_P5` → `BR_CNS_DV_P6`, in this order. It is made of 12 algorithms that repeat the same step position by position; each one is in Appendix A.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 120564125470008 | → | **113811042790001** |

## 11.2 Health identifiers

BR_SAUDE_IDENTIFICADOR

COLUMN NAME | CHARACTER MAPPING

WHAT IT IS

Medical record, visit, hospitalization (AIH), TISS form, health plan card, prescription and SINAN notification numbers.

HOW IT IS FOUND

COLUMN NAME — BR_SAUDE_IDENTIFICADOR — Path

85% `prontuario(_?(numero|num|nr|eletronico))?` `n(r|u|um|o|ro|umero)_?(prontuario|atendimento|internacao|aih|guia|carteirinha|beneficiario_?plano|registro_?paciente)` `aih` `autorizacao_?(de_?)?internacao` `numero_?(da_?)?guia` `guia_?(tiss|sadt|internacao)` `carteirinha_?(plano|convenio|saude)` `matricula_?(convenio|plano)` `cod_?(beneficiario|paciente)_?(ans|plano|convenio)` `registro_?(do_?)?paciente` `id_?paciente` `codigo_?paciente` `n(r|u)_?receita` `receita_?(numero|controlada)` `notificacao_?(sinan|numero)` `nu_?notific` `laudo_?(numero|nr)`

55% `atendimento` `internacao` `guia` `episodio`

HOW IT IS MASKED

Character by character, keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| PRONT-2024-00123 | → | **TVAQH-6080-27071** |

## 11.3 Diagnosis and ICD

BR_SAUDE_DIAGNOSTICO

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

WHAT IT IS

Diagnoses in words or by ICD-10 code, with the dot ( `F33.1` ) or without it, as in DATASUS ( `F331` ); causes of sick leave, hospitalization and death, comorbidities and allergies.

### HOW IT IS FOUND

**COLUMN NAME** BR_SAUDE_DIAGNOSTICO — Path

**85%** `diagnostico[a-z_]*` `cid(_?(10|11|principal|secundario|obito|afastamento|atestado|internacao))?` `(cd|co|ds)_?cid` `cid_?(cd|co|ds)` `hipotese_?diagnostica` `doenca[a-z_]*` `agravo` `comorbidade[a-z_]*` `patologia` `condicao_?(de_?)?saude` `causa_?(do_?)?(obito|afastamento|internacao|basica)` `motivo_?(do_?)?(afastamento|atestado|internacao|consulta)` `queixa_?principal` `alergias?` `doencas?_?pre_?existentes?` `historico_?(medico|clinico|familiar)` `diagnos(is|es)` `icd_?(10|11)?`

**VALUE FORMAT** BR_SAUDE_DIAGNOSTICO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `[A-TV-Za-tv-z]\d{2}(\.\d{1,2}\|\d)?` | 50% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

An ICD code becomes another common code in the same format, and a text a common diagnosis: `F33.1` may become `J06.9`. So an HIV or mental disorder diagnosis does not survive as a category.

`BR_DIAGNOSTICO` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_DIAGNOSTICO_TEXTO`.

> `BR_CID10_DECIMAL` **SECURE LOOKUP**
>
> Replaces the value with one of the 33 values in `br-cid10-decimal.txt`, chosen by a keyed hash.
>
> `BR_CID10` **SECURE LOOKUP**
>
> Replaces the value with one of the 37 values in `br-cid10.txt`, chosen by a keyed hash.
>
> `BR_FLAG` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_DIAGNOSTICO_TEXTO` **SECURE LOOKUP**
>
> Replaces the value with one of the 30 values in `br-diagnosticos.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| F33.1 | → | **K21.9** |
| B20 | → | **S60** |

## 11.4 Medicines

BR_SAUDE_MEDICAMENTO

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

Medicines prescribed, dispensed or taken continuously. An antiretroviral, an antipsychotic or a contraceptive reveals a diagnosis.

### HOW IT IS FOUND

**COLUMN NAME** BR_SAUDE_MEDICAMENTO — Path

85% `medicamentos?(_?(uso|prescrito|continuo|nome|dispensado))?` `remedios?` `farmacos?` `principio_?ativo` `prescricao(_?medicamento)?` `posologia` `receita_?(medicamento|item)` `uso_?continuo` `(cd|co|ds|no)_?medicamento` `medications?` `drugs?_?prescribed`

**VALUE FORMAT** BR_SAUDE_MEDICAMENTO — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 53 known values, ignoring case — Losartana potássica, Metformina, Omeprazol, Sinvastatina, Hidroclorotiazida, Dipirona, … | 70% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

Replaced with a common medicine from the list; a flag stays a flag.

`BR_MEDICAMENTO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_MEDICAMENTO_LISTA` .

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `BR_MEDICAMENTO_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 25 values in `br-medicamentos.txt` , chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Sertralina 50 mg` | → | **`Captopril`** |

## 11.5 Clinical text

BR_SAUDE_TEXTO_CLINICO

COLUMN NAME | DATA TYPE | FREE TEXT REDACTION

### WHAT IT IS

Medical history, progress notes, physical examination, reports, discharge summary and nursing notes.

### HOW IT IS FOUND

**COLUMN NAME** BR_SAUDE_TEXTO_CLINICO – Path

85% `anamnese` `evolucao(_?(clinica|medica|enfermagem))?` `historia_?(clinica|da_?doenca_?atual)` `hda` `exame_?fisico` `conduta(_?medica)?` `plano_?terapeutico` `sumario_?(de_?)?alta` `resumo_?(de_?)?alta` `laudo(_?(texto|descricao|medico))?` `prescricao_?texto` `notas?_?(clinicas|medicas|enfermagem)` `observac(ao|oes)_?(clinicas?|medicas?|enfermagem)` `triagem(_?texto)?` `epicrise` `parecer_?medico` `clinical_?notes?` `discharge_?summary`

**DATA TYPE** BR_SAUDE_TEXTO_CLINICO – Type

Accepts text (length 30 or more); any other type rules the data class out.

### HOW IT IS MASKED

The same Free Text Redaction as free text: CPF, contacts and names from the list are redacted. The clinical content stays — if it must not be in the copy, suppress the column.

`BR_TEXTO_LIVRE` FREE TEXT REDACTION

Replaces the text matching 12 patterns with `[DADO]`, and the 3,302 terms of `br-texto-nomes.txt` with `[NOME]`.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Paciente Souza, CPF 529.982.247-25, atendido por cefaleia.` | → | **`Paciente [NOME], CPF [DADO], atendido por cefaleia.`** |

## 11.6 Test results

BR_SAUDE_RESULTADO_EXAME

COLUMN NAME | CHARACTER MAPPING

### WHAT IT IS

Laboratory and test results — HIV, viral load, blood glucose, pregnancy, drug test —, also as numeric values.

### HOW IT IS FOUND

**COLUMN NAME** BR_SAUDE_RESULTADO_EXAME – Path

80% `resultado_?(do_?)?` `(exame|teste|laboratorio|laboratorial|hiv|covid|pcr|sorologia|biopsia|glicemia)` `exame_?(resultado|laboratorial)` `hiv(_?(status|resultado|teste))?` `carga_?viral` `cd4` `glicemia` `hemoglobina_?glicada` `hba1c` `colesterol` `imc` `pressao_?arterial` `pa_?(sistolica|diastolica)` `teste_?(de_?)?gravidez` `toxicologico` `antidoping` `sorologia` `lab_?results?` `test_?results?`

50% `resultado` `teste`

### HOW IT IS MASKED

Character by character, keeping the format: a number stays a number.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Reagente | → | **Suaqunwi** |

## 11.7 Blood group

BR_SAUDE_TIPO_SANGUINEO

COLUMN NAME · VALUE FORMAT · SECURE LOOKUP

WHAT IT IS

ABO group and Rh factor.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_TIPO_SANGUINEO – Path

**90%** `tipo_?sangu[ií]neo` `grupo_?sangu[ií]neo` `tipagem_?sangu[ií]nea` `fator_?rh` `abo(_?rh)?` `tipo_?(de_?)?sangue` `blood_?(type|group)`

VALUE FORMAT BR_SAUDE_TIPO_SANGUINEO – Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 18 known values, ignoring case — A, A-, B, B-, AB, AB-, … | 90% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

Replaced with another group from the list.

`dlpx-core:BloodType SL` SECURE LOOKUP

Comes with Delphix and is used as it ships. Replaces the blood group with one of the eight groups, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 0+ | → | **0–** |

## 11.8 Disability

BR_SAUDE_DEFICIENCIA

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The type of disability — physical, visual, hearing, intellectual, mental or psychosocial, multiple —, being a person with a disability (Lei 13.146/2015) and the disability report.

HOW IT IS FOUND

COLUMN NAME BR_SAUDE_DEFICIENCIA – Path

**90%** `deficiencia[a-z_]*` `(tp|cd|co|ds|in|fl)_?deficiencia` `pcd` `pessoa_?com_?deficiencia` `portador_?(de_?)?(deficiencia|necessidades_?especiais)` `necessidades?_?especia(l|is)` `laudo_?pcd` `mobilidade_?reduzida` `tea` `autismo` `disabilit(y|ies)`

VALUE FORMAT BR_SAUDE_DEFICIENCIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 23 known values, ignoring case — Física, Visual, Auditiva, Intelectual, Mental ou psicossocial, Múltipla, … | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced with a type from the list.

`BR_DEFICIENCIA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_DEFICIENCIA_LISTA`.

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `BR_DEFICIENCIA_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 7 values in `br-deficiencias.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Transtorno do espectro autista` | → | **`Auditiva`** |

## 11.9 Occupational health

BR_SAUDE_OCUPACIONAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

What HR keeps about an employee's health: the result of the occupational health certificate (ASO), the work accident report (CAT), medical certificates, sick leave and absences.

### HOW IT IS FOUND

**COLUMN NAME** BR_SAUDE_OCUPACIONAL — Path

85% `aso(_?(resultado|conclusao|parecer))?` `atestado_?(de_?)?saude_?ocupacional` `aptidao(_?(medica|laboral))?` `resultado_?(do_?)?aso` `parecer_?(medico_?)?ocupacional` `cat(_?(numero|tipo|acidente))?` `comunicacao_?(de_?)?acidente_?(de_?)?trabalho` `acidente_?(de_?)?trabalho` `afastamento_?(medico|inss|doenca|motivo)` `licenca_?(medica|saude|maternidade)` `atestado_?medico` `dias_?(de_?)?atestado` `motivo_?afastamento`

**VALUE FORMAT** BR_SAUDE_OCUPACIONAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 5 known values, ignoring case — apto, inapto, apto com restrições, apto com restricao, inapto temporariamente | 60% | |

A value that matches none counts 30% against the data class.

### HOW IT IS MASKED

A fitness result is replaced with another from the list — fit, unfit —; a flag stays a flag; the ICD of the leave falls under the diagnosis data class.

`BR_SAUDE_OCUPACIONAL` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SAUDE_OCUPACIONAL_LISTA`.

> `BR_FLAG` **REGEX DECOMPOSE**
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` **SECURE LOOKUP**
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_CM_ALFANUM` **CHARACTER MAPPING**
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `BR_SAUDE_OCUPACIONAL_LISTA` **SECURE LOOKUP**
>
> Replaces the value with one of the 4 values in `br-aptidao.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Inapto temporariamente` | → | **`Inapto`** |

## 11.10 Sexual and reproductive health

BR_SAUDE_REPRODUTIVA

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

Pregnancy, gestational age, last menstrual period, prenatal care, childbirth, abortion, contraception, fertility, sexually transmitted infections.

### HOW IT IS FOUND

COLUMN NAME — BR_SAUDE_REPRODUTIVA — Path

90% `gestante` `gravidez` `gestacao` `idade_?gestacional` `semanas_?(de_?)?gestacao` `dum` `data_?(da_?)?ultima_?menstruacao` `pre_?natal` `aborto` `abortamento` `parto(_?tipo)?` `tipo_?(de_?)?parto` `puerperio` `metodo_?contraceptivo` `contracepcao` `anticoncepcao` `planejamento_?familiar` `fertilidade` `infertilidade` `fertilizacao` `fiv` `ist` `dst` `infeccoes?_?sexualmente_?transmissive(l|is)` `pregnan(t|cy)` `contracepti(on|ve)`

### HOW IT IS MASKED

Flags keep their shape; any other value becomes `Não informado`.

`BR_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `BR_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 12 semanas de gestação | → | **Não informado** |
| S | → | **S** |

## 11.11 Mental health

BR_SAUDE_MENTAL

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

Mental disorders, psychiatric or psychological care, CAPS, suicide attempts, substance dependence.

### HOW IT IS FOUND

> COLUMN NAME BR_SAUDE_MENTAL — Path
>
> 90% `saude_?mental` `transtorno[a-z_]*` `psiquiatri[a-z_]*` `psicolog[a-z_]*` `depressao` `ansiedade` `suicid[a-z_]*` `autolesao` `automutilacao` `dependencia_?(quimica|alcool|drogas)` `uso_?(de_?)?(alcool|drogas|substancias)` `alcoolismo` `tabagismo` `caps(_?(ad|i))?` `internacao_?psiquiatrica` `mental_?health`

### HOW IT IS MASKED

Flags keep their shape; any other value becomes `Não informado`.

`BR_CATEGORIA_SUPRIMIDA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING
>
> Maps each character to another of its group (7 groups), keeping length and format.
>
> `BR_SUPRIMIR` SECURE LOOKUP
>
> Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Transtorno bipolar` | → | **`Não informado`** |

PART 12

# Financial and credit data

The LGPD does not call them sensitive, but a person's financial life is protected by bank secrecy (Lei Complementar 105/2001), credit databases have a law of their own (Lei 12.414/2011) and credit protection is a legal basis of its own (art. 7, X). And several of these data reveal sensitive data: the type of an INSS benefit tells whether a person is incapacitated, has a disability or had a child, and a Bolsa Família benefit or a Cadastro Único bracket tells that they are poor. Categories are replaced within their own notation; amounts are mapped into a plausible range unrelated to the real value.

## 12.1 Credit and negative records

BR_FIN_CREDITO

COLUMN NAME | REGEX DECOMPOSE

WHAT IT IS

Credit score (0 to 1,000), negative record, restriction, protest, days overdue and contract status.

HOW IT IS FOUND

COLUMN NAME BR_FIN_CREDITO — Path

90% `score(_?(credito|serasa|spc|boa_?vista|quod|interno))?` `pontuacao_?(de_?)?credito` `rating_?(de_?)?credito` `risco_?(de_?)?credito` `nota_?(de_?)?credito` `cadastro_?positivo` `negativad[oa]` `negativacao` `restricao(_?(cpf|credito|spc|serasa))?` `protesto` `inadimplen(te|cia)` `dias_?(de_?)?atraso` `atraso_?dias` `situacao_?(do_?)?(credito|contrato|pagamento)` `classificacao_?risco` `credit_?score` `credit_?rating`

HOW IT IS MASKED

A score gets other digits of the same length; a flag stays a flag; a status is replaced with another from the list.

`BR_CREDITO` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SITUACAO_CREDITO` .

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt` , chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt` , chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt` , chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt` , chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt` , chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt` , chosen by a keyed hash.

`BR_SITUACAO_CREDITO` SECURE LOOKUP

Replaces the value with one of the 10 values in `br-situacoes-credito.txt` , chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Negativado no Serasa | → | **Atraso de 90 dias** |
| 745 | → | **471** |

## 12.2 Debt

BR_FIN_DIVIDA

COLUMN NAME | DATA TYPE | NUMERIC MAPPING

WHAT IT IS

Outstanding balance, overdue amount, instalments and credit limits.

HOW IT IS FOUND

COLUMN NAME BR_FIN_DIVIDA – Path

85% `saldo_?devedor` `valor_?(da_?)?(divida|debito|parcela|financiamento|emprestimo|em_?atraso|atrasado)` `vl_?(divida|debito|parcela|saldo_?devedor|atraso)` `divida(_?total)?` `debito_?total` `limite_?(de_?)?credito` `limite_?cartao` `prestacao` `parcela(_?mensal)?` `loan_?(amount|balance)` `outstanding_?balance` `debt`

DATA TYPE BR_FIN_DIVIDA – Type

Accepts number; any other type rules the data class out.

HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 8500 | → | **5134** |

## 12.3 Income and salary

BR_FIN_RENDA

COLUMN NAME | DATA TYPE | NUMERIC MAPPING

WHAT IT IS

Salary, pay, earnings and monthly or household income.

HOW IT IS FOUND

COLUMN NAME BR_FIN_RENDA – Path

85% `salario(_?(base|bruto|liquido|mensal|contratual))?` `vl_?(salario|renda|remuneracao|provento)` `renda(_?(mensal|familiar|bruta|liquida|per_?capita|comprovada|presumida))?` `remuneracao` `proventos` `vencimentos?(_?basico)?` `faturamento_?(mensal|anual)_?(pf|pessoa)` `rendimento(_?(mensal|anual|tributavel))?` `ganho_?mensal` `salary` `income` `wages?`

**DATA TYPE** BR_FIN_RENDA – Type

Accepts number; any other type rules the data class out.

#### HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` **NUMERIC MAPPING**

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 3200 | → | **6046** |

---

## 12.4 Assets

BR_FIN_PATRIMONIO

**COLUMN NAME** **DATA TYPE** **NUMERIC MAPPING**

#### WHAT IT IS

Value of real estate and goods, account balances, savings, investments and FGTS.

#### HOW IT IS FOUND

**COLUMN NAME** BR_FIN_PATRIMONIO – Path

**85%**

`patrimonio(_?(liquido|total))?` `valor_?(do_?)?`

`(imovel|patrimonio|bens|veiculo|venal|investimento|aplicacao)` `vl_?`

`(imovel|patrimonio|bens|venal|investimento|aplicacao)` `saldo_?`

`(conta|poupanca|investimento|aplicacao|fgts)` `bens_?(e_?direitos|declarados)` `declaracao_?`

`(de_?)?bens` `saldo_?fgts` `net_?worth` `account_?balance`

**DATA TYPE** BR_FIN_PATRIMONIO – Type

Accepts number; any other type rules the data class out.

#### HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` **NUMERIC MAPPING**

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

#### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 185000 | → | **477515** |

---

## 12.5 Benefit and pension amount

BR_FIN_VALOR_BENEFICIO

COLUMN NAME | DATA TYPE | NUMERIC MAPPING

### WHAT IT IS

Amount of a retirement pension, survivor's pension, allowance, BPC and child support.

### HOW IT IS FOUND

> **COLUMN NAME** BR_FIN_VALOR_BENEFICIO – Path
>
> **90%** `valor_?(do_?)?(beneficio|aposentadoria|pensao|auxilio|bolsa|bpc)` `vl_?(beneficio|aposentadoria|pensao|auxilio|bpc)` `renda_?mensal_?inicial` `rmi` `pensao_?alimenticia` `aposentadoria_?valor` `pension_?amount`

> **DATA TYPE** BR_FIN_VALOR_BENEFICIO – Type
>
> Accepts number; any other type rules the data class out.

### HOW IT IS MASKED

Masked by CM Numeric, which comes with Delphix: each digit becomes another, and the amount keeps its number of digits, its sign and its decimal places — the same order of magnitude, with no link to the real value.

`dlpx-core:CM Numeric` NUMERIC MAPPING

Comes with Delphix and is used as it ships. Maps each digit to another, keeping the number of digits, the sign, the decimal places and the zeros on the left — an amount keeps its order of magnitude.

### EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| 1800 | → | **7193** |

---

## 12.6 INSS benefit type

BR_FIN_BENEFICIO_INSS

COLUMN NAME | VALUE FORMAT | REGEX DECOMPOSE

### WHAT IT IS

The type of social security or welfare benefit, in words or as a code: 31 and 91 (temporary incapacity), 32 and 92 (permanent incapacity), 80 (maternity pay), 87 and 88 (BPC for people with disabilities and for the elderly), 41 and 42 (retirement), 21 (survivor's pension). Several types reveal an illness, a disability or a pregnancy.

### HOW IT IS FOUND

> **COLUMN NAME** BR_FIN_BENEFICIO_INSS – Path
>
> **90%** `especie_?(do_?)?beneficio` `(cd|co|tp|ds)_?especie(_?beneficio)?` `tipo_?(de_?)?beneficio(_?(inss|previdenciario))?` `beneficio_?(inss|previdenciario)(_?(tipo|especie))?` `natureza_?beneficio` `b(31|91|32|92|87|88|41|42)`

VALUE FORMAT BR_FIN_BENEFICIO_INSS — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 27 known values, ignoring case — Aposentadoria por idade, Aposentadoria por tempo de contribuição, Aposentadoria especial, Aposentadoria por incapacidade permanente, Auxílio por incapacidade temporária, Auxílio-acidente, … | 80% | |
| One of 14 known values, ignoring case — 21, 31, 32, 36, 41, 42, … | 50% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A code is replaced with another benefit-type code; a text with another type from the list.

`BR_BENEFICIO_INSS` REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_BENEFICIO_INSS_LISTA`.

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_BENEFICIO_INSS_CODIGO` SECURE LOOKUP
>
> Replaces the value with one of the 14 values in `br-beneficios-inss-codigos.txt`, chosen by a keyed hash.
>
> `BR_BENEFICIO_INSS_LISTA` SECURE LOOKUP
>
> Replaces the value with one of the 11 values in `br-beneficios-inss.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Auxílio-reclusão | → | **BPC ao idoso** |
| 31 | → | **88** |

## 12.7 Benefit number

BR_FIN_NUMERO_BENEFICIO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The INSS benefit number (NB), `123.456.789-0`.

HOW IT IS FOUND

COLUMN NAME BR_FIN_NUMERO_BENEFICIO — Path

85% `n(r|u|um|o|ro|umero)_?(do_?)?beneficio` `nb(_?(inss|beneficio))?` `numero_?beneficio_?inss` `beneficio_?numero` `matricula_?(inss|beneficio)`

VALUE FORMAT BR_FIN_NUMERO_BENEFICIO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{3}\.\d{3}\.\d{3}-\d)` | 50% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Digit by digit, keeping format and separators.

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `123.456.789-0` | → | **`171.367.811-5`** |

## 12.8 Social programmes

BR_FIN_PROGRAMA_SOCIAL

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Taking part in social programmes and in the Cadastro Único — Bolsa Família, Benefício de Prestação Continuada, Auxílio Gás, Pé-de-Meia, Tarifa Social de Energia, Minha Casa, Minha Vida, Garantia-Safra, Seguro-Defeso —, often as an S/N flag.

HOW IT IS FOUND

COLUMN NAME BR_FIN_PROGRAMA_SOCIAL — Path

90% `bolsa_?familia` `beneficiario_?(bolsa|programa|bpc|auxilio)` `programa_?(social|governo|transferencia_?renda)` `auxilio_?(brasil|emergencial|gas)` `cad_?unico` `cadunico` `cadastro_?unico` `inscrito_?cadunico` `pe_?de_?meia` `tarifa_?social` `bpc` `loas` `minha_?casa_?minha_?vida` `mcmv` `seguro_?defeso` `garantia_?safra` `transferencia_?(de_?)?renda` `social_?programme`

VALUE FORMAT BR_FIN_PROGRAMA_SOCIAL — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 20 known values, ignoring case — Bolsa Família, Benefício de Prestação Continuada, Auxílio Gás dos Brasileiros, Pé-de-Meia, Tarifa Social de Energia Elétrica, Minha Casa Minha Vida, … | 80% | |
| One of 7 known values, ignoring case — s, n, sim, não, y, 0, … | 30% | |

A value that matches none counts 40% against the data class.

HOW IT IS MASKED

A flag stays a flag and any other value is replaced with a programme from the list.

`BR_PROGRAMA_SOCIAL` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_PROGRAMA_SOCIAL_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

`BR_FLAG_0_1` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_PROGRAMA_SOCIAL_LISTA` SECURE LOOKUP

Replaces the value with one of the 9 values in `br-programas-sociais.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| Auxílio Brasil | → | **Minha Casa, Minha Vida** |
| S | → | **S** |

## 12.9 Income bracket and social class

BR_FIN_FAIXA_RENDA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The Cadastro Único brackets (extreme poverty, poverty, low income), the Critério Brasil classes (A, B1, B2, C1, C2, DE), income deciles and quintiles.

HOW IT IS FOUND

COLUMN NAME BR_FIN_FAIXA_RENDA — Path

90% `faixa_?(de_?)?renda(_?(familiar|per_?capita|cadunico))?` `faixa_?salarial` `classe_?(social|economica|socioeconomica)` `criterio_?brasil` `nivel_?socioeconomico` `situacao_?(de_?)?(pobreza|vulnerabilidade)` `pobreza` `extrema_?pobreza` `vulnerabilidade_?social` `renda_?per_?capita_?faixa` `quintil` `decil` `income_?(bracket|quintile|decile)`

VALUE FORMAT BR_FIN_FAIXA_RENDA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 15 known values, ignoring case — extrema pobreza, pobreza, baixa renda, acima de meio salário mínimo, classe a, classe b, … | 40% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced within its own notation: a bracket with another bracket, a class with another class, a decile with another decile.

`BR_FAIXA_RENDA` REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_FAIXA_CADUNICO`.

`BR_QUINTIL` SECURE LOOKUP

Replaces the value with one of the 5 values in `br-quintil.txt`, chosen by a keyed hash.

`BR_DECIL_ALTO` SECURE LOOKUP

Replaces the value with one of the 5 values in `br-decil-alto.txt`, chosen by a keyed hash.

`BR_DECIL` SECURE LOOKUP

Replaces the value with one of the 10 values in `br-decil.txt`, chosen by a keyed hash.

`BR_CLASSE_SOCIAL` SECURE LOOKUP

Replaces the value with one of the 6 values in `br-classes-sociais.txt`, chosen by a keyed hash.

`BR_FAIXA_CADUNICO` SECURE LOOKUP

Replaces the value with one of the 4 values in `br-faixas-cadunico.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Extrema pobreza` | → | **`Extrema pobreza`** |
| `Classe C1` | → | **`Classe B2`** |

## 12.10 Housing

BR_FIN_MORADIA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

The tenure of the dwelling — owned outright or with a mortgage, rented, lent, occupation.

HOW IT IS FOUND

COLUMN NAME BR_FIN_MORADIA — Path

90% `condicao_?(de_?)?(ocupacao|moradia)` `situacao_?(do_?)?(domicilio|imovel|moradia)` `tipo_?(de_?)?moradia` `posse_?(do_?)?imovel` `imovel_?(proprio|alugado)` `reside_?em_?imovel` `moradia_?(propria|alugada)` `domicilio_?(proprio|alugado)` `housing_?tenure`

VALUE FORMAT BR_FIN_MORADIA — Values

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| One of 18 known values, ignoring case — Próprio quitado, Próprio em pagamento, Alugado, Cedido, Ocupação, Outra, … | 60% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

Replaced with another tenure from the list.

`BR_MORADIA` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_MORADIA_LISTA`.

`BR_FLAG` REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

`BR_FLAG_SIM_NAO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

`BR_FLAG_TRUE_FALSE` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

`BR_FLAG_YES_NO` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

`BR_FLAG_S_N` SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

`BR_FLAG_Y_N` SECURE LOOKUP

> Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
>
> `BR_FLAG_0_1` **SECURE LOOKUP**
>
> Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

`BR_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`BR_MORADIA_LISTA` **SECURE LOOKUP**

Replaces the value with one of the 6 values in `br-condicoes-moradia.txt`, chosen by a keyed hash.

**EXAMPLES**

| VALUE | | MASKED |
|---|---|---|
| Invasão | → | **Outra** |

PART 13

# Criminal records and court cases

Not on the list of art. 5, II, but among the data that most expose people to discrimination: requiring a job candidate's record without justification gives rise to presumed moral damages (TST, IRR Tema 1), and the LGPD only stays out of the way when processing is exclusively for public security or for the investigation and prosecution of criminal offences (art. 4, III). In a company, this data gets the same care as sensitive data.

## 13.1 Criminal records

BR_PENAL_ANTECEDENTES

COLUMN NAME | REGEX DECOMPOSE

### WHAT IT IS

Criminal record certificates, convictions, indictment, type of crime, prison status, sentence regime, electronic monitoring, socio-educational measures and arrest warrants.

### HOW IT IS FOUND

COLUMN NAME BR_PENAL_ANTECEDENTES — Path

95% `antecedentes?(_?(criminais|criminal|penais))?` `certidao_?(de_?)?antecedentes` `atestado_?(de_?)?antecedentes` `folha_?(de_?)?antecedentes` `ficha_?criminal` `registro_?criminal` `condenac(ao|oes)(_?(criminal|penal))?` `condenado` `crime(_?(tipo|natureza))?` `tipo_?(de_?)?crime` `delito` `infracao_?penal` `artigo_?(penal|cp)` `tipificacao_?penal` `indiciamento` `indiciado` `reincidente` `situacao_?(prisional|carceraria|penal)` `regime_?(prisional|de_?cumprimento)` `preso` `presidiario` `egresso_?(prisional|sistema)` `medida_?socioeducativa` `tornozeleira` `monitoramento_?eletronico` `mandado_?(de_?)?prisao` `criminal_?record`

55% `situacao_?processual` `resultado_?(do_?)?processo` `sentenca`

### HOW IT IS MASKED

Replaced with a generic status from the list — nothing on record, convicted, acquitted —, whatever the outcome; a flag stays a flag.

`BR_ANTECEDENTES` REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ANTECEDENTES_LISTA`.

> `BR_FLAG` REGEX DECOMPOSE
>
> Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.
>
> > `BR_FLAG_SIM_NAO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_TRUE_FALSE` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_YES_NO` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_S_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_Y_N` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.
> >
> > `BR_FLAG_0_1` SECURE LOOKUP
> >
> > Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.
>
> `BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

`BR_ANTECEDENTES_LISTA` SECURE LOOKUP

Replaces the value with one of the 8 values in `br-antecedentes.txt`, chosen by a keyed hash.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `Condenado por furto em 2019` | → | **`Não informado`** |

## 13.2 Case number

BR_PENAL_PROCESSO

COLUMN NAME · VALUE FORMAT · REGEX DECOMPOSE

WHAT IT IS

The unique case number of the Judiciary (Resolução CNJ 65/2008), `0001234-56.2023.8.26.0100` : sequence, check digits, year, branch of the judiciary, court and origin. Cases can be looked up on court portals and lead to the parties. Also police reports, inquiries and summary reports.

HOW IT IS FOUND

COLUMN NAME BR_PENAL_PROCESSO — Path

| | |
|---|---|
| 90% | `n(r\|u\|um\|o\|ro\|umero)_?(do_?)?processo(_?(judicial\|criminal\|penal\|cnj\|trabalhista))?` `processo_?(judicial\|criminal\|penal\|cnj\|numero\|trabalhista)` `numeracao_?unica` `(nr\|nu\|num)_?cnj` `n(r\|u\|um\|o\|ro\|umero)_?(bo\|boletim\|boletim_?ocorrencia\|inquerito\|ip\|termo_?circunstanciado\|tco\|execucao_?penal\|mandado)` `boletim_?(de_?)?ocorrencia` `inquerito(_?policial)?(_?numero)?` `autos_?(numero\|n)` |
| 50% | `processo` `protocolo_?judicial` |

VALUE FORMAT BR_PENAL_PROCESSO — Regex

| PATTERN | STRENGTH | NOTES |
|---|---|---|
| `0*(?:\d{7}-\d{2}\.\d{4}\.\d\.\d{2}\.\d{4})` | 90% | |

A value that matches none counts 30% against the data class.

HOW IT IS MASKED

The sequence and the check digits are mapped digit by digit; year, branch, court and origin stay. The check digits (modulus 97) are not recomputed. Other numbers are masked character by character.

`BR_PROCESSO` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM` .

`BR_CM_DIGITOS` CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

`BR_CM_ALFANUM` CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

| VALUE | | MASKED |
|---|---|---|
| `0001234-56.2023.8.26.0100` | → | **`0009707-72.2023.8.26.0100`** |

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# Check digits computed by the policy

Brazilian systems validate their identifiers. An invalid CPF in a test database breaks the registration, the integration with the Receita Federal or the issuing of an invoice. So where the check-digit rule is public, the policy recomputes it.

- **CPF.** Two Check Digit algorithms, one inside the other. The outer one receives the eleven digits and hands the first ten to the inner one, which hands the first nine to a Character Mapping and recomputes the first digit with weights 10 to 2; the outer one recomputes the second with weights 11 to 2. Check Digit writes a remainder of 0 or 1 as 0 — exactly the Receita's rule. Separators stay out of the computation, and a short number is padded with zeros on the left.
- **CNPJ, alphanumeric included.** The same design with weights 5 to 2 and 9 to 2, and with the character handling *ASCII value minus 48* — the rule of Instrução Normativa RFB 2.229/2024, in which `A` is worth 17. Only the root is mapped.
- **PIS and RENAVAM.** One Check Digit with weights 3 2 9 8 7 6 5 4 3 2.
- **Voter card.** Check Digit cannot follow the voter card's rule: the first digit is the remainder itself, it comes after the two state digits, and in São Paulo and Minas Gerais a remainder of 0 gives 1. The policy computes it with tables: the sequence is mapped and the old digits become `00`; a running sum travels through the value two digits per step, each step a Data Cleansing that turns *sum + two digits* into *two digits + new sum*; at the end, a table turns *sum + state + 00* into the state and the two right digits.
- **National Health Card.** The PIS inside a definitive CNS is masked by the PIS algorithm, and the same step tables carry the weighted sum to the end, where `0000` becomes `000` or `001` and the digit that makes the sum a multiple of 11.

The tables are Data Cleansing files generated with the policy — 1,100 lines per step —, and the steps are Regex Decompose algorithms joined in String Algorithm Chains. Every identifier stays one-to-one: two different CPFs never become the same CPF.

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# Where people live: municipalities and CEP

With the resident population of the 2022 Demographic Census (IBGE) and the location of each seat, the policy knows the size, state and position of the 5,570 municipalities.

- A **municipality** with fewer than 20,000 inhabitants becomes the nearest municipality of the same state with at least 20,000. There are 3,860 of them, with 32 million inhabitants — 15.8% of the population. Serra da Saudade (MG), the smallest, becomes São Gotardo; Borá (SP) becomes Paraguaçu Paulista; Uiramutã (RR) becomes Boa Vista.
- The **state** stays: the smallest, Roraima, had 636,707 inhabitants.
- Distance is measured between the coordinates of the seats, in a straight line. Fernando de Noronha, more than 350 km off the coast, becomes the nearest municipality of Pernambuco with at least 20,000.
- Two hundred and thirty-two names repeat — there are five Bom Jesus and five São Domingos —, and others differ only in an accent ( `Goianá` and `Goiana` ). A name on its own is generalized only when every municipality bearing it is small, and then it follows the largest of them. With the state ( `Bom Jesus – RS` , `Bom Jesus/RS` ) there is no ambiguity, and the municipality is always generalized.
- The **IBGE code**, of seven or six digits, follows the same table, and a person's name and code still agree.

- The **CEP** loses its last five digits: the postal sector stays, which gathers many streets and, outside the capitals, several municipalities.

Names are recognized as systems write them: with accents, without accents and in capitals. Municipality names that are also given names or surnames, and state names ( `Paraná` , `Goiás` ), are left out of the detection list, so that a column of surnames is not taken for municipalities.

---

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# Limits

- **Binary columns** — images of RG and CNH, proof-of-life selfies, fingerprints, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **Check digits not recomputed.** CNH (the second digit depends on whether the first went past 9), RG (one rule per state), certificates, bank accounts (one rule per bank) and CNJ case numbers (modulus 97) are masked digit by digit, and applications that validate those digits may reject the values.
- **Provisional CNS and numbers with spaces.** A CNS starting with 7, 8 or 9, and a CNS or voter card written with spaces ( `898 0012 3456 7890` ), are masked digit by digit, without a valid digit.
- **State, DDD and bank code stay.** In a column where only state and DDD remain, they identify no one; combined with a birth date and a small municipality, those have already been generalized.
- **Data Cleansing lets through what it does not know**: a municipality written in a way the table does not have — abbreviated, misspelled — passes unchanged.
- **Free text** is redacted word by word: a phone written with spaces is not found, and a name that is also a common word ( `Rosa` , `Luz` , `Paz` ) is redacted wherever it appears.
- **Children and adolescents** — a birth date moves at most 60 days, so few people cross their 12th or 18th birthday, but some do. Where that matters, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, neighbourhood, municipality and CEP in separate columns are masked independently and no longer form a coherent address — a masked CEP does not match the masked street.
- **The engine key** links masked values back to production for whoever holds both: in that case the result is pseudonymized data (art. 13, § 4), still personal data. Keep the key away from the users of the copy.

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# Verification

Before publication, the policy was checked outside Delphix:

- The check-digit rules were checked against published examples: the CPF `529.982.247-25` , the first alphanumeric CNPJ issued by the Receita Federal, `00.000.000/E08G-12` (July 2026), and the voter card `1023 8501 0671` of OBMEP.
- Every classifier configuration was reviewed against what Delphix accepts.
- Discovery was simulated on 164 columns with the whole policy at its assignment threshold — well named, badly named and 33 columns that are not personal data — and again with only the classifiers of the essential pack.
- The table that generalizes municipalities was audited for all 5,570 municipalities: by name, by name with the state and by IBGE code.
- The algorithms, run with the Delphix algorithm plugin, masked each data class's sample values and placeholders such as `N/I` , checking every output against its vocabulary and every CPF, CNPJ, PIS, voter

card, CNS and RENAVAM against its check-digit rule; 150 of each for collisions and validity; and 20 people whose PIS and CNS must still agree. Every algorithm also masked its own sample value.

Every column was assigned as expected and every value masked as expected. Before using the policy on real data, run a discovery job and a masking job on a representative copy.

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# Sources


- Lei 13.709 of 14 August 2018 — General Personal Data Protection Law, compiled text (Presidency of the Republic), with the amendments of Laws 13.853/2019, 14.010/2020, 14.460/2022 and 15.352/2026. Federal Constitution, art. 5, LXXIX (Constitutional Amendment 115/2022). Lei 15.211/2025 (Digital Statute of Children and Adolescents).
- Agência Nacional de Proteção de Dados: preliminary study *Anonimização e Pseudonimização para a Proteção de Dados Pessoais* (public consultation, January 2024); Resolução CD/ANPD 15/2024 (Security Incident Reporting Regulation); Resolução CD/ANPD 4/2023 (Sanction Calculation Regulation); Regulatory Agenda 2025–2026 progress report (second half of 2025).
- Lei Complementar 105/2001 (bank secrecy). Lei 12.414/2011 (positive credit register). Lei 14.534/2023 (CPF as the single number). Lei 13.146/2015 (Brazilian Inclusion Law). Superior Labour Court, IRR Tema 1 (requiring criminal record certificates).
- Resident population by municipality, 2022 Demographic Census (IBGE, SIDRA, table 4709). Most frequent given names and surnames — *Nomes no Brasil*, 2022 Census (IBGE). Colour or race, religion and indigenous peoples — 2022 Census (IBGE). Coordinates of municipal seats — `kelvins/municipios-brasileiros` (MIT licence).
- Registered political parties — Superior Electoral Court, September 2026. Voter card check digits — OBMEP Math Clubs. Alphanumeric CNPJ — Receita Federal, Instrução Normativa RFB 2.229/2024. CEP structure — Correios. National Health Card — Ministry of Health. Unique case numbering — Resolução CNJ 65/2008. Brazilian Classification of Occupations — Ministry of Labour and Employment.

APPENDIX A

# Algorithms

Every algorithm of the policy in alphabetical order, with its configuration. The one a data class names is the top of its tree; the others are building blocks.

## BR_ANO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_ANO_NASCIMENTO`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{3})(\d))` | keep · keep · `BR_CM_DIGITOS` |
| otherwise | keep |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## BR_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ANTECEDENTES_LISTA`.

Used by: `BR_PENAL_ANTECEDENTES`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_ANTECEDENTES_LISTA` |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## BR_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `br-antecedentes.txt`, chosen by a keyed hash.

Used by: `BR_ANTECEDENTES`

### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-antecedentes.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

## BR_BAIRRO SECURE LOOKUP

Replaces the value with one of the 40 values in `br-bairros.txt`, chosen by a keyed hash.

Used by: `BR_L2_BAIRRO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-bairros.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_BENEFICIO_INSS REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_BENEFICIO_INSS_LISTA`.

Used by: `BR_FIN_BENEFICIO_INSS`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*)(21\|31\|32\|36\|41\|42\|46\|80\|87\|88\|91\|92\|93\|94)` | keep · `BR_BENEFICIO_INSS_CODIGO` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_BENEFICIO_INSS_CODIGO` |
| otherwise | `BR_BENEFICIO_INSS_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_BENEFICIO_INSS_CODIGO SECURE LOOKUP

Replaces the value with one of the 14 values in `br-beneficios-inss-codigos.txt`, chosen by a keyed hash.

Used by: `BR_BENEFICIO_INSS`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-beneficios-inss-codigos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_BENEFICIO_INSS_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in `br-beneficios-inss.txt`, chosen by a keyed hash.

Used by: `BR_BENEFICIO_INSS`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-beneficios-inss.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_CARTAO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L1_CARTAO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d[\d\s\-]{11,22}\d))` | keep · `dlpx-core:Credit Card` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SUPRIMIR`.

Used by: `BR_L3_CONVICCAO_ORGANIZACAO`, `BR_L3_FILIACAO_SINDICAL`, `BR_L3_VIDA_SEXUAL`, `BR_SAUDE_REPRODUTIVA`, `BR_SAUDE_MENTAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(0*?)(?:(\d{1,6})))))))` | keep · keep · keep · keep · keep · `BR_CM_ALFANUM` |
| otherwise | `BR_SUPRIMIR` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CBO SECURE LOOKUP

Replaces the value with one of the 34 values in `br-cbo.txt`, chosen by a keyed hash.

Used by: `BR_OCUPACAO_OU_CBO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cbo.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_CEP REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_CEP`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{2})(\.?)(\d)(\d{2})(-?)(\d{3}))` | keep · keep · keep · keep · replace with `00` · keep · replace with `000` |
| `(0*?)(?:(\d{2})(\d{2})(\d{3}))` | keep · keep · replace with `00` · replace with `000` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |

| | |
|---|---|
| requireMask | no |

### BR_CHAVE_PIX REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_CHAVE_PIX`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([^@\s]+@[^@\s]+)` | `BR_EMAIL` |
| `(0*?)(?:(\+55\d{10,11}))` | keep · `BR_TELEFONE` |
| `(0*?)(?:(\d{3}\.\d{3}\.\d{3}-\d{2}\|\d{11}))` | keep · `BR_CPF` |
| `(0*?)(?:([0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2}\|[0-9A-Z]{12}\d{2}))` | keep · `BR_CNPJ` |
| `(0*?)(?:([0-9a-fA-F]{8}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{4}-[0-9a-fA-F]{12}))` | keep · `BR_CM_HEX` |
| otherwise | `BR_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CID10 SECURE LOOKUP

Replaces the value with one of the 37 values in `br-cid10.txt`, chosen by a keyed hash.

Used by: `BR_DIAGNOSTICO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cid10.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_CID10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in `br-cid10-decimal.txt`, chosen by a keyed hash.

Used by: `BR_DIAGNOSTICO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cid10-decimal.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_CLASSE_SOCIAL SECURE LOOKUP

Replaces the value with one of the 6 values in `br-classes-sociais.txt`, chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-classes-sociais.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: `BR_L1_CNH`, `BR_L1_CTPS`, `BR_L1_PASSAPORTE`, `BR_L1_DOCUMENTO_ESTRANGEIRO`, `BR_L1_CERTIDAO`, `BR_L1_ENDERECO_COMPLEMENTO`, `BR_L1_CHASSI`, `BR_L1_IMOVEL`, `BR_L1_CONTRATO`, `BR_L1_USUARIO`, `BR_SAUDE_IDENTIFICADOR`, `BR_SAUDE_RESULTADO_EXAME`, `BR_CATEGORIA_SUPRIMIDA`, `BR_NOMES`, `BR_SOBRENOMES`, `BR_NOME_COMPLETO`, `BR_EMAIL`, `BR_CHAVE_PIX`, `BR_PLACA`, `BR_ETNIA`, `BR_RELIGIAO`, `BR_OPINIAO_POLITICA`, `BR_ORIENTACAO_SEXUAL`, `BR_IDENTIDADE_GENERO`, `BR_MEDICAMENTO`, `BR_DEFICIENCIA`, `BR_SAUDE_OCUPACIONAL`, `BR_PROGRAMA_SOCIAL`, `BR_MORADIA`, `BR_ANTECEDENTES`, `BR_PROCESSO`

CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `AEIOU` `BCDFGHJKLMNPQRSTVWXYZ` `aeiou` `bcdfghjklmnpqrstvwxyz` `ÁÉÍÓÚÂÊÔÃÕÀ` `áéíóúâêôãõà` |
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### BR_CM_DIGITOS CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `BR_L1_RG`, `BR_L1_CONTA_BANCARIA`, `BR_L2_ZONA_SECAO_ELEITORAL`, `BR_FIN_NUMERO_BENEFICIO`, `BR_CPF`, `BR_CNPJ`, `BR_TITULO_CORPO`, `BR_TITULO`, `BR_PIS_CORPO`, `BR_PIS`, `BR_REGISTRO_PROFISSIONAL`, `BR_TELEFONE`, `BR_PLACA`, `BR_RENAVAM`, `BR_COORDENADA`, `BR_ANO`, `BR_IDADE`, `BR_OCUPACAO_OU_CBO`, `BR_CNS`, `BR_CREDITO`, `BR_PROCESSO`

CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` |
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### BR_CM_DIGITOS_LARGURA_FIXA CHARACTER MAPPING

Maps each character to another of its group (1 groups), keeping length and format.

Used by: `BR_RENAVAM_VALIDO`

CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` |
| caseSensitive | yes |
| minMaskedPositions | `0` |

### BR_CM_DIGITOS_LETRAS CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: `BR_PLACA`

CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `ABCDEFGHIJKLMNOPQRSTUVWXYZ` |

| | |
|---|---|
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### BR_CM_DIGITOS_LETRAS_LARGURA_FIXA CHARACTER MAPPING

Maps each character to another of its group (2 groups), keeping length and format.

Used by: `BR_CNPJ_RAIZ`

#### CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `ABCDEFGHIJKLMNOPQRSTUVWXYZ` |
| caseSensitive | yes |
| minMaskedPositions | `0` |

### BR_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: `BR_L1_DISPOSITIVO`, `BR_CHAVE_PIX`, `BR_IP`

#### CONFIGURATION

| | |
|---|---|
| characterGroups | `0123456789` `abcdef` `ABCDEF` `ghijklmnopqrstuvwxyz` `GHIJKLMNOPQRSTUVWXYZ` |
| caseSensitive | yes |
| minMaskedPositions | `0` |
| preserveLeadingZeros | yes |

### BR_CNPJ REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

Used by: `BR_L1_CNPJ`, `BR_CHAVE_PIX`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2}))` | keep · `BR_CNPJ_VALIDO` |
| `(0*?)(?:([0-9A-Z]{12}\d{2}))` | keep · `BR_CNPJ_VALIDO` |
| `(0*?)(?:(\d{3}\.\d{3}\.\d{3}-\d{2}\|\d{9}-\d{2}))` | keep · `dlpx-core:Brazil CPF` |
| `(0*?)(?:(\d{3,13}))` | keep · `BR_CNPJ_VALIDO_ZEROS` |
| otherwise | `BR_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CNPJ_DV1 CHECK DIGIT

Masks the digits with `BR_CNPJ_RAIZ` and recomputes a modulus 11 check digit with weights `5 4 3 2 9 8 7 6 5 4 3 2`.

Used by: `BR_CNPJ_VALIDO`, `BR_CNPJ_VALIDO_ZEROS`

#### CONFIGURATION

| | |
|---|---|
| weightList | `5` `4` `3` `2` `9` `8` `7` `6` `5` `4` `3` `2` |
| modulusNumber | `11` |
| checkDigitIndex | `12` |

| | |
|---|---|
| numDigitsForCheckdigitCalculation | `12` |
| calculateChecksumRightToLeft | no |
| numericAlgorithm | `BR_CNPJ_RAIZ` |
| alphaNumericAlgorithm | `BR_CNPJ_RAIZ` |
| preserveRegex | `[.\-/]` |
| inputHandlingConfig.characterHandling | `ASCII_VALUE_MINUS_48` |
| inputHandlingConfig.invalidInputHandling | `ERROR` |
| inputHandlingConfig.shortInputHandling | `FALLBACK` |
| inputHandlingConfig.padCharacter | `0` |
| inputHandlingConfig.trimWhitespace | yes |

### `BR_CNPJ_RAIZ` REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_CNPJ_DV1`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([0-9A-Z]{2}\.?[0-9A-Z]{3}\.?[0-9A-Z]{3})(/?[0-9A-Z]{4})` | `BR_CM_DIGITOS_LETRAS_LARGURA_FIXA` · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### `BR_CNPJ_VALIDO` CHECK DIGIT

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

Used by: `BR_CPF`, `BR_CNPJ`

#### CONFIGURATION

| | |
|---|---|
| weightList | `6 5 4 3 2 9 8 7 6 5 4 3 2` |
| modulusNumber | `11` |
| checkDigitIndex | `13` |
| numDigitsForCheckdigitCalculation | `13` |
| calculateChecksumRightToLeft | no |
| numericAlgorithm | `BR_CNPJ_DV1` |
| alphaNumericAlgorithm | `BR_CNPJ_DV1` |
| preserveRegex | `[.\-/]` |
| inputHandlingConfig.characterHandling | `ASCII_VALUE_MINUS_48` |
| inputHandlingConfig.invalidInputHandling | `ERROR` |
| inputHandlingConfig.shortInputHandling | `FALLBACK` |
| inputHandlingConfig.padCharacter | `0` |
| inputHandlingConfig.trimWhitespace | yes |

### `BR_CNPJ_VALIDO_ZEROS` CHECK DIGIT

Masks the digits with `BR_CNPJ_DV1` and recomputes a modulus 11 check digit with weights `6 5 4 3 2 9 8 7 6 5 4 3 2`.

Used by: `BR_CPF`, `BR_CNPJ`

#### CONFIGURATION

| | |
|---|---|
| weightList | `6` `5` `4` `3` `2` `9` `8` `7` `6` `5` `4` `3` `2` |
| modulusNumber | `11` |
| checkDigitIndex | `13` |
| numDigitsForCheckdigitCalculation | `13` |
| calculateChecksumRightToLeft | no |
| numericAlgorithm | `BR_CNPJ_DV1` |
| alphaNumericAlgorithm | `BR_CNPJ_DV1` |
| preserveRegex | `[.\-/]` |
| inputHandlingConfig.characterHandling | `ASCII_VALUE_MINUS_48` |
| inputHandlingConfig.invalidInputHandling | `ERROR` |
| inputHandlingConfig.shortInputHandling | `PAD_LEFT` |
| inputHandlingConfig.padCharacter | `0` |
| inputHandlingConfig.trimWhitespace | yes |

---

### BR_CNS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS` .

Used by: `BR_SAUDE_CNS`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([12]\d{14}))` | keep · `BR_CNS_VALIDO` |
| `(0*?)(?:([789])(\d{14}))` | keep · keep · `BR_CM_DIGITOS` |
| otherwise | `BR_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CNS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_CNS_VALIDO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(\d{11})(\d{4})` | `BR_PIS_VALIDO` · replace with `0000` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CNS_DV STRING ALGORITHM CHAIN

Applies `BR_CNS_DV_P1` → `BR_CNS_DV_P2` → `BR_CNS_DV_P3` → `BR_CNS_DV_P4` → `BR_CNS_DV_P5` → `BR_CNS_DV_P6` , in this order.

Used by: `BR_CNS_VALIDO`

#### CONFIGURATION

| | |
|---|---|
| algorithmReferences[1] | `BR_CNS_DV_P1` |
| algorithmReferences[2] | `BR_CNS_DV_P2` |
| algorithmReferences[3] | `BR_CNS_DV_P3` |

| | |
|---|---|
| algorithmReferences[4] | BR_CNS_DV_P4 |
| algorithmReferences[5] | BR_CNS_DV_P5 |
| algorithmReferences[6] | BR_CNS_DV_P6 |

### BR_CNS_DV_P1 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{0})(.{2})(.*) | keep · BR_CNS_DV_T1 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CNS_DV_P2 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{2})(.{3})(.*) | keep · BR_CNS_DV_T2 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CNS_DV_P3 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{4})(.{3})(.*) | keep · BR_CNS_DV_T3 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CNS_DV_P4 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{6})(.{3})(.*) | keep · BR_CNS_DV_T4 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |

| | |
|---|---|
| requireMask | no |

### BR_CNS_DV_P5 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{8})(.{3})(.*) | keep · BR_CNS_DV_T5 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CNS_DV_P6 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_CNS_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{10})(.{6})(.*) | keep · BR_CNS_DV_T6 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_CNS_DV_T1 DATA CLEANSING

Replaces each value found in br-cns-dv-1.txt (100 pairs) by its counterpart; values not in the file stay as they are.

Used by: BR_CNS_DV_P1

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-cns-dv-1.txt |
| delimiter | , |
| caseSensitive | yes |
| trimWhitespace | yes |

### BR_CNS_DV_T2 DATA CLEANSING

Replaces each value found in br-cns-dv-2.txt (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: BR_CNS_DV_P2

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-cns-dv-2.txt |
| delimiter | , |
| caseSensitive | yes |
| trimWhitespace | yes |

### BR_CNS_DV_T3 DATA CLEANSING

Replaces each value found in br-cns-dv-3.txt (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P3`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cns-dv-3.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_CNS_DV_T4 DATA CLEANSING

Replaces each value found in `br-cns-dv-4.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P4`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cns-dv-4.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_CNS_DV_T5 DATA CLEANSING

Replaces each value found in `br-cns-dv-5.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P5`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cns-dv-5.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_CNS_DV_T6 DATA CLEANSING

Replaces each value found in `br-cns-dv-6.txt` (110 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CNS_DV_P6`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-cns-dv-6.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_CNS_VALIDO STRING ALGORITHM CHAIN

Applies `BR_CNS_CORPO` → `BR_CNS_DV`, in this order.

Used by: `BR_CNS`

CONFIGURATION

| | |
|---|---|
| algorithmReferences[1] | `BR_CNS_CORPO` |
| algorithmReferences[2] | `BR_CNS_DV` |

---

### BR_CODIGO_MUNICIPIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_CODIGO_MUNICIPIO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*)([1-9]\d*)` | keep · `BR_CODIGO_MUNICIPIO_TABELA` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CODIGO_MUNICIPIO_TABELA DATA CLEANSING

Replaces each value found in `br-codigos-municipio-generalizados.txt` (7,720 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_CODIGO_MUNICIPIO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-codigos-municipio-generalizados.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_CONTAGEM_LIMITADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_DEPENDENTES`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([0-5]))` | keep · keep |
| `(0*?)(?:(\d+))` | keep · replace with `5` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L1_GEOLOCALIZACAO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(-?\d{1,3}[.,]\d)(\d+)\s*([,;])\s*(-?\d{1,3}[.,]\d)(\d+)` | keep · `BR_CM_DIGITOS` · keep · keep · `BR_CM_DIGITOS` |
| `(-?\d{1,3}[.,]\d)(\d+)` | keep · `BR_CM_DIGITOS` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CPF REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

Used by: `BR_L1_CPF`, `BR_CHAVE_PIX`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{3}\.?\d{3}\.?\d{3}-?\d{2}))` | keep · `dlpx-core:Brazil CPF` |
| `(0*?)(?:(\d{3,10}))` | keep · `dlpx-core:Brazil CPF` |
| `(0*?)(?:([0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2}))` | keep · `BR_CNPJ_VALIDO` |
| `(0*?)(?:([0-9A-Z]{12}\d{2}))` | keep · `BR_CNPJ_VALIDO` |
| `(0*?)(?:(\d{12,13}))` | keep · `BR_CNPJ_VALIDO_ZEROS` |
| otherwise | `BR_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_CREDITO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SITUACAO_CREDITO` .

Used by: `BR_FIN_CREDITO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{1,4}))` | keep · `BR_CM_DIGITOS` |
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| otherwise | `BR_SITUACAO_CREDITO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_DATA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: `BR_L2_DATA_EVENTO`

#### CONFIGURATION

| | |
|---|---|
| minRange | `-30` |
| maxRange | `30` |
| unit | `DAYS` |
| roll | no |

---

### BR_DATA_NASCIMENTO DATE SHIFT

Moves the date by -60 to 60 days.

Used by: `BR_L2_DATA_NASCIMENTO`

#### CONFIGURATION

| | |
|---|---|
| minRange | `-60` |
| maxRange | `60` |
| unit | `DAYS` |
| roll | no |

---

### BR_DECIL SECURE LOOKUP

Replaces the value with one of the 10 values in `br-decil.txt` , chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-decil.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_DECIL_ALTO SECURE LOOKUP

Replaces the value with one of the 5 values in `br-decil-alto.txt`, chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-decil-alto.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_DEFICIENCIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_DEFICIENCIA_LISTA`.

Used by: `BR_SAUDE_DEFICIENCIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_DEFICIENCIA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_DEFICIENCIA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `br-deficiencias.txt`, chosen by a keyed hash.

Used by: `BR_DEFICIENCIA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-deficiencias.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_DIAGNOSTICO_TEXTO`.

Used by: `BR_SAUDE_DIAGNOSTICO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([A-TV-Za-tv-z]\d{2}\.\d{1,2})` | `BR_CID10_DECIMAL` |
| `([A-TV-Za-tv-z]\d{2,3})` | `BR_CID10` |
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| otherwise | `BR_DIAGNOSTICO_TEXTO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_DIAGNOSTICO_TEXTO SECURE LOOKUP

Replaces the value with one of the 30 values in `br-diagnosticos.txt`, chosen by a keyed hash.

Used by: `BR_DIAGNOSTICO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-diagnosticos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_EMAIL`, `BR_CHAVE_PIX`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([^@\s]+)@([^@\s]+)\.([A-Za-z]{2,24}(?:\.[A-Za-z]{2})?)` | `BR_CM_ALFANUM` · `BR_CM_ALFANUM` · replace with `test` |
| otherwise | `BR_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_EMPREGADOR SECURE LOOKUP

Replaces the value with one of the 24 values in `br-empregadores.txt`, chosen by a keyed hash.

Used by: `BR_L2_EMPREGADOR`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-empregadores.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |

| | |
|---|---|
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ENDERECO SECURE LOOKUP

Replaces the value with one of the 4,000 values in `br-enderecos.txt`, chosen by a keyed hash.

Used by: `BR_L1_ENDERECO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-enderecos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ESCOLA SECURE LOOKUP

Replaces the value with one of the 12 values in `br-escolas.txt`, chosen by a keyed hash.

Used by: `BR_L2_ESCOLA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-escolas.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ESCOLARIDADE REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_ESCOLARIDADE_LISTA`.

Used by: `BR_L2_ESCOLARIDADE`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*)(01\|02\|03\|04\|05\|06\|07\|08\|09\|10\|11\|12)` | keep · `BR_ESCOLARIDADE_CODIGO` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_ESCOLARIDADE_CODIGO` |
| otherwise | `BR_ESCOLARIDADE_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_ESCOLARIDADE_CODIGO SECURE LOOKUP

Replaces the value with one of the 12 values in `br-escolaridade-codigos.txt`, chosen by a keyed hash.

Used by: `BR_ESCOLARIDADE`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-escolaridade-codigos.txt` |
| hashMethod | `SHA256` |

| | |
|---|---|
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ESCOLARIDADE_LISTA SECURE LOOKUP

Replaces the value with one of the 10 values in `br-escolaridade.txt`, chosen by a keyed hash.

Used by: BR_ESCOLARIDADE

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-escolaridade.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ESTADO_CIVIL REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: BR_ESTADO_CIVIL_LISTA.

Used by: BR_L2_ESTADO_CIVIL

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01]) | BR_FLAG |
| (0*)(1\|2\|3\|4\|5) | keep · BR_ESTADO_CIVIL_CODIGO |
| (0*?)(?:(\d{1,6})) | keep · BR_ESTADO_CIVIL_CODIGO |
| otherwise | BR_ESTADO_CIVIL_LISTA |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_ESTADO_CIVIL_CODIGO SECURE LOOKUP

Replaces the value with one of the 5 values in `br-estado-civil-codigos.txt`, chosen by a keyed hash.

Used by: BR_ESTADO_CIVIL

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-estado-civil-codigos.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ESTADO_CIVIL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `br-estado-civil.txt`, chosen by a keyed hash.

Used by: BR_ESTADO_CIVIL

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-estado-civil.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_ETNIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ETNIA_LISTA`.

Used by: `BR_L3_ETNIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_ETNIA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_ETNIA_LISTA SECURE LOOKUP

Replaces the value with one of the 30 values in `br-etnias.txt`, chosen by a keyed hash.

Used by: `BR_ETNIA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-etnias.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FAIXA_CADUNICO SECURE LOOKUP

Replaces the value with one of the 4 values in `br-faixas-cadunico.txt`, chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-faixas-cadunico.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FAIXA_RENDA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `BR_FAIXA_CADUNICO`.

Used by: `BR_FIN_FAIXA_RENDA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([1-5]))` | keep · `BR_QUINTIL` |
| `(0*?)(?:([6-9]\|10))` | keep · `BR_DECIL_ALTO` |
| `(?i)(decil\s*)(\d{1,2})` | keep · `BR_DECIL` |
| `(?i)(quintil\s*)([1-5])` | keep · `BR_QUINTIL` |
| `(?i)(classe\s*)?(A\|B1\|B2\|C1\|C2\|D-E\|DE\|D\|E)` | keep · `BR_CLASSE_SOCIAL` |
| otherwise | `BR_FAIXA_CADUNICO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_FLAG REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_CATEGORIA_SUPRIMIDA`, `BR_ESTADO_CIVIL`, `BR_ESCOLARIDADE`, `BR_RACA_COR`, `BR_ETNIA`, `BR_RELIGIAO`, `BR_OPINIAO_POLITICA`, `BR_PARTIDO`, `BR_ORIENTACAO_SEXUAL`, `BR_IDENTIDADE_GENERO`, `BR_DIAGNOSTICO`, `BR_MEDICAMENTO`, `BR_DEFICIENCIA`, `BR_SAUDE_OCUPACIONAL`, `BR_CREDITO`, `BR_BENEFICIO_INSS`, `BR_PROGRAMA_SOCIAL`, `BR_MORADIA`, `BR_ANTECEDENTES`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o)` | `BR_FLAG_SIM_NAO` |
| `(?i)(true\|false)` | `BR_FLAG_TRUE_FALSE` |
| `(?i)(yes\|no)` | `BR_FLAG_YES_NO` |
| `(?i)([sn])` | `BR_FLAG_S_N` |
| `(?i)([yn])` | `BR_FLAG_Y_N` |
| `([01])` | `BR_FLAG_0_1` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_FLAG_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-0-1.txt`, chosen by a keyed hash.

Used by: `BR_FLAG`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-flag-0-1.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FLAG_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-s-n.txt`, chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-flag-s-n.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FLAG_SIM_NAO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-sim-nao.txt`, chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-flag-sim-nao.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FLAG_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-true-false.txt`, chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-flag-true-false.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FLAG_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-y-n.txt`, chosen by a keyed hash.

Used by: `BR_FLAG`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-flag-y-n.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_FLAG_YES_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-flag-yes-no.txt`, chosen by a keyed hash.

Used by: `BR_FLAG`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-flag-yes-no.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_IDADE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_IDADE`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:([1-9]\d{2,}))` | keep · replace with `90` |
| `(0*?)(?:([1-9])(\d)([.,]\d+))` | keep · keep · `BR_CM_DIGITOS` · keep |
| `(0*?)(?:([1-9])(\d))` | keep · keep · `BR_CM_DIGITOS` |
| `(0*?)(?:(\d))` | keep · `BR_CM_DIGITOS` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_IDENTIDADE_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_IDENTIDADE_GENERO_LISTA`.

Used by: `BR_L3_IDENTIDADE_GENERO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_IDENTIDADE_GENERO_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_IDENTIDADE_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `br-identidades-genero.txt`, chosen by a keyed hash.

Used by: `BR_IDENTIDADE_GENERO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-identidades-genero.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_HEX`.

Used by: `BR_L1_IP`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{1,3})\.(\d{1,3})\.(\d{1,3})\.(\d{1,3}))` | keep · `BR_OCTETO` · `BR_OCTETO` · `BR_OCTETO` · `BR_OCTETO` |
| otherwise | `BR_CM_HEX` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_MEDICAMENTO_LISTA`.

Used by: `BR_SAUDE_MEDICAMENTO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_MEDICAMENTO_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 25 values in `br-medicamentos.txt`, chosen by a keyed hash.

Used by: `BR_MEDICAMENTO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-medicamentos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_MORADIA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_MORADIA_LISTA`.

Used by: `BR_FIN_MORADIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_MORADIA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |

| | |
|---|---|
| requireMask | no |

### BR_MORADIA_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `br-condicoes-moradia.txt`, chosen by a keyed hash.

Used by: `BR_MORADIA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-condicoes-moradia.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_MUNICIPIO DATA CLEANSING

Replaces each value found in `br-municipios-generalizados.txt` (55,203 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_L2_MUNICIPIO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-municipios-generalizados.txt` |
| delimiter | `\|` |
| caseSensitive | yes |
| trimWhitespace | yes |

### BR_NACIONALIDADE SECURE LOOKUP

Replaces the value with one of the 16 values in `br-nacionalidades.txt`, chosen by a keyed hash.

Used by: `BR_L2_NACIONALIDADE`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-nacionalidades.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_NOME NAME

Replaces the name with one of the 895 names in `br-nomes.txt`.

Used by: `BR_PALAVRA_NOME`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-nomes.txt` |
| maskedValueCase | `PRESERVE_INPUT` |
| filterAccent | yes |
| inputCaseSensitive | no |

### BR_NOME_COMPLETO REGEX DECOMPOSE

Tries 7 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_NOME_COMPLETO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([^,]+),\s*(.+)` | `BR_SOBRENOMES` · `BR_NOMES` |
| `(\S+)` | `BR_PALAVRA_NOME` |
| `(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_NOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_NOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_NOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| otherwise | `BR_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_NOMES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_NOME`, `BR_NOME_COMPLETO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(\S+)` | `BR_PALAVRA_NOME` |
| `(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_NOME` |
| `(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_NOME` · `BR_PALAVRA_NOME` · `BR_PALAVRA_NOME` |
| otherwise | `BR_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: `BR_IP`

#### CONFIGURATION

| | |
|---|---|
| minValue | `1` |
| maxValue | `254` |

---

### BR_OCUPACAO SECURE LOOKUP

Replaces the value with one of the 40 values in `br-ocupacoes.txt`, chosen by a keyed hash.

Used by: `BR_OCUPACAO_OU_CBO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-ocupacoes.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |

| | |
|---|---|
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_OCUPACAO_OU_CBO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_OCUPACAO`.

Used by: `BR_L2_OCUPACAO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{6}))` | keep · `BR_CBO` |
| `(0*?)(?:(\d{4})-(\d{2}))` | keep · `BR_CM_DIGITOS` · `BR_CM_DIGITOS` |
| otherwise | `BR_OCUPACAO` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_OPINIAO_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_OPINIAO_POLITICA_LISTA`.

Used by: `BR_L3_OPINIAO_POLITICA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_OPINIAO_POLITICA_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_OPINIAO_POLITICA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `br-opinioes-politicas.txt`, chosen by a keyed hash.

Used by: `BR_OPINIAO_POLITICA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-opinioes-politicas.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_ORIENTACAO_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_ORIENTACAO_SEXUAL_LISTA`.

Used by: `BR_L3_ORIENTACAO_SEXUAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |

| | |
|---|---|
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_ORIENTACAO_SEXUAL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_ORIENTACAO_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `br-orientacoes-sexuais.txt`, chosen by a keyed hash.

Used by: `BR_ORIENTACAO_SEXUAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-orientacoes-sexuais.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_PALAVRA_NOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_NOME`.

Used by: `BR_NOMES`, `BR_NOME_COMPLETO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(d[aeo]s?\|e\|di\|del\|della\|van\|von\|der\|du\|le\|la)` | keep |
| `(?i)(j[uú]nior\|jr\.?\|filho\|filha\|neto\|neta\|sobrinho\|segundo\|terceiro)` | keep |
| otherwise | `BR_NOME` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_PALAVRA_SOBRENOME REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SOBRENOME`.

Used by: `BR_SOBRENOMES`, `BR_NOME_COMPLETO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(d[aeo]s?\|e\|di\|del\|della\|van\|von\|der\|du\|le\|la)` | keep |
| `(?i)(j[uú]nior\|jr\.?\|filho\|filha\|neto\|neta\|sobrinho\|segundo\|terceiro)` | keep |
| otherwise | `BR_SOBRENOME` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_PARTIDO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_PARTIDO_NOME`.

Used by: `BR_L3_FILIACAO_PARTIDARIA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:([1-9]\d))` | keep · `BR_PARTIDO_NUMERO` |
| `(?i)(MDB\|PDT\|PT\|PCdoB\|PSB\|PSDB\|AGIR\|MOBILIZA\|CIDADANIA\|PV\|AVANTE\|PP\|PSTU\|PCB\|PRTB\|DC\|PCO\|PODE\|REPUBLICANOS\|PSOL\|PL\|PSD\|SOLIDARIEDADE\|NOVO\|REDE\|DEMOCRATA\|UP\|UNIAO\|PRD\|MISSAO\|UNIÃO\|MISSÃO\|PFL\|DEM\|PMDB\|PSL\|PTB\|PROS\|PSC\|PMN\|PTC\|PHS\|PPS\|PR\|PRB)` | `BR_PARTIDO_SIGLA` |
| `otherwise` | `BR_PARTIDO_NOME` |

#### CONFIGURATION

| | |
|---|---|
| `trimInput` | yes |
| `requireMask` | no |

---

### BR_PARTIDO_NOME SECURE LOOKUP

Replaces the value with one of the 31 values in `br-partidos-nomes.txt`, chosen by a keyed hash.

Used by: `BR_PARTIDO`

#### CONFIGURATION

| | |
|---|---|
| `lookupFile` | `br-partidos-nomes.txt` |
| `hashMethod` | `SHA256` |
| `maskedValueCase` | `PRESERVE_INPUT` |
| `inputCaseSensitive` | no |
| `trimWhitespaceFromInput` | yes |
| `trimWhitespaceInLookupFile` | yes |

---

### BR_PARTIDO_NUMERO SECURE LOOKUP

Replaces the value with one of the 30 values in `br-partidos-numeros.txt`, chosen by a keyed hash.

Used by: `BR_PARTIDO`

#### CONFIGURATION

| | |
|---|---|
| `lookupFile` | `br-partidos-numeros.txt` |
| `hashMethod` | `SHA256` |
| `maskedValueCase` | `PRESERVE_INPUT` |
| `inputCaseSensitive` | no |
| `trimWhitespaceFromInput` | yes |
| `trimWhitespaceInLookupFile` | yes |

---

### BR_PARTIDO_SIGLA SECURE LOOKUP

Replaces the value with one of the 30 values in `br-partidos-siglas.txt`, chosen by a keyed hash.

Used by: `BR_PARTIDO`

#### CONFIGURATION

| | |
|---|---|
| `lookupFile` | `br-partidos-siglas.txt` |

| | |
|---|---|
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_LOOKUP_FILE |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_PIS REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: BR_CM_DIGITOS .

Used by: BR_L1_PIS_NIS

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (0*?)(?:(\d{3}\.?\d{5}\.?\d{2}-?\d)) | keep · BR_PIS_VALIDO |
| (0*?)(?:(\d{9,10})) | keep · BR_PIS_VALIDO_ZEROS |
| otherwise | BR_CM_DIGITOS |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_PIS_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_PIS_VALIDO , BR_PIS_VALIDO_ZEROS

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (\d)(.*) | keep · BR_CM_DIGITOS |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_PIS_VALIDO CHECK DIGIT

Masks the digits with BR_PIS_CORPO and recomputes a modulus 11 check digit with weights 3 2 9 8 7 6 5 4 3 2 .

Used by: BR_PIS , BR_CNS_CORPO

#### CONFIGURATION

| | |
|---|---|
| weightList | 3 2 9 8 7 6 5 4 3 2 |
| modulusNumber | 11 |
| checkDigitIndex | 10 |
| numDigitsForCheckdigitCalculation | 10 |
| calculateChecksumRightToLeft | no |
| numericAlgorithm | BR_PIS_CORPO |
| alphaNumericAlgorithm | BR_PIS_CORPO |
| preserveRegex | [.\-/] |
| inputHandlingConfig.characterHandling | STANDARD |
| inputHandlingConfig.invalidInputHandling | ERROR |
| inputHandlingConfig.shortInputHandling | FALLBACK |
| inputHandlingConfig.padCharacter | 0 |

| | |
|---|---|
| inputHandlingConfig.trimWhitespace | yes |

### BR_PIS_VALIDO_ZEROS CHECK DIGIT

Masks the digits with `BR_PIS_CORPO` and recomputes a modulus 11 check digit with weights `3 2 9 8 7 6 5 4 3 2`.

Used by: `BR_PIS`

#### CONFIGURATION

| | |
|---|---|
| weightList | `3 2 9 8 7 6 5 4 3 2` |
| modulusNumber | `11` |
| checkDigitIndex | `10` |
| numDigitsForCheckdigitCalculation | `10` |
| calculateChecksumRightToLeft | no |
| numericAlgorithm | `BR_PIS_CORPO` |
| alphaNumericAlgorithm | `BR_PIS_CORPO` |
| preserveRegex | `[.\-/]` |
| inputHandlingConfig.characterHandling | `STANDARD` |
| inputHandlingConfig.invalidInputHandling | `ERROR` |
| inputHandlingConfig.shortInputHandling | `PAD_LEFT` |
| inputHandlingConfig.padCharacter | `0` |
| inputHandlingConfig.trimWhitespace | yes |

### BR_PLACA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_PLACA`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `([A-Za-z]{3})(-?)(\d)([A-Za-z0-9])(\d{2})` | `BR_CM_DIGITOS_LETRAS` · keep · `BR_CM_DIGITOS` · `BR_CM_DIGITOS_LETRAS` · `BR_CM_DIGITOS` |
| otherwise | `BR_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_PROCESSO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_PENAL_PROCESSO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{7})(-?)(\d{2})(\.?\d{4}\.?\d\.?\d{2}\.?\d{4}))` | keep · `BR_CM_DIGITOS` · keep · `BR_CM_DIGITOS` · keep |
| otherwise | `BR_CM_ALFANUM` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_PROGRAMA_SOCIAL_LISTA`.

Used by: `BR_FIN_PROGRAMA_SOCIAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_PROGRAMA_SOCIAL_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `br-programas-sociais.txt`, chosen by a keyed hash.

Used by: `BR_PROGRAMA_SOCIAL`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-programas-sociais.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_QUINTIL SECURE LOOKUP

Replaces the value with one of the 5 values in `br-quintil.txt`, chosen by a keyed hash.

Used by: `BR_FAIXA_RENDA`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-quintil.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_RACA_COR REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_RACA_COR_LISTA`.

Used by: `BR_L3_RACA_COR`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*)(1\|2\|3\|4\|5\|9)` | keep · `BR_RACA_COR_CODIGO` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_RACA_COR_CODIGO` |
| otherwise | `BR_RACA_COR_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_RACA_COR_CODIGO SECURE LOOKUP

Replaces the value with one of the 6 values in `br-raca-cor-codigos.txt`, chosen by a keyed hash.

Used by: `BR_RACA_COR`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-raca-cor-codigos.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_RACA_COR_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `br-raca-cor.txt`, chosen by a keyed hash.

Used by: `BR_RACA_COR`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-raca-cor.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_REDIGIR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L1_CREDENCIAL`, `BR_L3_GENETICO`, `BR_L3_BIOMETRICO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(.+)` | each character → `X` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_REGISTRO_PROFISSIONAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

Used by: `BR_L1_REGISTRO_PROFISSIONAL`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)([A-Z]{2,6}\s*[-/]?\s*[A-Z]{2}\s*[-/]?\s*)(\d[\d.\-]*)(.*)` | keep · `BR_CM_DIGITOS` · keep |
| `(0*?)(?:(\d[\d.\-]*)(\s*[-/]?\s*[A-Za-z]{2}))` | keep · `BR_CM_DIGITOS` · keep |
| otherwise | `BR_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_RELIGIAO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_RELIGIAO_LISTA`.

Used by: `BR_L3_RELIGIAO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_RELIGIAO_LISTA` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_RELIGIAO_LISTA SECURE LOOKUP

Replaces the value with one of the 13 values in `br-religioes.txt`, chosen by a keyed hash.

Used by: `BR_RELIGIAO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-religioes.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

---

### BR_RENAVAM REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

Used by: `BR_L1_RENAVAM`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{9,11}))` | keep · `BR_RENAVAM_VALIDO` |
| otherwise | `BR_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_RENAVAM_VALIDO CHECK DIGIT

Masks the digits with `BR_CM_DIGITOS_LARGURA_FIXA` and recomputes a modulus 11 check digit with weights `3 2 9 8 7 6 5 4 3 2`.

Used by: `BR_RENAVAM`

#### CONFIGURATION

| | |
|---|---|
| weightList | `3 2 9 8 7 6 5 4 3 2` |
| modulusNumber | `11` |
| checkDigitIndex | `10` |

| | |
|---|---|
| numDigitsForCheckdigitCalculation | `10` |
| calculateChecksumRightToLeft | no |
| numericAlgorithm | `BR_CM_DIGITOS_LARGURA_FIXA` |
| alphaNumericAlgorithm | `BR_CM_DIGITOS_LARGURA_FIXA` |
| preserveRegex | `[.\-/]` |
| inputHandlingConfig.characterHandling | `STANDARD` |
| inputHandlingConfig.invalidInputHandling | `ERROR` |
| inputHandlingConfig.shortInputHandling | `PAD_LEFT` |
| inputHandlingConfig.padCharacter | `0` |
| inputHandlingConfig.trimWhitespace | yes |

## BR_SAUDE_OCUPACIONAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `BR_SAUDE_OCUPACIONAL_LISTA` .

Used by: `BR_SAUDE_OCUPACIONAL`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(sim\|n[aã]o\|true\|false\|yes\|no\|[sny01])` | `BR_FLAG` |
| `(0*?)(?:(\d{1,6}))` | keep · `BR_CM_ALFANUM` |
| otherwise | `BR_SAUDE_OCUPACIONAL_LISTA` |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## BR_SAUDE_OCUPACIONAL_LISTA SECURE LOOKUP

Replaces the value with one of the 4 values in `br-aptidao.txt` , chosen by a keyed hash.

Used by: `BR_SAUDE_OCUPACIONAL`

### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-aptidao.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

## BR_SEXO REGEX DECOMPOSE

Tries 6 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `BR_L2_SEXO`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?i)(feminino\|masculino)` | `BR_SEXO_FEMININO_MASCULINO` |
| `(?i)(mulher\|homem)` | `BR_SEXO_MULHER_HOMEM` |
| `(?i)(fem\|masc)` | `BR_SEXO_FEM_MASC` |
| `(?i)(female\|male)` | `BR_SEXO_FEMALE_MALE` |
| `(?i)([fm])` | `BR_SEXO_F_M` |
| `(0*?)(?:([12]))` | keep · `BR_SEXO_1_2` |

| | |
|---|---|
| otherwise | keep |

CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_SEXO_1_2 SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-1-2.txt`, chosen by a keyed hash.

Used by: `BR_SEXO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-sexo-1-2.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SEXO_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-f-m.txt`, chosen by a keyed hash.

Used by: `BR_SEXO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-sexo-f-m.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SEXO_FEM_MASC SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-fem-masc.txt`, chosen by a keyed hash.

Used by: `BR_SEXO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-sexo-fem-masc.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_INPUT` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SEXO_FEMALE_MALE SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-female-male.txt`, chosen by a keyed hash.

Used by: `BR_SEXO`

CONFIGURATION

| | |
|---|---|
| lookupFile | `br-sexo-female-male.txt` |
| hashMethod | `SHA256` |

| | |
|---|---|
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SEXO_FEMININO_MASCULINO SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-feminino-masculino.txt`, chosen by a keyed hash.

Used by: BR_SEXO

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-sexo-feminino-masculino.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SEXO_MULHER_HOMEM SECURE LOOKUP

Replaces the value with one of the 2 values in `br-sexo-mulher-homem.txt`, chosen by a keyed hash.

Used by: BR_SEXO

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-sexo-mulher-homem.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SITUACAO_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `br-situacoes-credito.txt`, chosen by a keyed hash.

Used by: BR_CREDITO

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-situacoes-credito.txt |
| hashMethod | SHA256 |
| maskedValueCase | PRESERVE_INPUT |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

### BR_SOBRENOME NAME

Replaces the name with one of the 535 names in `br-sobrenomes.txt`.

Used by: BR_PALAVRA_SOBRENOME

#### CONFIGURATION

| | |
|---|---|
| lookupFile | br-sobrenomes.txt |
| maskedValueCase | PRESERVE_INPUT |

| | |
|---|---|
| filterAccent | yes |
| inputCaseSensitive | no |

## BR_SOBRENOMES REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_ALFANUM`.

Used by: `BR_L1_SOBRENOME`, `BR_NOME_COMPLETO`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(\S+)` | `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)` | `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| `(\S+)\s+(\S+)\s+(\S+)\s+(\S+)` | `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` · `BR_PALAVRA_SOBRENOME` |
| otherwise | `BR_CM_ALFANUM` |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

## BR_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `br-nao-informado.txt`, chosen by a keyed hash.

Used by: `BR_CATEGORIA_SUPRIMIDA`

### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-nao-informado.txt` |
| hashMethod | `SHA256` |
| maskedValueCase | `PRESERVE_LOOKUP_FILE` |
| inputCaseSensitive | no |
| trimWhitespaceFromInput | yes |
| trimWhitespaceInLookupFile | yes |

## BR_TELEFONE REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

Used by: `BR_L1_TELEFONE`, `BR_CHAVE_PIX`

### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\+?55\s?)?(\(?0?\d{2}\)?\s?)(9)(\d{4})([\s\-]?)(\d{4}))` | keep · keep · keep · keep · `BR_CM_DIGITOS` · keep · `BR_CM_DIGITOS` |
| `(0*?)(?:(\+?55\s?)?(\(?0?\d{2}\)?\s?)([2-5])(\d{3})([\s\-]?)(\d{4}))` | keep · keep · keep · keep · `BR_CM_DIGITOS` · keep · `BR_CM_DIGITOS` |
| `(0*?)(?:(9)(\d{4})([\s\-]?)(\d{4}))` | keep · keep · `BR_CM_DIGITOS` · keep · `BR_CM_DIGITOS` |
| `(0*?)(?:([2-5])(\d{3})([\s\-]?)(\d{4}))` | keep · keep · `BR_CM_DIGITOS` · keep · `BR_CM_DIGITOS` |
| otherwise | `BR_CM_DIGITOS` |

### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_TEXTO_LIVRE FREE TEXT REDACTION

Replaces the text matching 12 patterns with `[DADO]`, and the 3,302 terms of `br-texto-nomes.txt` with `[NOME]`.

Used by: `BR_L1_TEXTO_LIVRE`, `BR_SAUDE_TEXTO_CLINICO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(?<![\d.\-])\d{3}\.\d{3}\.\d{3}-\d{2}(?![\d\-])` | |
| `(?<!\d)\d{11}(?!\d)` | |
| `(?<![\dA-Z.])[0-9A-Z]{2}\.[0-9A-Z]{3}\.[0-9A-Z]{3}/[0-9A-Z]{4}-\d{2}(?!\d)` | |
| `(?<!\d)\d{14}(?!\d)` | |
| `[\w.+\-]+@[\w\-]+(\.[\w\-]+)+` | |
| `(?<![\d\-])\(?\d{2}\)?-?9?\d{4}-\d{4}(?![\d\-])` | |
| `(?<![\d\-])9\d{4}-\d{4}(?![\d\-])` | |
| `(?<!\d)(\+?55)?\d{10,11}(?!\d)` | |
| `(?<!\d)\d{13,19}(?!\d)` | |
| `(?<!\d)\d{5}-\d{3}(?!\d)` | |
| `(?<![A-Za-z])[A-Z]{3}-?\d[A-Z0-9]\d{2}(?![A-Za-z\d])` | |
| `(?<!\d)(\d{1,3}\.){3}\d{1,3}(?!\d)` | |

#### CONFIGURATION

| | |
|---|---|
| regExRedactValue | `[DADO]` |
| lookupFile | `br-texto-nomes.txt` |
| lookupFileRedactValue | `[NOME]` |
| isDenyList | yes |

---

### BR_TITULO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `BR_CM_DIGITOS`.

Used by: `BR_L1_TITULO_ELEITOR`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(0*?)(?:(\d{12}))` | keep · `BR_TITULO_VALIDO` |
| `(0*?)(?:(\d{11}))` | keep · `BR_TITULO_VALIDO_1_ZERO` |
| `(0*?)(?:(\d{10}))` | keep · `BR_TITULO_VALIDO_2_ZEROS` |
| otherwise | `BR_CM_DIGITOS` |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_TITULO_1_ZERO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_TITULO_VALIDO_1_ZERO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(\d)(\d{10})` | `BR_TITULO_1_ZERO_TABELA` · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |

| | |
|---|---|
| requireMask | no |

### BR_TITULO_1_ZERO_TABELA DATA CLEANSING

Replaces each value found in `br-zeros-1.txt` (10 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_1_ZERO`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-zeros-1.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

### BR_TITULO_2_ZEROS REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_TITULO_VALIDO_2_ZEROS`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(\d)(\d{9})` | `BR_TITULO_2_ZEROS_TABELA` · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_TITULO_2_ZEROS_TABELA DATA CLEANSING

Replaces each value found in `br-zeros-2.txt` (10 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_2_ZEROS`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-zeros-2.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

### BR_TITULO_CORPO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_TITULO_VALIDO`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(\d{8})(\d{2})(\d{2})` | `BR_CM_DIGITOS` · keep · replace with `00` |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

### BR_TITULO_DV STRING ALGORITHM CHAIN

Applies `BR_TITULO_DV_P1` → `BR_TITULO_DV_P2` → `BR_TITULO_DV_P3` → `BR_TITULO_DV_P4` → `BR_TITULO_DV_P5`, in this order.

Used by: `BR_TITULO_VALIDO`

#### CONFIGURATION

| | |
|---|---|
| algorithmReferences[1] | BR_TITULO_DV_P1 |
| algorithmReferences[2] | BR_TITULO_DV_P2 |
| algorithmReferences[3] | BR_TITULO_DV_P3 |
| algorithmReferences[4] | BR_TITULO_DV_P4 |
| algorithmReferences[5] | BR_TITULO_DV_P5 |

---

### BR_TITULO_DV_P1 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{0})(.{2})(.*) | keep · BR_TITULO_DV_T1 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_TITULO_DV_P2 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{2})(.{3})(.*) | keep · BR_TITULO_DV_T2 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_TITULO_DV_P3 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{4})(.{3})(.*) | keep · BR_TITULO_DV_T3 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| trimInput | yes |
| requireMask | no |

---

### BR_TITULO_DV_P4 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: BR_TITULO_DV

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| (.{6})(.{3})(.*) | keep · BR_TITULO_DV_T4 · keep |
| otherwise | keep |

#### CONFIGURATION

| | |
|---|---|
| `trimInput` | yes |
| `requireMask` | no |

### BR_TITULO_DV_P5 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `BR_TITULO_DV`

#### PATTERNS, TRIED IN ORDER

| | |
|---|---|
| `(.{8})(.{5})(.*)` | keep · `BR_TITULO_DV_T5` · keep |
| `otherwise` | keep |

#### CONFIGURATION

| | |
|---|---|
| `trimInput` | yes |
| `requireMask` | no |

### BR_TITULO_DV_T1 DATA CLEANSING

Replaces each value found in `br-titulo-dv-1.txt` (100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P1`

#### CONFIGURATION

| | |
|---|---|
| `lookupFile` | `br-titulo-dv-1.txt` |
| `delimiter` | `,` |
| `caseSensitive` | yes |
| `trimWhitespace` | yes |

### BR_TITULO_DV_T2 DATA CLEANSING

Replaces each value found in `br-titulo-dv-2.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P2`

#### CONFIGURATION

| | |
|---|---|
| `lookupFile` | `br-titulo-dv-2.txt` |
| `delimiter` | `,` |
| `caseSensitive` | yes |
| `trimWhitespace` | yes |

### BR_TITULO_DV_T3 DATA CLEANSING

Replaces each value found in `br-titulo-dv-3.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P3`

#### CONFIGURATION

| | |
|---|---|
| `lookupFile` | `br-titulo-dv-3.txt` |
| `delimiter` | `,` |
| `caseSensitive` | yes |
| `trimWhitespace` | yes |

### BR_TITULO_DV_T4 DATA CLEANSING

Replaces each value found in `br-titulo-dv-4.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P4`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-titulo-dv-4.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_TITULO_DV_T5 DATA CLEANSING

Replaces each value found in `br-titulo-dv-5.txt` (1,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: `BR_TITULO_DV_P5`

#### CONFIGURATION

| | |
|---|---|
| lookupFile | `br-titulo-dv-5.txt` |
| delimiter | `,` |
| caseSensitive | yes |
| trimWhitespace | yes |

---

### BR_TITULO_VALIDO STRING ALGORITHM CHAIN

Applies `BR_TITULO_CORPO` → `BR_TITULO_DV`, in this order.

Used by: `BR_TITULO_VALIDO_1_ZERO`, `BR_TITULO_VALIDO_2_ZEROS`, `BR_TITULO`

#### CONFIGURATION

| | |
|---|---|
| algorithmReferences[1] | `BR_TITULO_CORPO` |
| algorithmReferences[2] | `BR_TITULO_DV` |

---

### BR_TITULO_VALIDO_1_ZERO STRING ALGORITHM CHAIN

Applies `BR_TITULO_1_ZERO` → `BR_TITULO_VALIDO`, in this order.

Used by: `BR_TITULO`

#### CONFIGURATION

| | |
|---|---|
| algorithmReferences[1] | `BR_TITULO_1_ZERO` |
| algorithmReferences[2] | `BR_TITULO_VALIDO` |

---

### BR_TITULO_VALIDO_2_ZEROS STRING ALGORITHM CHAIN

Applies `BR_TITULO_2_ZEROS` → `BR_TITULO_VALIDO`, in this order.

Used by: `BR_TITULO`

#### CONFIGURATION

| | |
|---|---|
| algorithmReferences[1] | `BR_TITULO_2_ZEROS` |
| algorithmReferences[2] | `BR_TITULO_VALIDO` |

---

APPENDIX B

# Files

The lookup and detection files the policy reads. In Delphix each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| `br-antecedentes.txt` | 8 | `BR_ANTECEDENTES_LISTA` | Nada consta · Consta registro · Indiciado · Denunciado |
| `br-aptidao.txt` | 4 | `BR_SAUDE_OCUPACIONAL_LISTA` | Apto · Apto com restrições · Inapto · Não avaliado |
| `br-bairros.txt` | 40 | `BR_BAIRRO` | Centro · Jardim América · Vila Nova · Jardim Primavera |
| `br-beneficios-inss-codigos.txt` | 14 | `BR_BENEFICIO_INSS_CODIGO` | 21 · 31 · 32 · 36 |
| `br-beneficios-inss.txt` | 11 | `BR_BENEFICIO_INSS_LISTA` | Aposentadoria por idade · Aposentadoria por tempo de contribuição · Aposentadoria especial · Aposentadoria por incapacidade permanente |
| `br-cbo.txt` | 34 | `BR_CBO` | 411005 · 411010 · 521110 · 421125 |
| `br-cid10-decimal.txt` | 33 | `BR_CID10_DECIMAL` | E11.9 · E78.5 · J45.9 · J06.9 |
| `br-cid10.txt` | 37 | `BR_CID10` | I10 · E11 · E78 · J45 |
| `br-classes-sociais.txt` | 6 | `BR_CLASSE_SOCIAL` | A · B1 · B2 · C1 |
| `br-cns-dv-1.txt` | 100 | `BR_CNS_DV_T1` | 00,000 · 01,013 · 02,026 · 03,039 |
| `br-cns-dv-2.txt` | 1,100 | `BR_CNS_DV_T2` | 000,000 · 001,011 · 002,022 · 003,033 |
| `br-cns-dv-3.txt` | 1,100 | `BR_CNS_DV_T3` | 000,000 · 001,01A · 002,029 · 003,038 |
| `br-cns-dv-4.txt` | 1,100 | `BR_CNS_DV_T4` | 000,000 · 001,018 · 002,025 · 003,032 |
| `br-cns-dv-5.txt` | 1,100 | `BR_CNS_DV_T5` | 000,000 · 001,016 · 002,021 · 003,037 |
| `br-cns-dv-6.txt` | 110 | `BR_CNS_DV_T6` | 000000,00000 · 010000,10006 · 020000,20001 · 030000,30007 |
| `br-codigos-municipio-generalizados.txt` | 7,720 | `BR_CODIGO_MUNICIPIO_TABELA` | 1200013,1200450 · 120001,120045 · 1200054,1200104 · 120005,120010 |
| `br-condicoes-moradia.txt` | 6 | `BR_MORADIA_LISTA` | Próprio quitado · Próprio em pagamento · Alugado · Cedido |
| `br-decil-alto.txt` | 5 | `BR_DECIL_ALTO` | 6 · 7 · 8 · 9 |
| `br-decil.txt` | 10 | `BR_DECIL` | 1 · 2 · 3 · 4 |
| `br-deficiencias.txt` | 7 | `BR_DEFICIENCIA_LISTA` | Física · Visual · Auditiva · Intelectual |
| `br-detectar-nacionalidades.txt` | 58 | `BR_L2_NACIONALIDADE – List` | Brasileira · Venezuelana · Haitiana · Boliviana |
| `br-detectar-nomes.txt` | 1,046 | `BR_L1_NOME – List` `BR_L1_NOME_COMPLETO – List` | Maria · Ana · Francisca · Júlia |

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| br-detectar-sexo-letra.txt | 2 | BR_L2_SEXO – List | f · m |
| br-detectar-sexo.txt | 8 | BR_L2_SEXO – List | feminino · masculino · fem · masc |
| br-detectar-sobrenomes.txt | 636 | BR_L1_SOBRENOME – List BR_L1_NOME_COMPLETO – List | Silva · Santos · Oliveira · Souza |
| br-diagnosticos.txt | 30 | BR_DIAGNOSTICO_TEXTO | Hipertensão essencial · Diabetes mellitus tipo 2 · Asma · Lombalgia |
| br-empregadores.txt | 24 | BR_EMPREGADOR | Comercial Horizonte Ltda. · Construtora Vale Verde Ltda. · Transportes Rota Sul Ltda. · Supermercados Bom Preço Ltda. |
| br-enderecos.txt | 4,000 | BR_ENDERECO | Avenida Juscelino Kubitschek, 1340 - Jardim Brasil · Rua Santa Catarina, 936 · Rua Minas Gerais, 16 · Avenida Beira-Rio, 551, Apto 172 |
| br-escolaridade-codigos.txt | 12 | BR_ESCOLARIDADE_CODIGO | 01 · 02 · 03 · 04 |
| br-escolaridade.txt | 10 | BR_ESCOLARIDADE_LISTA | Sem instrução · Fundamental incompleto · Fundamental completo · Médio incompleto |
| br-escolas.txt | 12 | BR_ESCOLA | Escola Estadual Professora Maria José · Escola Municipal Monteiro Lobato · Colégio Estadual Rui Barbosa · Escola Estadual Castro Alves |
| br-estado-civil-codigos.txt | 5 | BR_ESTADO_CIVIL_CODIGO | 1 · 2 · 3 · 4 |
| br-estado-civil.txt | 6 | BR_ESTADO_CIVIL_LISTA | Solteiro(a) · Casado(a) · União estável · Divorciado(a) |
| br-etnias.txt | 30 | BR_ETNIA_LISTA | Tikúna · Kokama · Makuxí · Guarani Kaiowá |
| br-faixas-cadunico.txt | 4 | BR_FAIXA_CADUNICO | Extrema pobreza · Pobreza · Baixa renda · Acima de meio salário mínimo |
| br-flag-0-1.txt | 2 | BR_FLAG_0_1 | 0 · 1 |
| br-flag-s-n.txt | 2 | BR_FLAG_S_N | S · N |
| br-flag-sim-nao.txt | 2 | BR_FLAG_SIM_NAO | Sim · Não |
| br-flag-true-false.txt | 2 | BR_FLAG_TRUE_FALSE | true · false |
| br-flag-y-n.txt | 2 | BR_FLAG_Y_N | Y · N |
| br-flag-yes-no.txt | 2 | BR_FLAG_YES_NO | Yes · No |
| br-identidades-genero.txt | 8 | BR_IDENTIDADE_GENERO_LISTA | Mulher cisgênero · Homem cisgênero · Mulher trans · Homem trans |
| br-medicamentos.txt | 25 | BR_MEDICAMENTO_LISTA | Losartana potássica · Metformina · Omeprazol · Sinvastatina |
| br-municipios-generalizados.txt | 55,203 | BR_MUNICIPIO | Acrelândia\|Senador Guiomard · ACRELÂNDIA\|SENADOR GUIOMARD · Acrelandia\|Senador Guiomard · ACRELANDIA\|SENADOR GUIOMARD |
| br-nacionalidades.txt | 16 | BR_NACIONALIDADE | Brasileira · Venezuelana · Haitiana · Boliviana |
| br-nao-informado.txt | 1 | BR_SUPRIMIR | Não informado |

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| br-nomes.txt | 895 | BR_NOME | Maria · Ana · Francisca · Júlia |
| br-ocupacoes.txt | 40 | BR_OCUPACAO | Auxiliar administrativo · Assistente administrativo · Vendedor de comércio varejista · Operador de caixa |
| br-opinioes-politicas.txt | 7 | BR_OPINIAO_POLITICA_LISTA | Esquerda · Centro-esquerda · Centro · Centro-direita |
| br-orientacoes-sexuais.txt | 8 | BR_ORIENTACAO_SEXUAL_LISTA | Heterossexual · Gay · Lésbica · Bissexual |
| br-partidos-nomes.txt | 31 | BR_PARTIDO_NOME | Movimento Democrático Brasileiro · Partido Democrático Trabalhista · Partido dos Trabalhadores · Partido Comunista do Brasil |
| br-partidos-numeros.txt | 30 | BR_PARTIDO_NUMERO | 15 · 12 · 13 · 65 |
| br-partidos-siglas.txt | 30 | BR_PARTIDO_SIGLA | MDB · PDT · PT · PCdoB |
| br-programas-sociais.txt | 9 | BR_PROGRAMA_SOCIAL_LISTA | Bolsa Família · Benefício de Prestação Continuada · Auxílio Gás dos Brasileiros · Pé-de-Meia |
| br-quintil.txt | 5 | BR_QUINTIL | 1 · 2 · 3 · 4 |
| br-raca-cor-codigos.txt | 6 | BR_RACA_COR_CODIGO | 1 · 2 · 3 · 4 |
| br-raca-cor.txt | 6 | BR_RACA_COR_LISTA | Branca · Preta · Parda · Amarela |
| br-religioes.txt | 13 | BR_RELIGIAO_LISTA | Católica Apostólica Romana · Evangélica · Evangélica pentecostal · Evangélica de missão |
| br-sexo-1-2.txt | 2 | BR_SEXO_1_2 | 1 · 2 |
| br-sexo-f-m.txt | 2 | BR_SEXO_F_M | F · M |
| br-sexo-fem-masc.txt | 2 | BR_SEXO_FEM_MASC | Fem · Masc |
| br-sexo-female-male.txt | 2 | BR_SEXO_FEMALE_MALE | Female · Male |
| br-sexo-feminino-masculino.txt | 2 | BR_SEXO_FEMININO_MASCULINO | Feminino · Masculino |
| br-sexo-mulher-homem.txt | 2 | BR_SEXO_MULHER_HOMEM | Mulher · Homem |
| br-situacoes-credito.txt | 10 | BR_SITUACAO_CREDITO | Em dia · Atraso de 30 dias · Atraso de 60 dias · Atraso de 90 dias |
| br-sobrenomes.txt | 535 | BR_SOBRENOME | Silva · Santos · Oliveira · Souza |
| br-texto-nomes.txt | 3,302 | BR_TEXTO_LIVRE | Maria · MARIA · Ana · ANA |
| br-titulo-dv-1.txt | 100 | BR_TITULO_DV_T1 | 00,000 · 01,013 · 02,026 · 03,039 |
| br-titulo-dv-2.txt | 1,100 | BR_TITULO_DV_T2 | 000,000 · 001,015 · 002,02A · 003,034 |
| br-titulo-dv-3.txt | 1,100 | BR_TITULO_DV_T3 | 000,000 · 001,017 · 002,023 · 003,03A |
| br-titulo-dv-4.txt | 1,100 | BR_TITULO_DV_T4 | 000,000 · 001,019 · 002,027 · 003,035 |

| FILE | LINES | USED BY | FIRST LINES |
|---|---|---|---|
| `br-titulo-dv-5.txt` | 1,100 | `BR_TITULO_DV_T5` | 00000,0000 · 00100,0116 · 00200,0213 · 00300,0302 |
| `br-zeros-1.txt` | 10 | `BR_TITULO_1_ZERO_TABELA` | 0,00 · 1,01 · 2,02 · 3,03 |
| `br-zeros-2.txt` | 10 | `BR_TITULO_2_ZEROS_TABELA` | 0,000 · 1,001 · 2,002 · 3,003 |

APPENDIX C

# Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

**E1** `produto` `prod` `item` `artigo` `empresa` `emp` `fantasia` `razao` `social` `comercial` `arquivo` `arq` `rua` `logradouro` `lograd` `bairro` `municipio` `mun` `cidade` `estado` `uf` `pais` `banco` `agencia` `plano` `campanha` `projeto` `servico` `tabela` `coluna` `campo` `usuario` `host` `servidor` `dominio` `marca` `modelo` `categoria` `tipo` `escola` `colegio` `instituicao` `curso` `disciplina` `materia` `documento` `doc` `centro` `unidade` `clinica` `hospital` `area` `departamento` `depto` `setor` `cargo` `funcao` `sistema` `app` `aplicacao` `grupo` `lista` `relatorio` `processo` `evento` `tarefa` `perfil` `papel` `classe` `objeto` `conta` `medicamento` `remedio` `diagnostico` `estudo` `programa` `beneficio` `partido` `sindicato` `religiao` `etnia` `povo` `idioma` `lingua` `lugar` `local` `loja` `fornecedor` `forn` `convenio` `contrato` `apolice` `seguradora` `operadora` `org` `organizacao` `orgao` `entidade` `parametro` `variavel` `metodo` `indice` `job` `script` `fila` `recurso` `rede` `dispositivo` `template` `imagem` `foto` `pagina` `site` `modulo` `menu` `opcao` `acao` `permissao` `imposto` `tributo` `moeda` `emissor` `nota` `fatura` `boleto` `pagamento` `formulario` `zona` `rota` `linha` `predio` `edificio` `tela` `chave` `tag` `rotulo` `titulo` `codigo` `config` `empregador` `imovel` `filial` `matriz` `loja` `evento` `exame` `procedimento` `vacina` `turma` `sala` `equipe` `time` `projeto` `natureza` `motivo` `situacao` `status` `fase` `etapa` `origem` `destino` `canal` `campanha` `arquivo` `anexo` `file` `table` `column` `field` `user` `server` `product` `company` `brand` `category` `type` `school` `course` `document` `department` `system` `group` `report` `role` `account` `drug` `place` `store` `supplier` `contract` `policy` `device` `image` `page` `module` `action` `key` `label` `code` `org` `entity` `project` `city` `country` `street` `bank` `cliente` `cli` `paciente` `pac` `funcionario` `func` `empregado` `colaborador` `servidor` `segurado` `beneficiario` `benef` `titular` `dependente` `dep` `mae` `pai` `responsavel` `resp` `conjuge` `companheiro` `contato` `avalista` `fiador` `devedor` `tomador` `requerente` `solicitante` `testemunha` `representante` `procurador` `proprietario` `locatario` `locador` `inquilino` `vitima` `reu` `autor` `acusado` `investigado` `doador` `herdeiro` `pessoa` `pes` `motorista` `condutor` `cidadao` `eleitor` `contribuinte` `aluno` `estudante` `crianca` `medico` `enfermeiro` `professor` `socio` `tutor` `curador` `pagador` `favorecido` `destinatario` `remetente` `candidato` `associado` `membro` `cooperado` `consumidor` `comprador` `corretor` `interessado` `parte` `advogado` `usuario` `atendente` `vendedor` `gerente` `supervisor` `avo` `filho` `filha` `irmao` `parente` `customer` `patient` `employee` `member` `holder` `mother` `father` `spouse` `contact` `person` `student` `driver` `owner` `completo` `social` `civil` `sobrenome` `ultimo` `abreviado` `reduzido` `guerra` `usuario`

**E2** produto prod item artigo empresa emp fantasia razao social comercial arquivo arq rua logradouro lograd bairro municipio mun cidade estado uf pais banco agencia plano campanha projeto servico tabela coluna campo usuario host servidor dominio marca modelo categoria tipo escola colegio instituicao curso disciplina materia documento doc centro unidade clinica hospital area departamento depto setor cargo funcao sistema app aplicacao grupo lista relatorio processo evento tarefa perfil papel classe objeto conta medicamento remedio diagnostico estudo programa beneficio partido sindicato religiao etnia povo idioma lingua lugar local loja fornecedor forn convenio contrato apolice seguradora operadora org organizacao orgao entidade parametro variavel metodo indice job script fila recurso rede dispositivo template imagem foto pagina site modulo menu opcao acao permissao imposto tributo moeda emissor nota fatura boleto pagamento formulario zona rota linha predio edificio tela chave tag rotulo titulo codigo config empregador imovel filial matriz loja evento exame procedimento vacina turma sala equipe time projeto natureza motivo situacao status fase etapa origem destino canal campanha arquivo anexo file table column field user server product company brand category type school course document department system group report role account drug place store supplier contract policy device image page module action key label code org entity project city country street bank cliente cli paciente pac funcionario func empregado colaborador servidor segurado beneficiario benef titular dependente dep mae pai responsavel resp conjuge companheiro contato avalista fiador devedor tomador requerente solicitante testemunha representante procurador proprietario locatario locador inquilino vitima reu autor acusado investigado doador herdeiro pessoa pes motorista condutor cidadao eleitor contribuinte aluno estudante crianca medico enfermeiro professor socio tutor curador pagador favorecido destinatario remetente candidato associado membro cooperado consumidor comprador corretor interessado parte advogado usuario atendente vendedor gerente supervisor avo filho filha irmao parente customer patient employee member holder mother father spouse contact person student driver owner

**E3** produto prod item artigo empresa emp fantasia razao social comercial arquivo arq rua logradouro lograd bairro municipio mun cidade estado uf pais banco agencia plano campanha projeto servico tabela coluna campo usuario host servidor dominio marca modelo categoria tipo escola colegio instituicao curso disciplina materia documento doc centro unidade clinica hospital area departamento depto setor cargo funcao sistema app aplicacao grupo lista relatorio processo evento tarefa perfil papel classe objeto conta medicamento remedio diagnostico estudo programa beneficio partido sindicato religiao etnia povo idioma lingua lugar local loja fornecedor forn convenio contrato apolice seguradora operadora org organizacao orgao entidade parametro variavel metodo indice job script fila recurso rede dispositivo template imagem foto pagina site modulo menu opcao acao permissao imposto tributo moeda emissor nota fatura boleto pagamento formulario zona rota linha predio edificio tela chave tag rotulo titulo codigo config empregador imovel filial matriz loja evento exame procedimento vacina turma sala equipe time projeto natureza motivo situacao status fase etapa origem destino canal campanha arquivo anexo file table column field user server product company brand category type school course document department system group report role account drug place store supplier contract policy device image page module action key label code org entity project city country street bank cliente cli paciente pac funcionario func empregado colaborador servidor segurado beneficiario benef titular dependente dep mae pai responsavel resp conjuge companheiro contato avalista fiador devedor tomador requerente solicitante testemunha representante procurador proprietario locatario locador inquilino vitima reu autor acusado investigado doador herdeiro pessoa pes motorista condutor cidadao eleitor contribuinte aluno estudante crianca medico enfermeiro professor socio tutor curador pagador favorecido destinatario remetente candidato associado membro cooperado consumidor comprador corretor interessado parte advogado usuario atendente vendedor gerente supervisor avo filho filha irmao parente customer patient employee member holder mother father spouse contact person student driver owner full complete last first

**E4** ip mac email e_?mail eletronico web url tipo id cobranca_?id

| | |
|---|---|
| **E5** | `ip` `mac` `email` `e_?mail` `web` `url` `type` `id` |
| **E6** | `id` `cd` `cod` `codigo` `tp` `tipo` `dt` `data` `st` `status` `qt` `nr` `nu` |
| **E7** | `id` `tipo` `codigo` `data` `status` `fila` `log` |

APPENDIX D

# Discovery policy and classifiers

The discovery policy `BR – LGPD – v3` runs these 150 classifiers, with an assignment threshold of 60%.

LGPD - dados pessoais e sensíveis

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| `BR_L1_CPF – Path` | COLUMN NAME | `BR_L1_CPF` |
| `BR_L1_CPF – Type` | DATA TYPE | `BR_L1_CPF` |
| `BR_L1_CPF – Regex` | VALUE FORMAT | `BR_L1_CPF` |
| `BR_L1_CNPJ – Path` | COLUMN NAME | `BR_L1_CNPJ` |
| `BR_L1_CNPJ – Type` | DATA TYPE | `BR_L1_CNPJ` |
| `BR_L1_CNPJ – Regex` | VALUE FORMAT | `BR_L1_CNPJ` |
| `BR_L1_RG – Path` | COLUMN NAME | `BR_L1_RG` |
| `BR_L1_RG – Regex` | VALUE FORMAT | `BR_L1_RG` |
| `BR_L1_CNH – Path` | COLUMN NAME | `BR_L1_CNH` |
| `BR_L1_TITULO_ELEITOR – Path` | COLUMN NAME | `BR_L1_TITULO_ELEITOR` |
| `BR_L1_TITULO_ELEITOR – Regex` | VALUE FORMAT | `BR_L1_TITULO_ELEITOR` |
| `BR_L1_PIS_NIS – Path` | COLUMN NAME | `BR_L1_PIS_NIS` |
| `BR_L1_PIS_NIS – Regex` | VALUE FORMAT | `BR_L1_PIS_NIS` |
| `BR_L1_CTPS – Path` | COLUMN NAME | `BR_L1_CTPS` |
| `BR_L1_PASSAPORTE – Path` | COLUMN NAME | `BR_L1_PASSAPORTE` |
| `BR_L1_PASSAPORTE – Regex` | VALUE FORMAT | `BR_L1_PASSAPORTE` |
| `BR_L1_DOCUMENTO_ESTRANGEIRO – Path` | COLUMN NAME | `BR_L1_DOCUMENTO_ESTRANGEIRO` |
| `BR_L1_CERTIDAO – Path` | COLUMN NAME | `BR_L1_CERTIDAO` |
| `BR_L1_CERTIDAO – Regex` | VALUE FORMAT | `BR_L1_CERTIDAO` |
| `BR_L1_REGISTRO_PROFISSIONAL – Path` | COLUMN NAME | `BR_L1_REGISTRO_PROFISSIONAL` |
| `BR_L1_NOME – Path` | COLUMN NAME | `BR_L1_NOME` |
| `BR_L1_NOME – List` | KNOWN VALUES | `BR_L1_NOME` |
| `BR_L1_SOBRENOME – Path` | COLUMN NAME | `BR_L1_SOBRENOME` |
| `BR_L1_SOBRENOME – List` | KNOWN VALUES | `BR_L1_SOBRENOME` |
| `BR_L1_NOME_COMPLETO – Path` | COLUMN NAME | `BR_L1_NOME_COMPLETO` |
| `BR_L1_NOME_COMPLETO – List` | KNOWN VALUES | `BR_L1_NOME_COMPLETO` |
| `BR_L1_EMAIL – Path` | COLUMN NAME | `BR_L1_EMAIL` |
| `BR_L1_EMAIL – Type` | DATA TYPE | `BR_L1_EMAIL` |
| `BR_L1_EMAIL – Regex` | VALUE FORMAT | `BR_L1_EMAIL` |
| `BR_L1_TELEFONE – Path` | COLUMN NAME | `BR_L1_TELEFONE` |
| `BR_L1_TELEFONE – Regex` | VALUE FORMAT | `BR_L1_TELEFONE` |
| `BR_L1_ENDERECO – Path` | COLUMN NAME | `BR_L1_ENDERECO` |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| BR_L1_ENDERECO – Type | DATA TYPE | BR_L1_ENDERECO |
| BR_L1_ENDERECO – Regex | VALUE FORMAT | BR_L1_ENDERECO |
| BR_L1_ENDERECO_COMPLEMENTO – Path | COLUMN NAME | BR_L1_ENDERECO_COMPLEMENTO |
| BR_L1_CONTA_BANCARIA – Path | COLUMN NAME | BR_L1_CONTA_BANCARIA |
| BR_L1_CARTAO – Path | COLUMN NAME | BR_L1_CARTAO |
| BR_L1_CARTAO – Regex | VALUE FORMAT | BR_L1_CARTAO |
| BR_L1_CHAVE_PIX – Path | COLUMN NAME | BR_L1_CHAVE_PIX |
| BR_L1_CHAVE_PIX – Regex | VALUE FORMAT | BR_L1_CHAVE_PIX |
| BR_L1_IP – Path | COLUMN NAME | BR_L1_IP |
| BR_L1_IP – Regex | VALUE FORMAT | BR_L1_IP |
| BR_L1_DISPOSITIVO – Path | COLUMN NAME | BR_L1_DISPOSITIVO |
| BR_L1_DISPOSITIVO – Regex | VALUE FORMAT | BR_L1_DISPOSITIVO |
| BR_L1_PLACA – Path | COLUMN NAME | BR_L1_PLACA |
| BR_L1_PLACA – Regex | VALUE FORMAT | BR_L1_PLACA |
| BR_L1_RENAVAM – Path | COLUMN NAME | BR_L1_RENAVAM |
| BR_L1_CHASSI – Path | COLUMN NAME | BR_L1_CHASSI |
| BR_L1_CHASSI – Regex | VALUE FORMAT | BR_L1_CHASSI |
| BR_L1_IMOVEL – Path | COLUMN NAME | BR_L1_IMOVEL |
| BR_L1_CONTRATO – Path | COLUMN NAME | BR_L1_CONTRATO |
| BR_L1_USUARIO – Path | COLUMN NAME | BR_L1_USUARIO |
| BR_L1_CREDENCIAL – Path | COLUMN NAME | BR_L1_CREDENCIAL |
| BR_L1_CREDENCIAL – Regex | VALUE FORMAT | BR_L1_CREDENCIAL |
| BR_L1_GEOLOCALIZACAO – Path | COLUMN NAME | BR_L1_GEOLOCALIZACAO |
| BR_L1_GEOLOCALIZACAO – Regex | VALUE FORMAT | BR_L1_GEOLOCALIZACAO |
| BR_L1_TEXTO_LIVRE – Path | COLUMN NAME | BR_L1_TEXTO_LIVRE |
| BR_L1_TEXTO_LIVRE – Type | DATA TYPE | BR_L1_TEXTO_LIVRE |
| BR_L1_TEXTO_LIVRE – Regex | VALUE FORMAT | BR_L1_TEXTO_LIVRE |
| BR_L2_DATA_NASCIMENTO – Path | COLUMN NAME | BR_L2_DATA_NASCIMENTO |
| BR_L2_DATA_NASCIMENTO – Type | DATA TYPE | BR_L2_DATA_NASCIMENTO |
| BR_L2_ANO_NASCIMENTO – Path | COLUMN NAME | BR_L2_ANO_NASCIMENTO |
| BR_L2_ANO_NASCIMENTO – Type | DATA TYPE | BR_L2_ANO_NASCIMENTO |
| BR_L2_IDADE – Path | COLUMN NAME | BR_L2_IDADE |
| BR_L2_IDADE – Type | DATA TYPE | BR_L2_IDADE |
| BR_L2_DATA_EVENTO – Path | COLUMN NAME | BR_L2_DATA_EVENTO |
| BR_L2_DATA_EVENTO – Type | DATA TYPE | BR_L2_DATA_EVENTO |
| BR_L2_SEXO – Path | COLUMN NAME | BR_L2_SEXO |
| BR_L2_SEXO – List | KNOWN VALUES | BR_L2_SEXO |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| BR_L2_MUNICIPIO – Path | COLUMN NAME | BR_L2_MUNICIPIO |
| BR_L2_MUNICIPIO – Values | VALUE FORMAT | BR_L2_MUNICIPIO |
| BR_L2_CODIGO_MUNICIPIO – Path | COLUMN NAME | BR_L2_CODIGO_MUNICIPIO |
| BR_L2_CODIGO_MUNICIPIO – Regex | VALUE FORMAT | BR_L2_CODIGO_MUNICIPIO |
| BR_L2_CEP – Path | COLUMN NAME | BR_L2_CEP |
| BR_L2_CEP – Regex | VALUE FORMAT | BR_L2_CEP |
| BR_L2_BAIRRO – Path | COLUMN NAME | BR_L2_BAIRRO |
| BR_L2_ZONA_SECAO_ELEITORAL – Path | COLUMN NAME | BR_L2_ZONA_SECAO_ELEITORAL |
| BR_L2_NACIONALIDADE – Path | COLUMN NAME | BR_L2_NACIONALIDADE |
| BR_L2_NACIONALIDADE – List | KNOWN VALUES | BR_L2_NACIONALIDADE |
| BR_L2_ESTADO_CIVIL – Path | COLUMN NAME | BR_L2_ESTADO_CIVIL |
| BR_L2_ESTADO_CIVIL – Values | VALUE FORMAT | BR_L2_ESTADO_CIVIL |
| BR_L2_OCUPACAO – Path | COLUMN NAME | BR_L2_OCUPACAO |
| BR_L2_EMPREGADOR – Path | COLUMN NAME | BR_L2_EMPREGADOR |
| BR_L2_ESCOLARIDADE – Path | COLUMN NAME | BR_L2_ESCOLARIDADE |
| BR_L2_ESCOLARIDADE – Values | VALUE FORMAT | BR_L2_ESCOLARIDADE |
| BR_L2_ESCOLA – Path | COLUMN NAME | BR_L2_ESCOLA |
| BR_L2_DEPENDENTES – Path | COLUMN NAME | BR_L2_DEPENDENTES |
| BR_L2_DEPENDENTES – Type | DATA TYPE | BR_L2_DEPENDENTES |
| BR_L3_RACA_COR – Path | COLUMN NAME | BR_L3_RACA_COR |
| BR_L3_RACA_COR – Values | VALUE FORMAT | BR_L3_RACA_COR |
| BR_L3_ETNIA – Path | COLUMN NAME | BR_L3_ETNIA |
| BR_L3_ETNIA – Values | VALUE FORMAT | BR_L3_ETNIA |
| BR_L3_RELIGIAO – Path | COLUMN NAME | BR_L3_RELIGIAO |
| BR_L3_RELIGIAO – Values | VALUE FORMAT | BR_L3_RELIGIAO |
| BR_L3_CONVICCAO_ORGANIZACAO – Path | COLUMN NAME | BR_L3_CONVICCAO_ORGANIZACAO |
| BR_L3_OPINIAO_POLITICA – Path | COLUMN NAME | BR_L3_OPINIAO_POLITICA |
| BR_L3_OPINIAO_POLITICA – Values | VALUE FORMAT | BR_L3_OPINIAO_POLITICA |
| BR_L3_FILIACAO_PARTIDARIA – Path | COLUMN NAME | BR_L3_FILIACAO_PARTIDARIA |
| BR_L3_FILIACAO_PARTIDARIA – Values | VALUE FORMAT | BR_L3_FILIACAO_PARTIDARIA |
| BR_L3_FILIACAO_SINDICAL – Path | COLUMN NAME | BR_L3_FILIACAO_SINDICAL |
| BR_L3_FILIACAO_SINDICAL – Values | VALUE FORMAT | BR_L3_FILIACAO_SINDICAL |
| BR_L3_GENETICO – Path | COLUMN NAME | BR_L3_GENETICO |
| BR_L3_GENETICO – Regex | VALUE FORMAT | BR_L3_GENETICO |
| BR_L3_BIOMETRICO – Path | COLUMN NAME | BR_L3_BIOMETRICO |
| BR_L3_BIOMETRICO – Type | DATA TYPE | BR_L3_BIOMETRICO |
| BR_L3_ORIENTACAO_SEXUAL – Path | COLUMN NAME | BR_L3_ORIENTACAO_SEXUAL |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| BR_L3_ORIENTACAO_SEXUAL – Values | VALUE FORMAT | BR_L3_ORIENTACAO_SEXUAL |
| BR_L3_VIDA_SEXUAL – Path | COLUMN NAME | BR_L3_VIDA_SEXUAL |
| BR_L3_IDENTIDADE_GENERO – Path | COLUMN NAME | BR_L3_IDENTIDADE_GENERO |
| BR_L3_IDENTIDADE_GENERO – Values | VALUE FORMAT | BR_L3_IDENTIDADE_GENERO |
| BR_SAUDE_CNS – Path | COLUMN NAME | BR_SAUDE_CNS |
| BR_SAUDE_CNS – Regex | VALUE FORMAT | BR_SAUDE_CNS |
| BR_SAUDE_IDENTIFICADOR – Path | COLUMN NAME | BR_SAUDE_IDENTIFICADOR |
| BR_SAUDE_DIAGNOSTICO – Path | COLUMN NAME | BR_SAUDE_DIAGNOSTICO |
| BR_SAUDE_DIAGNOSTICO – Regex | VALUE FORMAT | BR_SAUDE_DIAGNOSTICO |
| BR_SAUDE_MEDICAMENTO – Path | COLUMN NAME | BR_SAUDE_MEDICAMENTO |
| BR_SAUDE_MEDICAMENTO – Values | VALUE FORMAT | BR_SAUDE_MEDICAMENTO |
| BR_SAUDE_TEXTO_CLINICO – Path | COLUMN NAME | BR_SAUDE_TEXTO_CLINICO |
| BR_SAUDE_TEXTO_CLINICO – Type | DATA TYPE | BR_SAUDE_TEXTO_CLINICO |
| BR_SAUDE_RESULTADO_EXAME – Path | COLUMN NAME | BR_SAUDE_RESULTADO_EXAME |
| BR_SAUDE_TIPO_SANGUINEO – Path | COLUMN NAME | BR_SAUDE_TIPO_SANGUINEO |
| BR_SAUDE_TIPO_SANGUINEO – Values | VALUE FORMAT | BR_SAUDE_TIPO_SANGUINEO |
| BR_SAUDE_DEFICIENCIA – Path | COLUMN NAME | BR_SAUDE_DEFICIENCIA |
| BR_SAUDE_DEFICIENCIA – Values | VALUE FORMAT | BR_SAUDE_DEFICIENCIA |
| BR_SAUDE_OCUPACIONAL – Path | COLUMN NAME | BR_SAUDE_OCUPACIONAL |
| BR_SAUDE_OCUPACIONAL – Values | VALUE FORMAT | BR_SAUDE_OCUPACIONAL |
| BR_SAUDE_REPRODUTIVA – Path | COLUMN NAME | BR_SAUDE_REPRODUTIVA |
| BR_SAUDE_MENTAL – Path | COLUMN NAME | BR_SAUDE_MENTAL |
| BR_FIN_CREDITO – Path | COLUMN NAME | BR_FIN_CREDITO |
| BR_FIN_DIVIDA – Path | COLUMN NAME | BR_FIN_DIVIDA |
| BR_FIN_DIVIDA – Type | DATA TYPE | BR_FIN_DIVIDA |
| BR_FIN_RENDA – Path | COLUMN NAME | BR_FIN_RENDA |
| BR_FIN_RENDA – Type | DATA TYPE | BR_FIN_RENDA |
| BR_FIN_PATRIMONIO – Path | COLUMN NAME | BR_FIN_PATRIMONIO |
| BR_FIN_PATRIMONIO – Type | DATA TYPE | BR_FIN_PATRIMONIO |
| BR_FIN_VALOR_BENEFICIO – Path | COLUMN NAME | BR_FIN_VALOR_BENEFICIO |
| BR_FIN_VALOR_BENEFICIO – Type | DATA TYPE | BR_FIN_VALOR_BENEFICIO |
| BR_FIN_BENEFICIO_INSS – Path | COLUMN NAME | BR_FIN_BENEFICIO_INSS |
| BR_FIN_BENEFICIO_INSS – Values | VALUE FORMAT | BR_FIN_BENEFICIO_INSS |
| BR_FIN_NUMERO_BENEFICIO – Path | COLUMN NAME | BR_FIN_NUMERO_BENEFICIO |
| BR_FIN_NUMERO_BENEFICIO – Regex | VALUE FORMAT | BR_FIN_NUMERO_BENEFICIO |
| BR_FIN_PROGRAMA_SOCIAL – Path | COLUMN NAME | BR_FIN_PROGRAMA_SOCIAL |
| BR_FIN_PROGRAMA_SOCIAL – Values | VALUE FORMAT | BR_FIN_PROGRAMA_SOCIAL |

| CLASSIFIER | KIND | DATA CLASS |
|---|---|---|
| BR_FIN_FAIXA_RENDA – Path | COLUMN NAME | BR_FIN_FAIXA_RENDA |
| BR_FIN_FAIXA_RENDA – Values | VALUE FORMAT | BR_FIN_FAIXA_RENDA |
| BR_FIN_MORADIA – Path | COLUMN NAME | BR_FIN_MORADIA |
| BR_FIN_MORADIA – Values | VALUE FORMAT | BR_FIN_MORADIA |
| BR_PENAL_ANTECEDENTES – Path | COLUMN NAME | BR_PENAL_ANTECEDENTES |
| BR_PENAL_PROCESSO – Path | COLUMN NAME | BR_PENAL_PROCESSO |
| BR_PENAL_PROCESSO – Regex | VALUE FORMAT | BR_PENAL_PROCESSO |